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Beyond IID: How General Are Tabular Foundation Models, Really?

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Abstract

Foundation models for predictive machine learning on tabular data have recently gained significant traction in academia and industry. Research communities across disciplines are increasingly evaluating tabular foundation models on diverse datasets and tasks. However, these task- and discipline-specific evaluations remain largely inaccessible to model researchers because benchmark software and evaluation protocols are fragmented. As a result, model researchers rely on standard benchmarks, which are mostly defined for tasks where tabular foundation models already excel. The most challenging scenarios are excluded, limiting meaningful progress in the field by focusing on marginal improvements on IID data rather than on broader, more demanding challenges. To overcome this, we introduce `BeyondArena`, the first unified holistic benchmark for tabular data that supports diverse task types (IID, temporal, grouped), across sample size and feature dimensionality scales, with diverse feature types (with text, with high cardinality) from a broad range of disciplines. To enable unified benchmarking beyond standard benchmarks, we introduce `DataFoundry`, a Python framework and metadata schema for curating tabular datasets for predictive machine learning. Our results across 11 models and 142 curated datasets show that existing tabular foundation models excel on tiny- to medium-sized IID data, while traditional tree-based and deep learning models still dominate on non-IID, large, and high-dimensional datasets. `BeyondArena` guides model research for the most demanding challenges in tabular data, enabling progress towards truly foundational tabular models.

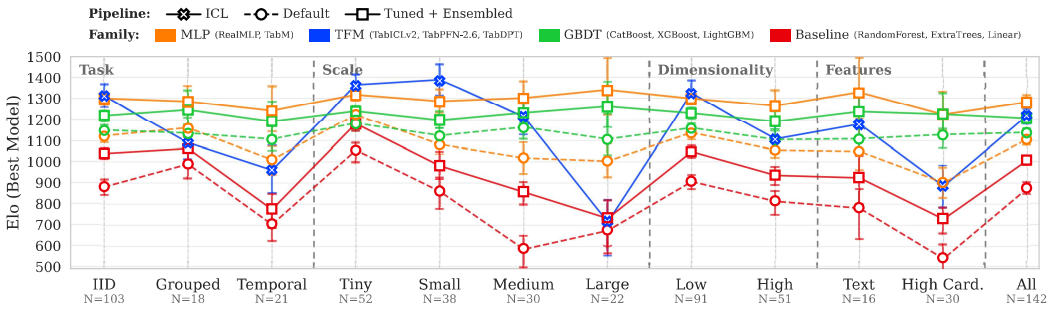

Figure 1: **BeyondArena Results.** We show the Elo of the best model per family across sub-benchmarks, comparing state-of-the-art MLPs, gradient-boosted decision trees (GBDT), tabular foundation models (TFM), and simple baselines. For TFMs, we evaluate in-context learning (ICL) performance. For traditional models, we evaluate performance with default hyperparameters and with tuning plus post-hoc ensembling. We show results per model in Figure 5.

1 Introduction

Tabular foundation models (TFMs) have been evaluated across many research communities, ranging from graph learning [1–4] or surrogate modeling [5–7], to real-world applications such as Cybersecurity [8], soil mapping [9], or clinical predictions [10]. While the broader research community evaluates TFMs across diverse tasks and datasets, the tabular research community is so far limited to evaluations on IID tasks [11–13]. Moreover, historically, projects that broaden evaluation diversity were often developed with independent benchmarking infrastructure and scientific standards, such as PMLB-Mini [14] for tiny data, TabReD [15] for tabular-temporal tasks, or TextTabBench [16] for tabular-text data. The current state of benchmarking for machine learning on tabular data is not only confusing to practitioners but also impeding researchers.

We aim to unify benchmarking and accelerate research on TFMs. Thus, we introduce `BeyondArena`, a holistic benchmark for machine learning on tabular data that unifies independent and identically distributed (IID) and non-IID task types (IID, temporal, grouped) across datasets with varying sample scales (100-1k, 1k-10k, 10k-100k, 100k to 1 million), dimensionality scales ($\leq$100, $>$100), with challenging feature types (high-cardinality categories, or text) from a broad range of applied disciplines. To create `BeyondArena`, we curate a large collection of representative datasets and provide reliable baselines. Furthermore, we unify the tabular benchmarking community by integrating our benchmark into TabArena's [11] open-source ecosystem. To summarize, **our contributions are:**

- ❖ We manually curate 142 datasets from 1128 – following rigorous protocols – to obtain high-quality tabular datasets across scales and feature types, and publish `DataFoundry`, a Python framework and metadata schema for reproducible tabular data curation;
- ❖ We systematically evaluate whether 3 state-of-the-art open-source TFMs generalize beyond IID settings by comparing them with 8 strong traditional baselines across diverse tasks;
- ❖ We integrate all our contributions into TabArena's open-source benchmark ecosystem, thereby avoiding fragmentation of the research community: code at `https://tabarena.ai/code`, data at `https://github.com/TabArena/data-foundry`.

Our driving motivation behind `BeyondArena` is to understand how well TFMs would perform when practitioners use them in real-world predictive applications, thereby closing the gap between evaluations in academia and real applications. In this paper, we detail our dataset curation, standardize benchmarking across task types, and evaluate the strengths and weaknesses of existing TFMs.

BeyondArena focuses on evaluating predictive machine learning models for tabular classification and regression on non-IID data, ranging from tiny to large (100-1M), with diverse feature types (high-cardinality categorical and text features). Out of scope are few-shot predictions ($<$100), images as features, and other tasks such as survival analysis or relational learning.

Our results demonstrate that state-of-the-art TFMs dominate on tiny, small, and IID data, while they fail to compete with traditional tree-based and deep learning models on non-IID (temporal, grouped), large-scale, high-dimensional, and high-cardinality categorical datasets. We further ablate our experimental setup to assess the validity of our claims and show the importance of (i) appropriate outer test splits for grouped data; (ii) appropriate inner validation splits for non-IID or tiny data; (iii) preprocessing for grouped data and text features; (iv) and calibrating probabilities.

2 Background: IID and non-IID Tabular Data

Tabular foundation models and, in extension, tabular deep learning have been evaluated in the broader research community across countless applications, from computer science disciplines (cf. [1–7, 17–36]) to applied science disciplines (cf. [8–10, 37–74]). To categorize such broad application data, we adopt an application-dependent definition of independent and identically distributed (IID) and non-IID data. We determine whether a dataset is IID or non-IID based on the appropriate train-test split. We deem the split appropriate that most closely mirrors the original real-world application.

Illustration. Two practitioners could work with the same data but for fundamentally different applications. Practitioner Alice uses the data to predict whether past transactions were fraudulent for follow-up fraud investigations. Practitioner Bob uses the data to predict whether new transactions are fraudulent, aiming to prevent fraud in real time. To estimate model performance, Alice uses a random IID split, while Bob uses a non-IID temporal split to simulate the expected distribution of unseen transactions when the model is deployed. If Bob were to use a random split, it would lead

to overestimating the performance of models that can exploit temporal leakage [15, 75]. On the contrary, Alice cannot use a temporal split, as this would underestimate the performance of such models. The appropriate split depends on the practitioner's application. `BeyondArena` extends this line of thinking from practitioners to data curation and academic benchmarking.

IID Tabular Data. We define data as IID if the test samples in the associated application do not follow a particular structure, allowing a random split. That is, we hold out randomly sampled data.

Non-IID Tabular Data. We define data as non-IID when the application requires a temporal or grouped split. In a temporal split, a time index is required, and test samples occur strictly after the training data, reflecting prediction for observations in the future (e.g., future transactions). In a grouped split, a group index is required, and all samples with the same index are kept together, so no group appears in both the training and test sets. Such applications aim to generalize to samples associated with unseen entities (groups). Grouped tasks can be of type `label-per-group`, where all samples in a group share the same label, or `label-per-sample`, where each sample has its own label. For `label-per-group`, the objective is to predict a group-level label (e.g., if a customer is fraudulent based on a collection of transactions). For `label-per-sample`, the objective is to predict individual sample labels for data sources not seen during training (e.g., a new country or hospital).
For some datasets, either a temporal or a grouped split is plausible. Using a temporal split does not remove group structures, and using a grouped split does not make the task time-invariant. The split only determines the importance of grouped or temporal dependencies – it does not remove them.
`BeyondArena` does not include data from time-series forecasting applications, despite the data being tabular-like and non-IID [76, 77]. During data curation, we distinguish data stemming from temporal tabular regression and time-series forecasting tasks, as they have fundamentally different assumptions, necessitating different validation procedures; see Appendix A.1. Nevertheless, tabular models can be used for time-series forecasting tasks [21, 78–80], and can be competitive [81–84].

3 Related Work

As a unified tabular benchmark, `BeyondArena` relates to IID and non-IID tabular benchmarks.

IID Tabular Benchmarks. The vast majority of prior benchmarks for predictive machine learning on tabular data focused on IID tasks, often inappropriately treating all data as stemming from IID applications. Benchmarks were created during model development [85–88], for tabular deep learning [89–96], to develop AutoML systems [97–102], or from a data-centric view [103–110], all aiming to understand and compare models. Recent benchmark projects have focused on broadening evaluation diversity by supporting few-shot or tiny data [14, 111], including diverse domains and large data [12, 112], or by curating datasets representative of real-world tasks [11]. `BeyondArena` builds on prior benchmarking efforts by extending their high-quality work beyond IID data.
In parallel, the research community developed benchmarks for multimodal tabular data. These efforts concentrated on tabular data with text [16, 113–121] or to benchmark LLMs solving predictive tasks by treating tables as text [122–135]. Tabular data with images has received little benchmarking attention [116, 136, 137], yet there have been application-specific model comparisons [115, 117, 138–147]. Time-series classification or regression benchmarks [148–151] are also multimodal IID benchmarks. They use random splits, and the state-of-the-art includes tabular models with feature engineering [152–154] or pretrained encoders [155, 156]. `BeyondArena` makes a first step toward unifying multimodal benchmarking by incorporating IID and non-IID tabular data with text. We leave tabular data with images and data from time-series classification or regression for future work.

Non-IID Tabular Benchmarks. Benchmarks for non-IID tabular data focused mainly on temporal data. While the broader research community worked on non-IID tabular data for decades [157–164], only recently Rubachev et al. [15] identified temporal tabular data as a gap in tabular deep learning benchmarks and introduced TabReD, enabling new research [165, 166]. `BeyondArena` builds upon TabReD by supporting more temporal datasets, more baselines, and tabular foundation models. Moreover, `BeyondArena` enables research on IID and temporal tabular data in a single framework.
There have been no dedicated benchmarks for tabular deep learning on grouped data. Many applications of grouped tabular data originate from longitudinal data [56, 167–170], algorithm selection problems [171–173], or distribution shift benchmarks [174–179]. `BeyondArena` is the first benchmark to incorporate grouped data from all of these fields. Thus, `BeyondArena` fills a gap in non-IID benchmarking and enables new research on models that generalize across diverse application domains.

To summarize, compared to related work, we unify IID and non-IID benchmarking in `BeyondArena` by curating representative datasets in one shared metadata format (Section 4) and enabling rigorous evaluation of state-of-the-art tabular models on all tasks and multimodal data types (Section 5).

4 Dataset Curation

Our curation process aims to extend tabular benchmarking to better represent more challenging real-world predictive machine learning applications. Therefore, we overhaul the selection criteria from TabArena [11] and extend them to `BeyondArena`'s focus by supporting: (**I**) non-IID data (temporal and grouped); (**II**) data with less than 500 or more than 100 000 training samples; and (**III**) tabular data with text and date features. To ensure selected datasets are high-quality and representative of `BeyondArena`'s focus, we opted for a manual curation process in which all results were verified by humans[1]. While this choice was extremely labor-intensive, it is crucial to the scientific rigor, as the data quality directly affects the correctness of the benchmark's conclusions [15, 75, 109, 110].

Data Foundry. To enable such extensive dataset curation and make it reproducible, we introduce `DataFoundry`, a Python framework and metadata schema for tabular datasets and predictive machine learning: `https://github.com/TabArena/data-foundry`. `DataFoundry` contains: (1) a mature Python package for dataset curation, and (2) the exact code we used to process each dataset in easy-to-understand Python notebooks (.ipynb). The Python package delivers code for dataset checks, train-test splits, and an extensible metadata schema. In addition, `DataFoundry` has one Python notebook for each dataset in `BeyondArena`. Each notebook contains reproducible code for metadata edits, preprocessing, dataset checks, train-test splits, and exporting the artifact for consumption by benchmarks, data repositories (such as OpenML, Kaggle, or Hugging Face), or other applications.

Dataset Sources. We gathered datasets from 21 tabular benchmark studies and searched for new datasets in public data repositories. In particular, we re-evaluated 304 datasets from 14 prior tabular benchmarks considered for the TabArena curation, comprising all accepted datasets and those rejected for being non-IID, too small, or for other data issues. We further collected datasets from 7 non-IID or multimodal tabular benchmarks: TabReD [15], TableShift [177], the string vectorizing benchmark [114], the CARTE benchmark [118], TextTabBench [16], the TabSTAR benchmark [180], and the AutoGluon Multimodal benchmark [116, 137, 181]. Furthermore, we searched for new datasets by browsing data repositories or competition websites (UCI Machine Learning Repository [182], OpenML [183, 184], Hugging Face [185], Kaggle [186], Zindi [187], ASlib [172]) or public domain government websites. Our search was guided by popularity, task, and application domain, while avoiding obvious duplicates, to provide diverse input for our curation process.

Dataset Selection Criteria. A dataset had to fulfill the following criteria to become part of `BeyondArena`: (**1**) The dataset and its predictive machine learning task are unique within our benchmark; (**2**) The dataset is not a few-shot prediction task, that is, it has at least 100 train samples; (**3**) The dataset is from a tabular IID or non-IID task, that is, a random, temporal, or grouped split is the appropriate validation protocol; (**4**) The dataset was published explicitly for a predictive classification or regression task; (**5**) The dataset and its task are representative of a problem in a real-world application that a practitioner would solve with tabular machine learning models, that is, the dataset stems from a real random distribution (e.g., the dataset is not generated from a deterministic function), and tabular models are a suitable, competitive choice (e.g., the dataset is not a vectorized version of ImageNet). (**6**) The dataset and its predictive task do not raise (obvious) ethical concerns.
Compared to the criteria by Erickson et al. [11], we made bigger but also nuanced changes. The bigger changes are that we allow non-IID data, relax the modality requirements, and allow the use of larger datasets with subsampling, making any dataset with at least 100 training samples in scope. Nuanced changes include a more precise definition of duplicates, a clearer distinction from tabular-adjacent predictive tasks, and a sharper definition of what constitutes a representative dataset for predictive classification or regression tasks. Furthermore, although some of our datasets overlap with the inclusion criteria of the TabArena datasets, we show that ours are more challenging (Appendix D.3). We provide a detailed description in Appendix B.2 and share our curation insights per dataset at `https://tabarena.github.io/data-foundry/`.

Dataset Processing. Each selected dataset was thoroughly investigated and integrated into `DataFoundry`. By doing so, we unified the data format (e.g., .csv, .parquet, or .xlsx) and feature

[1] We investigated (semi-)automated agentic procedures with limited success; see Appendix B.1 for details.

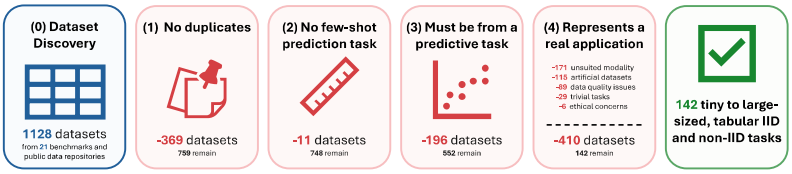

Figure 2: **Dataset Selection Results.** We present the main reasons for filtering datasets (1-4). Within (4), we also show sub-categories. Roughly 12.6% of all investigated datasets were selected.

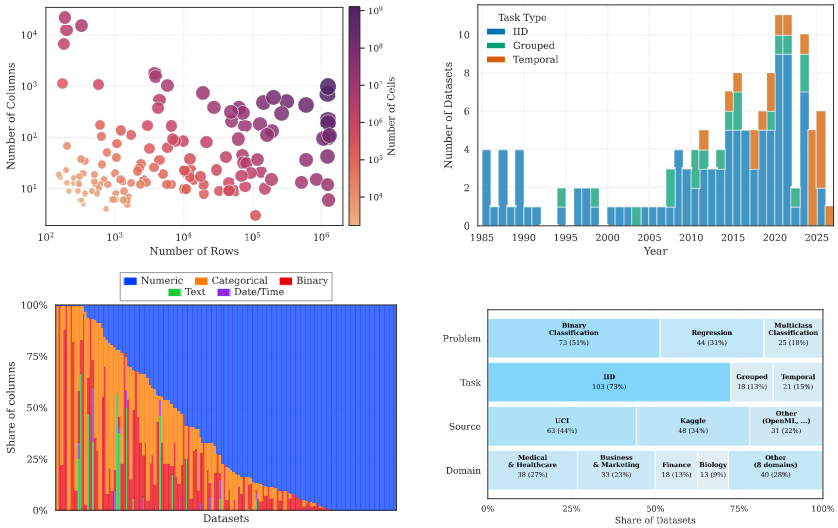

Figure 3: **Dataset Dashboard. (Top Left):** Dataset sizes (w.r.t. rows, columns, and cells). **(Top Right):** Age distribution. **(Bottom Left):** Distribution of feature types per dataset. **(Bottom Right):** The share of datasets from a specific problem type, task type, dataset source, or application domain.

type annotations (numbers, categorical, dates, strings). Moreover, we store all necessary metadata for the task (e.g., train-test splits, target column, columns for the group or time index), as well as annotations from our curation process (e.g., notes on data anomalies), in a machine-readable data schema. Whenever necessary, we also apply feature engineering to create the appropriate task, following the task description in the paper or the top solutions on competition websites. We share details and some general guidelines that we developed during the curation process in Appendix B.3.

Curation Outcome. We collected 1128 datasets, of which 142 met our selection criteria. As shown in Figure 2, the majority of datasets were duplicates, did not represent a real application, or were not originally from a predictive task. We summarize the characteristics of our dataset collection in Figure 3, with extended overviews in Appendix B.4. `BeyondArena` includes datasets that range from tiny to large-scale, are recent, and are diverse across feature types, problem types, task types, sources, and application domains. We provide details per dataset in Appendix B.5 and share their `DataFoundry` artifacts at `https://huggingface.co/datasets/TabArena/BeyondArena`.

5 Experimental Design

We follow best practices for evaluating tabular machine learning models. We compare state-of-the-art models with verified open-source implementations, perform reliable validation and tuning, use industry-standard preprocessing, employ robust metrics, and rigorously repeat our experiments.

BeyondArena Models. We compare 11 tabular machine learning models: Linear / Logistic Regression, Random Forest [188], Extremely Randomized Trees [189], CatBoost [190], LightGBM [191], XGBoost [192], RealMLP [85], TabM [87], TabDPT [193], TabPFN-2.6 [194, 195], and TabICLv2 [196]. We select models that performed competitively on the TabArena live leaderboard [197], TALENT [112], and TabReD [15], including a linear model, tree-based models, neural

networks, and TFMs. We run TabPFN-2.6 and TabDPT only for datasets with up to $100k$ training samples. For TabPFN-2.6, we stay within its designated pretraining limits. We limit TabDPT due to the infeasible cost of retrieval-based inference for benchmarking: for a dataset with 1 million samples under cross-validation, TabDPT needs 8 million forward passes, 8 per sample. We imputed missing results for TabPFN-2.6 and TabDPT using the performance of a default Random Forest.

Open-source Implementation. We build `BeyondArena` on top of TabArena [11], using its model implementations and benchmark infrastructure. As a result, all models we use were unit-tested, implemented in the standardized AutoGluon-based model framework [181], and are open-source – ensuring high-quality, reproducible benchmarking. We modified implementations to scale them to larger data and ensure proper behavior for non-IID tasks (e.g., handling unseen categories). Moreover, we added non-IID validation protocols and new (multimodal) preprocessing to TabArena.

Outer Splits. We create train-test splits for a dataset differently across sample sizes to guard against noise biasing the results while remaining conservative with the compute budget. We split and subsample datasets with more than 1.25 million samples into 1 million training and 250k test samples. For IID tasks, we use repeated 3-fold cross-validation for datasets with fewer samples. We perform n-repeated cross-validation with $n = 3$ for 2500-250k training samples; $n = 10$ for 500-2500; and $n = 20$ for fewer than 500. For grouped tasks, we do the same but utilize group-based cross-validation. We stratify splits on the target variable for IID and grouped classification tasks. For temporal tasks, we manually create application-specific temporal splits. The application determines the appropriate prediction time horizon for the test data. We roll back the time horizon to create multiple split time points for multiple train-test splits. At each time point, we use all data before for training and all data after within the time horizon for testing. We create the same number of splits as for IID or grouped tasks, but we do not use splits with $< 50\%$ of the original data as training data. We deem a split with $< 50\%$ too unrepresentative of the original application.

Inner Validation and Tuning Protocol. After the outer train-test split, we split the training data into 8 folds for cross-validation (CV) to estimate model performance during tuning. For datasets with < 500 training samples, we use 5-repeated 5-fold CV to avoid overtuning [198, 199]. We stratify the splits for classification. For IID tasks, we use random splits. For grouped tasks, we use grouped splits based on the group index. For temporal tasks, we bin the time index and create intervals used as folds for CV, a popular procedure on Kaggle, and following Bergmeir and Benítez [159].
We tune models using the search spaces by Erickson et al. [11], due to a limited compute budget, with 25 configurations from random search. For TFMs (TabDPT, TabPFN-2.6, TabICLv2), we evaluate only in-context learning without (fine-)tuning to assess their pretrained generality using the default preprocessing pipeline and no dataset-specific weight updates.
We set a 4-hour time limit for CV, plus 1 hour of overhead for predicting on test data. We increase the time limit to 12 hours for TabPFN-2.6 and TabICLv2, as the 5-hour limit was insufficient for larger datasets. We also had to reduce TabICLv2's ensemble member count to 1 for the 5 largest datasets. Unlike traditional models, existing TFMs do not support early stopping to respect predefined time constraints [200]. Only $\sim 0.01\%$ of jobs exceed 3.5 hours; see Appendix C.1 for details.

Multimodal and non-IID Preprocessing. We adjust TabArena's preprocessing pipeline to handle multimodal non-IID data. We use a single general preprocessing pipeline (model-agnostic) and extensions to it per model (model-specific). We retain the model-specific pipelines from TabArena, which handle, when needed, steps such as scaling or categorical encoding. Our new model-agnostic preprocessing pipeline handles date encoding, text encoding, and preprocessing for grouped data.
We use skrub [201] to convert datetime features into 10 numerical features, representing the weekday or a spline-based periodic encoding. We encode text features using `Qwen3-Embedding-8B` [202], the best zero-shot multilingual text encoding model based on the MMTEB leaderboard [203, 204], into a 32-dimensional vector using Qwen3's smallest Matryoshka representation learning [205] slice. Lastly, we add preprocessing for grouped non-IID data. For `label-per-sample` datasets, we follow common practice and drop the group-index column to prevent models from overfitting to it during training. For `label-per-group` datasets, we replace the group index with a 50-dimensional group-encoding vector, similar to common practice in Kaggle competitions (e.g., see the top solutions in the AMEX competition [206]). For more details on all preprocessing steps, see Appendix C.2.

Compute Resources. We used CPU/GPU virtual machines via GCP [207]; for specifications, see Appendix C.2. The estimated cost of `BeyondArena` was $\sim \$50k$ and took ~ 16.25 wall-clock years.

Metrics. We chose metrics aligning with prior benchmarks [11, 15, 101] and machine learning literature [208–210]. We use ROC AUC for binary classification, log-loss for multiclass classification, and RMSE for regression. We chose ROC AUC because it is invariant to threshold tuning for binary classification, unlike accuracy as used by TALENT [12]. For multiclass classification, we use log-loss because it is a proper scoring rule [209, 211]. We use RMSE, a point prediction metric, for regression, and do not use probabilistic prediction metrics [23, 212, 213]. Point predictions are closer to the application's objectives across almost all datasets we investigated. Thus, RMSE is more representative for practitioners. We normalized and aggregated the model performances with Elo [214] and mean Improvability [11], using implementations from TabArena. Improvability ($0\% - 100\%$) measures error relative to the best method on each outer split, making it sensitive to performance gaps. Elo ratings are computed by modeling pairwise comparisons and reflect the probability of winning. We always calibrate 1000 Elo to be the performance of a default XGBoost.

6 Results

`BeyondArena` aims to understand how well tabular foundation models would perform when practitioners use them in real-world predictive applications. Therefore, we first assess the extent to which tabular foundation models are truly general, and then investigate the validity of our results through ablation studies. To investigate the performance with respect to dataset characteristics, we define 12 **sub-benchmarks**: one per task type (**IID**, **Grouped**, **Temporal**); four for dataset scale (**Tiny** for datasets with $100 \leq n < 1k$ training rows, **Small** for $1k \leq n < 10k$, **Medium** for $10k \leq n < 100k$, **Large** for $100k \leq n$); two for dataset dimensionality (**Low** for datasets with $m \leq 100$ columns after preprocessing, **High** for $m > 100$); two for special feature types (**Text** for datasets with text features, **High Card.** for datasets with at least one categorical column that has more than 50 categories); and finally the full benchmark containing **All** datasets.
Figure 4 shows the Elo and Improvability for all methods on all datasets. Figure 5 presents the Elo per sub-benchmark, and Figure 1 shows the Elo of the best model per family.

When do tabular foundation models outperform traditional models? TFMs perform best on small, tiny, and IID datasets (Figure 1). Moreover, TFMs are also the best on average when ranked by Improvability, or when comparing default performance (Figure 4, Leaderboard as table in Appendix D.2). However, traditional approaches – particularly deep learning models such as RealMLP – benefit substantially from tuning and ensembling. Consequently, the in-context learning performance of all TFMs fails to compete with traditional models on non-IID (temporal and grouped), large-scale, high-dimensional, and high-cardinality datasets (Figure 5). In general, as sub-benchmarks are not independent (Section 4), bad performance on one (e.g., large) might also explain bad performance on another (e.g., temporal). Lastly, we observe that the default training costs, including cross-validation, of TFMs remain significantly lower than those of their tuned competitors (Figure 4, right), and continue to have large inference costs (Appendix D.1). We present results per dataset in Appendix G. To ground our findings in formal statistical evidence, we complement them with a suite of non-parametric tests in Appendix E. In general, our analysis shows that the global ranking and pairwise comparisons across models and sub-benchmarks are significant. In detail, we answer 5 concrete questions about our results, such as whether the methods really perform differently (§E.1) or which method a practitioner should pick for each data regime (§E.5).

When are TFMs all you need for peak-performance? In Figure 6, we investigate the ability of TFMs to achieve peak performance (rank 1) on individual datasets, showing that (**1**) TabICLv2 ranks first on 19% of all datasets, followed by TabPFN-2.6 (10.5%). (**2**) TFMs significantly outperform other models on 49% of datasets, and at least one of the TFMs performs as well as the best non-TFM on an additional 21%, making TFMs a viable solution to get peak performance on 70% of all datasets. (**3**) On grouped, temporal, and large datasets, TFMs attain peak performance only on a small fraction of datasets. Overall, TFMs are clearly outperformed on 42 datasets, which are large, high-dimensional, non-IID, or include high-cardinality categorical features, illustrating the need for unified, holistic benchmarks to further improve TFMs across a wider variety of datasets.

BeyondArena Ablations

To better understand `BeyondArena` and inform future benchmarks, we ablate our experiment setup as described in Section 5. We summarize the results for the majority of our ablations in Table 1.

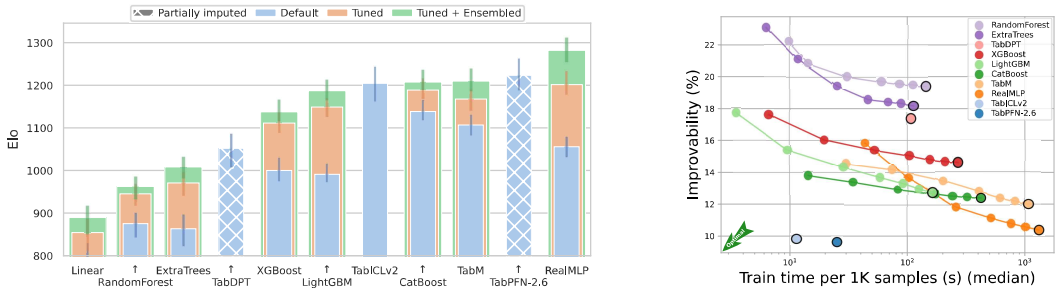

Figure 4: **BeyondArena Leaderboard (Left=Elo, Right=Improvability).** We show the performance of 11 models across 142 datasets with the default hyperparameters, with tuning, and with tuning and post-hoc ensembling. The default performance of TFMs equals their in-context learning performance.

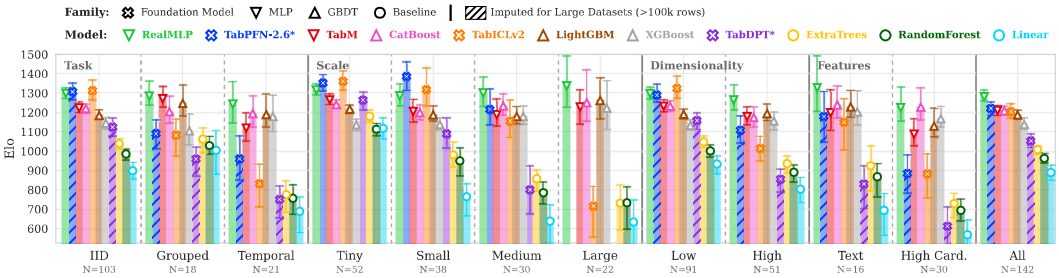

Figure 5: **Elo per model across sub-benchmarks.** For each sub-benchmark of `BeyondArena`, we show the performance of each traditional model with tuning and post-hoc ensembling. For TFMs, we show their in-context learning performance. We aggregate results per model family in Figure 1.

Different Outer Splits. **(A.1)** We use up to 60 outer validation splits in `BeyondArena`, which drastically increases compute costs. To reduce environmental impact and enable compute-constrained researchers, we would like to use fewer outer splits. Erickson et al. [11] introduced `TabArena-Lite`, saving compute by using only the first split per dataset, which produces a comparable ranking *on average*. We instead introduce `BeyondArena-Core`, by automatically selecting a subset of splits per dataset, ensuring that the ranking *per dataset* on the subset is comparable to that obtained with all splits. Compared to `BeyondArena`, `-Lite` would be $\times 9$ and `-Core` $\times 5$ faster. Yet, the per dataset win rates of `-Core` are $\times 2.3$ more stable than `-Lite`; see Appendix F.1 for details.
(A.2) Besides using fewer outer splits, we ablate using IID splits for non-IID data. Prior work has shown that inappropriate splits confound the evaluation of temporal data, cf. [15, 75]. In Appendix F.2, we show that, for two exemplary grouped datasets, inappropriate outer splits can drastically distort model rankings ($\tau = 0.49$ and $\tau = 0.60$) and raw performance estimates. Hence, it is pivotal to use appropriate non-IID splits when evaluating grouped data.

Different Inner Splits. **(B.1)** `BeyondArena` uses 5-repeated 5-fold inner cross-validation (5×5 CV) for datasets with fewer than 500 training samples, and 8-fold CV otherwise, to avoid overtuning hurting test performance [198, 199]. Table 1 and Appendix F.3 show that 5×5 CV consistently achieves a higher test performance, verifying the correctness of our setup.
(B.2) We use non-IID inner splits for non-IID tasks. Table 1 shows that using IID inner splits results in similar ranks, but non-IID splits perform better on average and can reach significantly higher peaks (see Appendix F.4). Thus, using non-IID inner splits for temporal and grouped data was appropriate.

Different Preprocessing. We use preprocessing for grouped data to handle the group index column. We ablate using only the raw data in Table 1 and Appendix F.5. **(C.1a)** For `L-P-G` data, the model rankings are mostly stable, while **(C.1b)** the rank ordering is mixed up for `L-P-S` data. In both cases, `BeyondArena`'s preprocessing is better on average, yet TabPFN-2.6 improves substantially when preprocessing is disabled (F.5). The results invite future work. For `BeyondArena`, we conclude that the performance of most models is representative, whereas TabPFN-2.6 is negatively biased.
(C.2.a/b) We used `Qwen3-Embedding-8B` to encode text features. We compare Qwen3 to TF-IDF from skrub [201], a classical text encoder in Table 1 and Appendix F.6. For short-text datasets, Qwen3

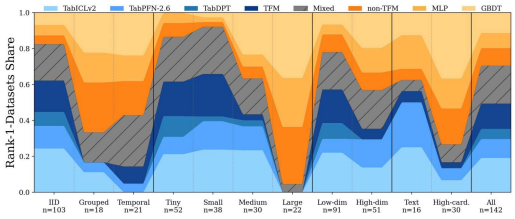

Figure 6: **Rank-1 share across sub-benchmarks**. We show the share of datasets where each model (family) significantly outperforms the others. Significance is measured by a Wilcoxon signed-rank test with $\alpha = 0.05$ for datasets with >3 outer splits and by whether a model wins all splits otherwise. **MLP**, **GBDT**, win for the family; **TFM**, win for the family, no winner within the family; **non-TFM** no clear winner for the best GBDT and MLP; **Mixed** no clear winner for the best TFM and non-TFM.

Table 1: **Ablation Results.** We show Kendall's tau (τ) [215] and the win rate when ablating our experimental setup. $\tau \in [-1, 1]$ measures the difference in ranking, with 1.0 being identical ranks. Win rate shows how often `BeyondArena`'s setting leads to a higher average rank across models. We split ablations for grouped data: `label-per-group` (`L-P-G`) / `label-per-sample` (`L-P-S`); and text data: ≤ 50 characters on average (short) / > 50 (long). We share details in Appendix F.

Part	Ablation	Setting (`BeyondArena` → new)	Kendall τ	Win Rate
Inner Splits	**(B.1)**	5×5-fold CV → 8-fold CV	0.93	100%
	(B.2)	Non-IID → IID	1.00	100%
Grouped Data Pre-processing	**(C.1a)**	(L-P-G) Agg. Index → N/A	0.81	71%
	(C.1b)	(L-P-S) Drop Index → N/A	0.43	71%
Text Data Pre-processing	**(C.2a)**	(Short Text) Qwen3 → TF-IDF	1.00	78%
	(C.2b)	(Long Text) Qwen3 → TF-IDF	0.89	6%
Post-processing	**(D)**	N/A → Probability Calibration	0.85	18%

performs better than TF-IDF. For long texts, TF-IDF generally yields better predictive performance. Yet, the model rankings are only slightly affected; thus, our results are still representative.
In the future, we recommend including preprocessing in the search space for grouped and text data.

Probability Calibration for Log-loss. **(D)** We use log-loss as a metric for multiclass classification. Log-loss is affected by probability calibration, and thus, post hoc calibration methods can often improve performance. `BeyondArena` does not use post hoc calibration by default; we ablate adding it based on the saved predictions. In Table 1, we show that calibration would have performed better most of the time. Across all models, only TabPFN-2.6 and RealMLP perform worse with calibration (Appendix F.7). Thus, we recommend using calibration by default for other models going forward.

7 Conclusion

We introduced `BeyondArena`, a unified benchmark for predictive modeling on tabular data that consolidates diverse task types (IID, temporal, grouped), dataset scales, feature dimensionality, and feature types. Together with `DataFoundry`, a framework for reproducible dataset curation, we enable standardized evaluation across 142 datasets and 11 models. Our results show that tabular foundation models (TFMs) perform well on small- to medium-scale IID data, but are outperformed by traditional methods in non-IID, large-scale, and high-dimensional settings, highlighting key gaps for future research. In summary, TFMs generalize well to existing paradigms and show very promising in-context learning performance, but they are not yet fully general for tabular data.

Limitations and Societal Impact. The conclusions of `BeyondArena` are limited by: (**1**) We tune models for 25 random configurations due to compute constraints. (**2**) We evaluate in-context learning performance of TFMs, but do not investigate fine-tuning TFMs. (**3**) We include the top three TFMs from TabArena, but do not explore the performance of other open- or closed-source models. (**4**) Across the 142 datasets, some sub-benchmarks have more datasets than others, leading to sub-benchmarks being under- and over-representative in the leaderboard computed across all datasets. (**5**) We make the first step towards a unified pipeline for preprocessing, validation, and tuning across task types and feature types. Yet, the best practices for validation protocols for temporal and grouped data, as well as for preprocessing grouped and text data, remain to be explored. On the societal side, `BeyondArena` improves rigor and reproducibility, narrowing the gap between academics and practitioners at non-trivial computational cost – which we trade off using `BeyondArena-Core`.

Future Work. To broaden the scope of evaluations of TFMs and further explore their generalizability, future directions could extend `BeyondArena` to few-shot predictions, more multimodal tabular data (e.g., tabular-image-text), and other tasks such as relational learning or survival analysis.

To conclude, `BeyondArena` surfaces critical limitations of current TFMs and establishes a new evaluation standard for tabular machine learning on broader, more demanding challenges.

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Author Contributions

L.P. led the code development, led the experiments, led the writing of the paper, led the evaluation and visualizations, led dataset curation, and led the management of the collaboration. A.T. contributed to the writing of the paper, contributed to evaluation and visualizations, and contributed to dataset curation. N.E. contributed to the writing of the paper and contributed to the evaluation and visualizations. G.B. helped with writing the paper, contributed to the evaluation and visualizations, helped with dataset curation, and helped with reviewing and editing. D.H. contributed to writing the paper, helped with the management of the collaboration, and helped with supervising the project. A.A. helped with writing the paper, and contributed to reviewing and editing. A.P. helped with dataset curation. M.T. helped with dataset curation. G.V. helped with evaluation and visualizations, helped with management of the collaboration, and supervised G.B. during the project. F.H. supervised L.P. during the project.

Competing Interests

D.H. is one of the authors of RealMLP. L.P. and F.H. are a subset of the authors of TabPFN-2.6. D.H. and G.V. are a subset of the authors of TabICLv2. L.P., A.T., N.E., and D.H. are the maintainers of TabArena. L.P. and N.E. are maintainers of AutoGluon. G.V. is a core contributor to skrub. G.V. is (co)founder of scikit-learn. L.P., N.E., A.A., and F.H. are affiliated with Prior Labs, a company focused on developing tabular foundation models. G.V. is affiliated with probabl, a company focused on developing software for industrial-grade data science. The authors declare no other competing interests.

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Appendices

Table of Contents.

A Background **33**

A.1 Time-series Forecasting Tasks vs. Temporal Tabular Tasks 33

B Dataset Curation **34**

B.1 Towards Automated Data Curation? 34

B.2 Details on Dataset Selection Criteria 34

B.3 Details on Dataset Processing 35

B.4 Details on Curation Outcome 36

B.5 BeyondArena Datasets Overview 36

C Experimental Setup **41**

C.1 Time Limit Impact 41

C.2 Additional Details 41

D Results **43**

D.1 Inference Time Overview 43

D.2 BeyondArena Leaderboard as a Table 43

D.3 TabArena-v0.1 Datasets vs. BeyondArena Datasets in the Same Scope 43

E Statistical Analysis **46**

E.1 Test 1: Are the methods really different from each other? 46

E.2 Test 2: At what training-set size does using a TFM stop being a reasonable choice? 46

E.3 Test 3: How much do tuning and ensembling help non-TFM models? 47

E.4 Test 4: What dataset properties predict that GBDTs will beat TFMs? 48

E.5 Test 5: Which method should a practitioner pick, given their data regime? 48

F Ablations **51**

F.1 (Outer Splits, A.1) Using Fewer Splits with BeyondArena-Core 51

F.2 (Outer Splits, A.2) IID Splits for non-IID Grouped Data 52

F.3 (Inner Splits, B.1) Using 8-fold Cross-Validation for Tiny Data 52

F.4 (Inner Splits, B.2) Using IID Splits for non-IID Data 52

F.5 (Grouped Data Preprocessing, C.1a/b) Disabling Preprocessing for Grouped Data. . 54

F.6 (Text Data Preprocessing, C.2a/b) TF-IDF for Text Encoding. 54

F.7 (Post-processing, D) Using Probability Calibration for Log-loss. 54

G Per Dataset Results **57**

H NeurIPS Paper Checklist **93**

A Background

A.1 Time-series Forecasting Tasks vs. Temporal Tabular Tasks

We argue that there is a fundamental difference between the definitions of time-series forecasting and temporal tabular data tasks, yielding distinct validation requirements. In short: time-series forecasting tasks predict *for* the future; temporal tabular tasks predict *in* the future – thus, when validating a model, it cannot or can use data from the future depending on the task.
To illustrate, assume, we have a time index $t \in T$ and are given training data $D_{\text{past}} = (\{x_s \in X : s \leq t\}, \{y_s \in Y : s \leq t\})$ with the goal to learn to predict Y. In time-series forecasting tasks with a forecast horizon h, we fit on D_{past} and predict $Y_{\text{future}} = \{\, y_s \in Y \;:\; t < s \leq t + h \,\}$ – at time point t. In tabular temporal data tasks, we fit on D_{past} at time point t and predict $Y_{\text{future}} = \{\, y_s \in Y \;:\; t < s \,\}$ at time point s, with $s \in T,\ s > t$, using D_{past} and $X_{\text{future}} = \{\, x_s \,\}$. That is, in temporal tabular tasks, we fit a model at time point t and deploy it to predict at time point s using data from s. Moreover, such tasks do not have a forecast horizon but a refit horizon, which determines the time until we refit the model on a new D_{past}.

While the many nuances and language differences between the two fields of structured data are out of scope here, the key difference for us lies in how the task affects the data. When naively using data from time-series forecasting tasks, one would violate the assumptions underlying the data and its associated validation process. For example, if we were to use a forecasting dataset and validate a model as if we were predicting at time points s instead of t, the validation would be unrepresentative of the dataset's task. Moreover, it becomes questionable what the validation would tell us; for many forecasting tasks, one could just read off the target variable (e.g., the stock price) at time points s. When this is the original intended task, we argue that it is a temporal tabular task and not a forecasting problem.

For us, the distinction between time-series forecasting data and temporal tabular data is important during data curation. If we naively use time-series forecasting data for temporal tabular regression, we ignore the real-world context of the data. Moreover, the validation procedure would not accurately reflect how well a real-world tabular temporal model performs, but instead estimates how well the model can solve a task, with at best questionable relevance in reality. As we aim to curate data and benchmark models such that the results are as representative of the real-world as possible, we create and use this distinction between data from time-series forecasting and temporal tabular task.

B Dataset Curation

B.1 Towards Automated Data Curation?

We investigated (semi-)automated procedures to aid dataset selection, but found that most of our selection criteria cannot be faithfully applied – not due to a lack of tooling, but to a significant lack of documented, structured metadata required for curation. In most cases, the data and its context from the data repository do not contain the necessary information to make decisions. Moreover, asking agentic LLMs to determine the metadata led to hallucination and faulty judgment. Instead, subjective human judgment and collaborative discussion were required to obtain the metadata for our final decisions. We found agentic tooling most useful for documenting metadata created by humans. In particular, we used Agent Skills [217] via Claude Code [218] to fill metadata templates from human input. Nevertheless, the future of high-quality automated data curation is promising, and we strongly believe that human-curated metadata and standardized schema, like in `DataFoundry`, will enable us to codify data curation. Therefore, we also share our agent skill markdown file as part of `DataFoundry`.

B.2 Details on Dataset Selection Criteria

Below, we provide a detailed description of each criterion used in `BeyondArena`. Our selection criteria are based on the work by Erickson et al. [11]. Moreover, our criteria involve subjective human interpretation. Thus, we publicly share our curation insights and decisions for each dataset at `https://tabarena.github.io/data-foundry/`.

Unique Dataset: We ensure datasets in `BeyondArena` are unique to avoid biasing our results towards a particular distribution. To do so, we ensure that each dataset in `BeyondArena` has a unique original data source.
When we started evaluating new datasets from OpenML and Kaggle, we quickly realized that the only way to identify duplicates was to determine the original source of each dataset, because many datasets were re-uploaded to OpenML and Kaggle under different names and without proper attribution to the original source. In particular, on Kaggle, we noticed that gamified incentives (such as upvotes and medals) might have even prompted users to avoid making their data look like duplicates, so that new, original contributions result in a larger reward. At the same time, Kaggle offers a discussion board for datasets, which enables the community to call out bad behavior and data quality issues. As a result, given a dataset from any data repository, our first priority is to determine its original source, or if it's an original contribution. In many cases, this is trivial; in others, it requires prolonged investigation, including comparing data statistics and comparing the dataset to popular datasets from the same domain. When the source remains unclear even after a thorough investigation, we leave it to the curator team's subjective judgment to assess the dataset's authenticity and uniqueness.

No Few-shot Prediction Tasks: We exclude datasets that have fewer than 100 train samples after creating the train-test split. Such data, often called few-shot prediction tasks, require a fundamentally different validation protocol, as data scarcity creates unique challenges, such as the inability to split the data for cross-validation. At the same time, it enables unique solutions to be highly competitive, such as using LLMs for tabular machine learning [111, 123, 127, 131, 134]. While our benchmark focuses on 1 000 000 training samples, we do not exclude larger datasets; instead, we sub-sample them.

IID and non-IID Tabular Datasets: We include any tabular dataset whose original task requires a random, temporal, or grouped split. We exclude datasets from time-series forecasting tasks because benchmarking such data in a representative way for their real-world context would require a fundamentally different validation protocol than data from temporal tabular tasks (see Appendix A.1). We leave benchmarking tabular machine learning models for time-series forecasting tasks to benchmarks from the time-series forecasting community [81–84].

Predictive Machine Learning Tasks for Classification or Regression: We exclude datasets that do not originally stem from a predictive machine learning task for classification or regression. Following TabArena, we exclude data from scientific discovery tasks (e.g., survey data or

non-predictive tables). For BeyondArena, we also exclude data from other tabular-adjacent tasks, in particular, click-through rate prediction tasks [219–221] and ranking prediction or information retrieval tasks used in recommender systems [158, 222–224]. Like for data from time-series forecasting tasks, datasets from such tasks require specific validation protocols to represent their real-world context appropriately. Furthermore, such tasks have their own benchmarking communities.

Representative for Real-world Applications of Tabular Machine learning: BeyondArena aims to understand how well tabular machine learning models would perform when practitioners use them in real-world applications. Consequently, we exclude datasets that represent tasks for which tabular machine learning models would not be used. Moreover, we exclude datasets that do not represent real-world applications or cannot be preprocessed to be representative. Practically speaking, this means we exclude datasets that **(A)** are from non-tabular modalities, where modality-specific models are clearly superior; **(B)** do not stem from a real random distribution; **(C)** are trivial; **(D)** have irreversible data quality issues; **(E)** and for which we did not find sufficient information to make an informed decision.
(A) We exclude datasets from non-tabular modalities, where modality-specific models are clearly superior. However, unlike TabArena, we determine whether modality-specific models are superior for each dataset rather than excluding all datasets from non-tabular modalities. Consequently, we allow the inclusion of vectorized image, text, audio, or time-series data if tabular machine learning models are competitive with modality-specific solutions. We investigate recent benchmarks related to the application of the dataset or introductory paper to determine if modality-specific solutions are clearly superior. Even if we're not domain experts, we believe our curation team is sufficiently well-versed in interpreting academic papers to determine whether tabular models are competitive with modality-specific solutions.
(B) We exclude datasets that do not stem from a real random distribution. In line with the TabArena definition, we exclude artificial data and data generated by a deterministic function. Unlike TabArena, we believe that simulated data (e.g., the Higgs bosons dataset [225]) represent real random distributions from important real-world applications. Nevertheless, we do not include such datasets in BeyondArena, as such tasks usually have dedicated domain-specific benchmarks [226, 227].
(C) We exclude datasets that are trivial because practitioners would not use machine learning to solve trivial datasets. A dataset is deemed trivial if all models, without tuning, consistently achieve the same better-than-random score. That is, there is no variance, or all models solve the dataset perfectly (error of 0). We utilize the performance of models from BeyondArena.
(D) We exclude datasets that have irreversible data quality issues. Many datasets shared online are already preprocessed; some have been processed in ways that make them unusable for benchmarking. Thus, we follow TabArena and exclude datasets where irreversible preprocessing leaks the target or test feature distribution (e.g., PCA-transformed data).
(E) We exclude datasets for which we did not find sufficient information to determine whether the datasets would pass our selection criteria. Datasets shared online often come with only limited information about their source and significant ambiguity about their authenticity. In such cases, if a prolonged investigation failed to identify any source information, it was up to the curation team to decide whether the data should be used for benchmarking.

Ethically Unambiguous Tasks: We exclude datasets with tasks that pose ethical concerns. As part of BeyondArena, we extend the scope of ethical concerns to include cases in which data subjects or data creators request that the data not be used for machine learning research, such as the Pima Indian Diabetes Dataset (PIDD) [228].

B.3 Details on Dataset Processing

All datasets underwent a processing procedure designed to create consistency and add reproducibility. Across datasets, we observed some recurring processing patterns, which we share below. Case-by-case details and comments for the processing can be found in DataFoundry.

General: Many datasets come with unique sample identifiers. We drop all uninformative sample identifiers. Informative sample identifiers, such as a time index, were kept and processed to represent their original meaning (e.g., time) where possible. In many cases, sample identifiers were an artifact of data storage.
For similar reasons, many datasets contained missing values represented by proxy values

(e.g., `999`, `-1`). We converted these to explicit missing values (`NA`) whenever such encoding could be reliably inferred from the data description and task context.
For numerical target variables with strong skewness and heavy tails (e.g., housing prices), logarithmic scaling was considered on a case-by-case basis.
Dataset names were standardized using `snake_case` naming conventions.
We always shuffle the sample order for both IID and grouped data to avoid methods exploiting implicit order leakage. For temporal data, samples were sorted chronologically by their time index.

Feature Type Assignments (Dtypes): We transformed features with object or string representations to categorical or string type. We used a categorical type when the number of possible values was fixed and finite (as inferred from the data description or task context). Otherwise, we used a string type.
We converted date features to standardized `YYYY-MM-DD` datetime representations whenever possible. For date features originally encoded numerically, we reconstructed as much of the date information as possible.
All other features were encoded as numerical types.

Creating Temporal Tasks and Splits: For temporal prediction tasks, we manually define the prediction horizon and the associated test time points. Then, we verified that each feature was valid to use, i.e., the information from the feature would have been available at prediction time, thereby preventing target leakage from future information. We filtered the samples and features as needed to avoid any temporal leakage
Datasets were additionally analyzed for grouped temporal structure, such as repeated observations from the same entities over time (e.g., multiple transactions from a single customer). In such cases, using a group split was carefully considered during split construction.
We ensure that the first split of a temporal task is always the one with the most recent time test point. Thus, it is also the split with the most training data, and we deem it the most representative split. We order all other splits by descending time test points.

B.4 Details on Curation Outcome

We investigate additional characteristics of our curated dataset.
In Figure B.1, we show how many datasets we have per domain, and in Figure B.2, we show how the dataset scales (in terms of number of rows and columns) varies across domains. Our datasets cover a broad range of domains with a large focus on some of the most common applied disciplines such as healthcare, marketing and finance. Interestingly, we noticed that medical and healthcare datasets are generally smaller, whereas finance datasets are several orders of magnitude larger on average.
In Figure B.3, we show the distribution of tabular datasets with text in our collection. Our text datasets span a wide range of scales, with multiple text columns and varying average text lengths.
In Figure B.4, we show the distribution of the cardinality across all categorical features. We see a clear long-tail for high cardinality, making up 14% of all categorical columns.
In Figure B.5, we investigate how the scale and dimensionality of our datasets change based on the task type and problem type. We observe that temporal data tends to have larger average sizes, while multiclass classification often stems from smaller problems.
In Figure B.6, we show the distribution of time horizons for temporal tasks. In most cases, the time horizon is days or months. Two temporal datasets have only a time index, decoupled from dates. Both used higher step sizes for the prediction horizon.
TabArena Reinvestigation. We re-investigated all accepted datasets from TabArena during our data curation process and filtered 2 out of the 51 datasets. We removed the `anneal` dataset because it is trivial (all models achieved perfect accuracy). Moreover, we removed the `diabetes` dataset due to ethical concerns, already raised by Radin [228] in 2017. Finally, while ingesting the datasets into `DataFoundry`, we fixed several minor issues, including dropping constant columns, ensuring correct feature types, and using log scaling for price prediction tasks when appropriate.

B.5 BeyondArena Datasets Overview

Table B.1 provides a per-dataset overview of all datasets used in `BeyondArena`.

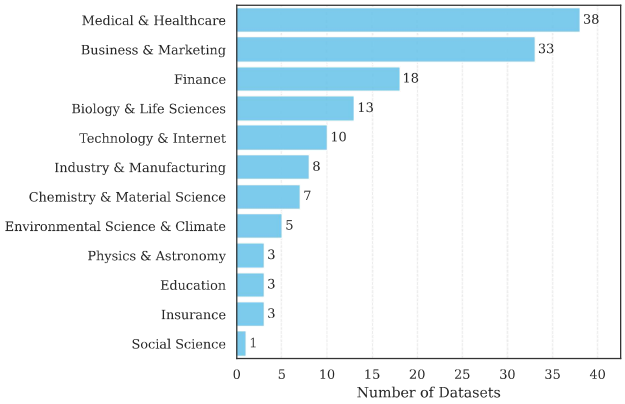

Figure B.1: **Dataset Domain Breakdown.** We use the domain categories from TALENT [112] and indicate how many out of our 142 datasets stem from each domain.

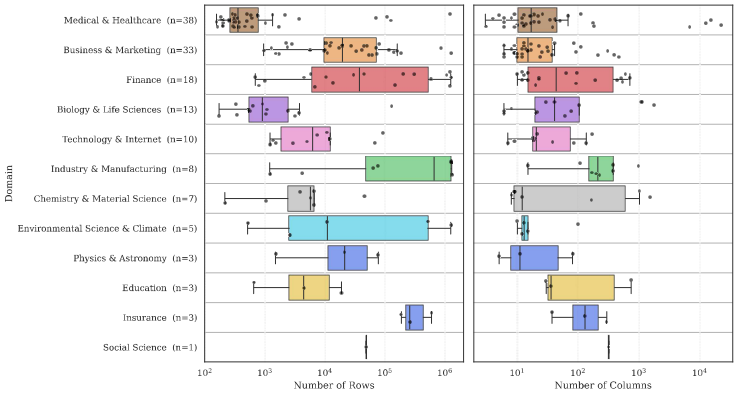

Figure B.2: **Dataset Scale Per Domain.** We show the number of rows and columns for each dataset binned by their domain.

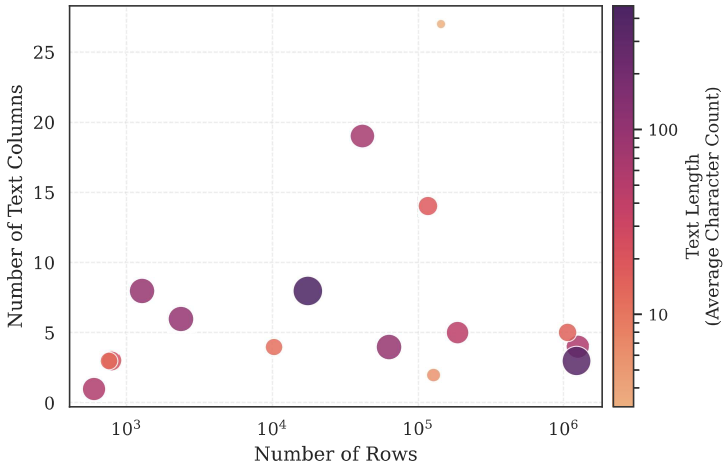

Figure B.3: **Characteristics Tabular Data with Text Features.** We show the number of rows and the number of text columns. The hue highlights the average text length. We have an even mix of short and long text datasets, spanning several orders of magnitude in sample size and a diverse number of text columns.

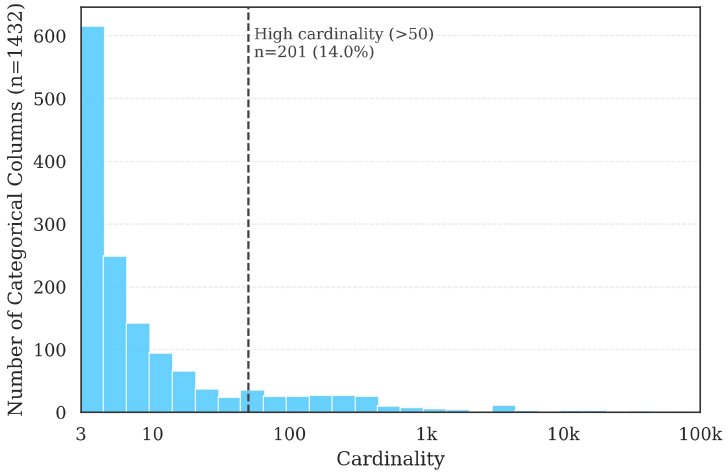

Figure B.4: **Distribution Cardinality of Categorical.** The distribution of the cardinality of all categorical columns across all datasets in our dataset collection.

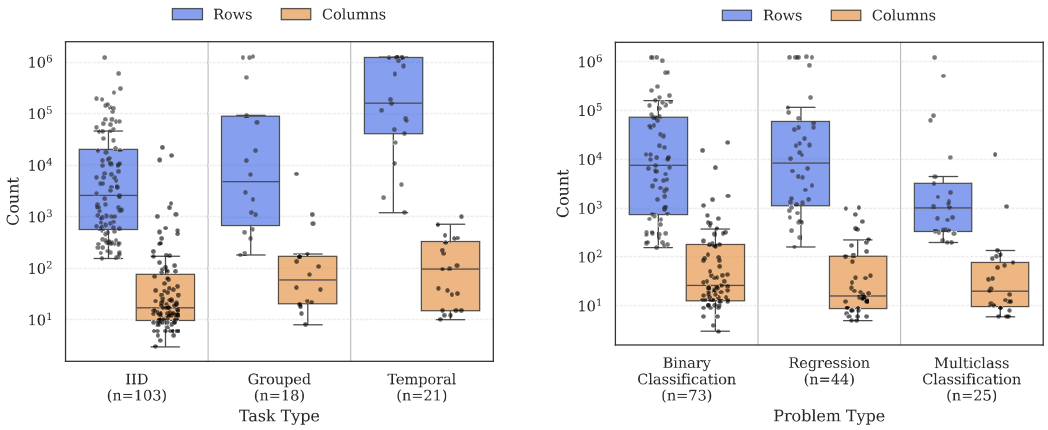

Figure B.5: **(Left) Data Scales Per Task.** The row and column count for each dataset, binned by task type. We see a trend towards larger temporal datasets. **(Right) Data Scales Per Problem.** Datasets binned by problem type show that multiclass classification problems are smaller on average, while binary classification and regression are similarly distributed in size.

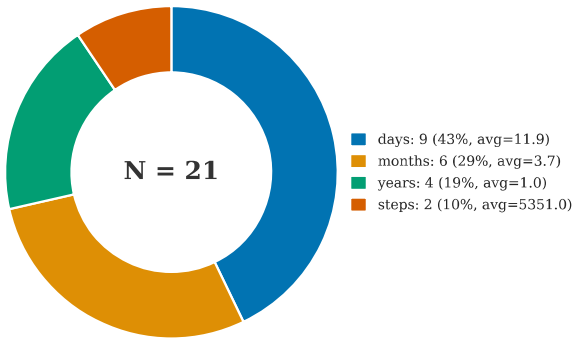

Figure B.6: **Distribution of Time Horizons for Temporal Dataset.** We show the count of time horizon units (days, months, years, steps) and the average number of values per unit (e.g. 11.9 days or 3.7 months). The time horizon indicates the time until we refit or get new labeled data.

Table B.1: Per-dataset metadata for `BeyondArena` benchmark, sorted by number of rows (N). Columns: N rows, d columns before preprocessing, C classes (regression: —), problem type (**Binary** classification / **Multi**class / **Reg**ression), task type (IID / Temporal / Grouped). Domain abbreviations: M & H = Medical & Healthcare; B & M = Business & Marketing; B & L = Biology & Life Sciences; T & I = Technology & Internet; I & M = Industry & Manufacturing; C & M = Chemistry & Material Science; E & C = Environmental Science & Climate; P & A = Physics & Astronomy.

Dataset	Domain	Source	Year	Age	N	d	C	Prob.	Task
hepatitis_survival_prediction [229]	M & H	UCI	1981	45	155	19	2	Binary	IID
cirrhosis_patient_survival_prediction [230]	M & H	UCI	1984	42	161	17	–	Reg	IID
clock_protein_toxicity [231]	B & L	UCI	2021	5	171	1,117	2	Binary	IID
pancreatic_cancer_mouse_detection [232]	M & H	Other	2003	23	181	6,771	2	Binary	Grouped
lung_cancer_epithelial_genexp [233]	M & H	GOV Website	2006	20	187	22,215	2	Binary	IID
parkinsons_biomedical_voice_measurements [234]	M & H	UCI	2007	19	195	23	2	Binary	Grouped
lung_cancer [235]	M & H	Other	2001	25	197	12,600	4	Multi	IID
audiology_diagnosis [236]	M & H	UCI	1987	39	199	68	3	Multi	IID
heart_disease_va_long_beach [237]	M & H	UCI	1989	37	200	13	2	Binary	IID
forensic_glass_identification [238]	C & M	UCI	1987	39	214	9	6	Multi	IID
early_stage_diabetes_risk_prediction [239]	M & H	UCI	2019	7	251	16	2	Binary	IID
body_density_prediction [240]	M & H	Kaggle	1985	41	252	13	–	Reg	IID
ljubljana_breast_cancer [241]	M & H	UCI	1988	38	286	9	2	Binary	IID
heart_disease_hungary [237]	M & H	UCI	1989	37	294	13	2	Binary	IID
heart_failure_followup_survival [242]	M & H	UCI	2020	6	299	12	2	Binary	IID
ljubljana_primary_tumor [243]	M & H	UCI	1987	39	302	17	11	Multi	IID
heart_disease_cleveland [237]	M & H	UCI	1989	37	303	13	2	Binary	IID
biomechanical_orthopaedic_prediction [244]	M & H	UCI	2011	15	310	6	3	Multi	IID
gallstone_disease [245]	M & H	UCI	2023	3	319	38	2	Binary	IID
prostate_cancer_detection [246]	M & H	Other	2002	24	322	15,154	2	Binary	IID
ecoli_proteins [247]	B & L	UCI	1996	30	327	6	5	Multi	IID
horse_colic_survival [248]	B & L	UCI	1989	37	344	20	3	Multi	IID
blood_tests_drink_prediction [249]	M & H	UCI	1996	30	345	5	–	Reg	IID
eryhemato_squamous_disease [250]	M & H	UCI	1997	29	366	34	6	Multi	IID
dementia_prediction [251]	M & H	Other	2010	16	370	8	3	Multi	Grouped
south_africa_coronary_heart_disease [252]	M & H	Kaggle	1983	43	462	9	2	Binary	IID
obesity_estimation [253]	M & H	UCI	2019	7	498	14	–	Reg	IID
telemonitoring_parkinsons_biomedical_voice_measurements [254]	M & H	UCI	2007	19	502	19	–	Reg	Grouped
forest_fires [255]	E & C	UCI	2008	18	517	12	–	Reg	IID
qsar_aquatic_toxicity [256]	B & L	UCI	2014	12	546	8	–	Reg	IID
micro_mass [257]	B & L	UCI	2013	13	571	1,082	20	Multi	Grouped
indian_liver_patient_dataset [258]	M & H	UCI	2012	14	583	10	2	Binary	IID
drug_induced_autoimmunity_prediction [259]	M & H	UCI	2025	1	597	177	2	Binary	IID
hepatitis_c_prediction [260]	M & H	UCI	2018	8	608	12	4	Multi	IID
biogeographical_ancestry_prediction [58, 261, 262]	B & L	GitHub	2025	1	635	104	10	Multi	IID
student_portuguese_performance [263]	Education	UCI	2008	18	649	30	–	Reg	IID
credit_approval [264]	Finance	UCI	1987	39	690	15	2	Binary	IID
blood_transfusion [265]	M & H	UCI	2008	18	748	4	2	Binary	IID
regensburg_pediatric_appendicitis [266]	M & H	Other	2021	5	763	51	2	Binary	IID
mutual_funds_india [267]	Finance	Kaggle	2023	3	793	12	–	Reg	IID
qsar_fish_toxicity [268]	B & L	UCI	2015	11	908	6	–	Reg	IID
tour_travels_churn [269]	B & M	Kaggle	2021	5	954	6	2	Binary	IID
credit_g [270]	Finance	UCI	1994	32	1,000	20	2	Binary	IID
maternal_health_risk [271]	M & H	UCI	2020	6	1,014	6	3	Multi	IID
concrete_compressive_strength [272]	C & M	UCI	1998	28	1,030	8	–	Reg	IID
qsar_biodeg [273]	B & L	UCI	2013	13	1,054	41	2	Binary	IID
mice_protein_trisomy_discriminant [274]	B & L	UCI	2015	11	1,080	76	8	Multi	Grouped
garments_worker_productivity [275]	I & M	UCI	2020	6	1,197	15	–	Reg	Temporal
asp_potassco_classification [276, 277]	T & I	ASlib	2014	12	1,212	136	11	Multi	Grouped
wine_world_cost [278]	B & M	Kaggle	2023	3	1,279	14	–	Reg	IID
healthcare_insurance_expenses [279]	M & H	Kaggle	2023	3	1,338	6	–	Reg	IID
website_phishing [280]	T & I	UCI	2014	12	1,353	9	3	Multi	IID
fitness_club [281]	B & M	Kaggle	2023	3	1,500	6	2	Binary	IID
airfoil_self_noise [282]	P & A	UCI	2014	12	1,503	5	–	Reg	IID
fiat_500 [283]	T & I	Kaggle	2020	6	1,538	7	–	Reg	IID
mic [284]	M & H	UCI	2020	6	1,699	111	8	Multi	IID
bad_customer_detection [285]	B & M	Kaggle	2020	6	1,723	13	2	Binary	IID
cardiotocography [286]	M & H	UCI	2010	16	2,126	22	3	Multi	Grouped
marketing_campaign [287]	B & M	Kaggle	2020	6	2,240	25	2	Binary	IID
coffee_rating_prediction [288]	B & M	Kaggle	2023	3	2,369	12	–	Reg	Temporal
hazelnut_spread_contaminant_detection [289]	B & L	OpenML	2020	6	2,400	30	2	Binary	IID
seismic_bumps [290]	E & C	UCI	2013	13	2,584	15	2	Binary	IID
iranian_churn [291]	B & M	UCI	2011	15	2,850	13	2	Binary	IID
sat11_hand_algo_runtime [216, 277, 292]	T & I	ASlib	2011	15	2,960	169	–	Reg	Grouped
splice [293]	B & L	UCI	1991	35	3,190	60	3	Multi	IID
thyroid_discordant [264]	M & H	UCI	1986	40	3,711	26	2	Binary	IID
bioresponse [294]	B & L	Kaggle	2012	14	3,751	1,776	2	Binary	IID
hiva_agnostic [295]	C & M	Other	2007	19	3,845	1,518	2	Binary	IID
mercedes_benz_greener_manufacturing [296]	I & M	Kaggle	2017	9	4,204	371	–	Reg	Temporal
predict_students_dropout_and_academic_success [297]	Education	UCI	2021	5	4,424	36	3	Multi	IID
santander_transaction_value [298]	Finance	Kaggle	2018	8	4,447	540	–	Reg	IID

Table B.1: (continued) Per-dataset metadata for `BeyondArena` benchmark.

Dataset	Domain	Source	Year	Age	N	d	C	Prob.	Task
churn [299]	T & I	OpenML	2005	21	5,000	19	2	Binary	IID
homeq_default_prediction [300]	B & M	Other	2016	10	5,708	12	2	Binary	IID
qsar_tid_11 [301]	C & M	OpenML	2015	11	5,741	1,024	–	Reg	IID
polish_companies_bankruptcy [302]	Finance	UCI	2010	16	5,790	64	2	Binary	IID
wine_quality [303]	C & M	UCI	2009	17	6,497	12	–	Reg	IID
musk [304]	C & M	UCI	1994	32	6,598	166	2	Binary	Grouped
taiwanese_bankruptcy_prediction [305]	Finance	UCI	2009	17	6,819	92	2	Binary	IID
naticusdroid_android_permissions_dataset [306]	T & I	UCI	2021	5	7,491	85	2	Binary	IID
coil_2000 [307]	B & M	UCI	2000	26	9,822	85	2	Binary	IID
bank_customer_churn [308]	B & M	Kaggle	2020	6	10,000	10	2	Binary	IID
immoscout_german_house_prices [309, 310]	B & M	Kaggle	2019	7	10,317	23	–	Reg	IID
heloc [311]	Finance	Kaggle	2021	5	10,459	23	2	Binary	IID
jm1 [312]	T & I	OpenML	2004	22	10,885	21	2	Binary	IID
ghanas_indigenous_intel [313]	E & C	Zindi	2025	1	10,928	10	4	Multi	Temporal
ecommerce_shipping [314]	B & M	Kaggle	2021	5	10,999	10	2	Binary	IID
video_game_fps_prediction [315]	T & I	OpenML	2020	6	12,288	38	–	Reg	Grouped
online_shoppers_purchasing_intention_dataset [316]	B & M	UCI	2017	9	12,330	17	2	Binary	IID
in_vehicle_coupon_recommendation [317]	B & M	UCI	2017	9	12,684	24	2	Binary	IID
miami_housing [318]	Finance	Kaggle	2016	10	13,776	15	–	Reg	IID
emscad [319]	B & M	Other	2014	12	17,460	17	2	Binary	IID
early_learning_predictors [320]	Education	Other	2023	3	18,874	743	–	Reg	Grouped
hr_analytics [321]	B & M	Kaggle	2021	5	19,158	12	2	Binary	IID
houses [322]	B & M	Other	1990	36	19,675	8	–	Reg	IID
superconductivity [323]	P & A	UCI	2018	8	21,263	81	–	Reg	IID
sberbank_housing_market_forecasting [324]	B & M	Kaggle	2017	9	27,195	386	–	Reg	Temporal
credit_card_clients_default [325]	Finance	UCI	2009	17	30,000	23	2	Binary	IID
amazon_employee_access [326]	B & M	Kaggle	2010	16	32,769	9	2	Binary	IID
california_house_prices_2020 [327]	B & M	Kaggle	2021	5	41,528	41	–	Reg	Temporal
bank_marketing [328]	Finance	UCI	2012	14	45,211	13	2	Binary	IID
food_delivery_time [329]	B & M	Kaggle	2023	3	45,451	9	–	Reg	IID
physiochemical_protein [330]	C & M	UCI	2013	13	45,730	9	–	Reg	IID
anes_voting_2026 [331]	Social Science	Other	2026	0	48,587	318	2	Binary	Temporal
kdd_cup_09_appetency [332]	B & M	Other	2008	18	50,000	212	2	Binary	IID
diamonds [333]	B & M	Other	2015	11	53,940	9	–	Reg	IID
otto_group_product_classification_challenge [334]	B & M	Kaggle	2015	11	61,878	93	9	Multi	IID
labour_inspection_compliance [335]	I & M	Other	2019	7	63,634	376	2	Binary	IID
video_transcoding_time_prediction [336]	T & I	UCI	2015	11	68,784	18	–	Reg	Grouped
santander_customer_satisfaction [337]	B & M	Kaggle	2016	10	71,080	307	2	Binary	IID
diabetes_130_us [338]	M & H	UCI	2014	12	71,518	44	2	Binary	IID
kick [339]	B & M	Kaggle	2011	15	72,983	32	2	Binary	Temporal
aps_failure [340]	I & M	UCI	2016	10	76,000	170	2	Binary	IID
sdss_17 [341]	P & A	Kaggle	2022	4	78,053	11	3	Multi	IID
hotel_booking_demand [342]	B & M	Other	2019	7	81,418	31	2	Binary	Temporal
5g_energy_consumption [343]	T & I	HuggingFace	2023	3	92,629	20	–	Reg	Grouped
sepsis_survival_minimal_clinical_records [344]	M & H	UCI	2020	6	110,204	3	2	Binary	IID
sf_permit_time [345]	B & M	GOV Website	2025	1	116,954	37	–	Reg	Temporal
wids_diabetes_mellitus [346]	M & H	Kaggle	2021	5	127,358	181	2	Binary	IID
customer_satisfaction_in_airline [347]	B & L	Kaggle	2023	3	129,880	21	2	Binary	IID
pva_revenue_prediction_kddcup98 [348]	B & M	Other	1997	29	144,095	477	2	Binary	IID
give_me_some_credit [349]	Finance	Kaggle	2011	15	150,000	10	2	Binary	IID
acquire_valued_shoppers_challenge [350]	B & M	Kaggle	2014	12	160,057	111	2	Binary	Temporal
kickstarter [351]	B & M	Other	2025	1	187,118	15	2	Binary	Temporal
allstate_claims_severity [352]	Insurance	Kaggle	2016	10	188,317	130	–	Reg	IID
santander_customer_transaction_prediction [353]	Finance	Kaggle	2019	7	200,000	600	2	Binary	IID
homesite_quote_conversion [354]	Insurance	Kaggle	2015	11	260,753	295	2	Binary	IID
home_credit_default_risk [355]	Finance	Kaggle	2018	8	307,507	504	2	Binary	IID
covertype [356]	E & C	UCI	1998	28	512,625	13	3	Multi	Grouped
ieee_fraud_detection [357]	Finance	Kaggle	2019	7	590,540	435	2	Binary	Temporal
porto_seguro [358]	Insurance	Kaggle	2017	9	595,206	37	2	Binary	IID
rossmann_store_sales [359]	B & M	Kaggle	2015	11	844,392	15	–	Reg	Temporal
lending_club_1m [360]	Finance	Kaggle	2018	8	1,064,751	96	2	Binary	Temporal
home_credit_default_stability_1m [324]	Finance	Kaggle	2024	2	1,224,927	711	2	Binary	Temporal
consumer_complaints_1m [361]	Finance	GOV Website	2025	1	1,226,140	12	3	Multi	Temporal
sepsis_prediction_1m [362]	M & H	Other	2019	7	1,228,686	42	2	Binary	Grouped
amex_non_iid_1m [363]	Finance	Kaggle	2022	4	1,249,605	189	2	Binary	Grouped
delivery_eta_1m [364]	I & M	Kaggle	2024	2	1,250,000	225	–	Reg	Temporal
cooking_time_1m [364]	I & M	Kaggle	2024	2	1,250,000	196	–	Reg	Temporal
climate_model_weather_forecasting_1m [364]	E & C	Kaggle	2024	2	1,250,000	100	–	Reg	Temporal
maps_router_eta_1m [364]	I & M	Kaggle	2024	2	1,250,000	988	–	Reg	Temporal
mercari_price_suggestion_1m [365]	B & M	Kaggle	2018	8	1,250,000	6	–	Reg	IID
electric_motor_temperature_prediction [366]	I & M	Kaggle	2021	5	1,296,316	109	–	Reg	Grouped

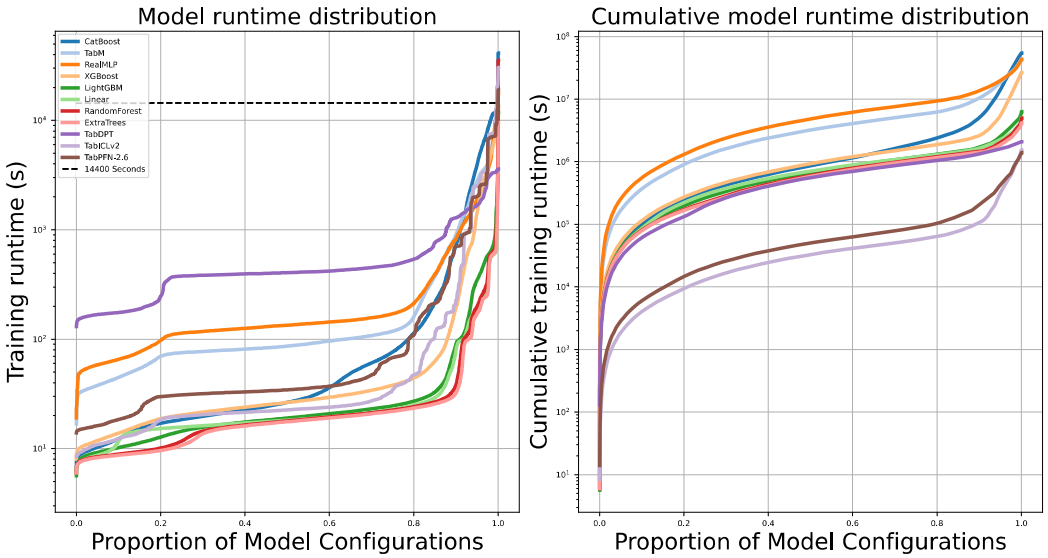

Figure C.1: **Training runtime distribution across hyperparameter configurations.** Each point shows the training runtime, in seconds, for a hyperparameter configuration evaluated on one train split of a dataset, aggregated across all models. The x-axis shows the cumulative proportion of configurations, and the y-axis shows training time in seconds. The vast majority of configurations finish well below the 4-hour time limit: only $\sim$1.72% exceed 1 hour, $\sim$0.01% exceed 3.5 hours, and $\sim$0.003% (31 out of 785,208) exceed the 4-hour limit.

C Experimental Setup

C.1 Time Limit Impact

In our experiments, we limit the evaluation of a single configuration on a single train split of a dataset to 4 hours. Thus, a model must complete training across all inner folds within 4 hours; otherwise, training is gracefully stopped early. We use a 4-hour limit instead of the 1-hour limit used in TabArena because our benchmark includes larger datasets.

Figure C.1 shows the training runtime distribution across all hyperparameter configurations and models, visualizing the proportion of configurations (x-axis) that required a given training time in seconds (y-axis).

We find that the time limit is rarely reached. Only $\sim$1.72% of jobs exceeded 1 hour, $\sim$0.01% exceeded 3.5 hours, and only $\sim$0.003% of jobs, corresponding to 31 out of 785,208, exceeded the 4-hour time limit.

C.2 Additional Details

Below, we provide more details on specific parts of our experimental setup.

Validation Splits & Insufficient Samples per Class. To avoid errors when evaluating validation metrics due to missing classes, we dynamically reduce the number of folds to match the minimum number of samples per class when a dataset has an insufficient number of minority-class samples for cross-validation (i.e., fewer than 5 or 8).

Date Encoding We use the `DatetimeEncoder` from Skrub [201] to convert datetime features into 10 numerical features that represent values such as the year or weekday, as well as a spline-based periodic encoding of the timestamp.

Text Encoding We use Qwen3-Embedding-8B to transform each text feature into a 4096-dimensional vector, from which we take the first 32 dimensions using Qwen's supported minimal Matryoshka representation learning [205] slice. We also add caching, so we only need to compute the text encoding once.

Grouped Preprocessing We add preprocessing for grouped non-IID data, executed at the end of the model-agnostic preprocessing. For `label-per-group` datasets, we replace the group index with a transductive 50-dimensional group-encoding.
Per group: we (a) sort all samples by a group time index, when available; (b) aggregate all features across samples (mean/std/min/max/last for numeric, count/last/nunique for non-numeric); (c) select the top 50 aggregations by variance when fitting the preprocessing; and (d) join the top 50 aggregated features back to all original samples. At test time, when given a new group as input, we replace the group's index with these 50 aggregations computed over the group's samples. This preprocessing assumes a predictive machine learning task in which all samples from a group are available when predicting. This is the case for all `label-per-group` datasets in `BeyondArena`.

Compute Hardware. We run all our experiments on the Google Cloud Platform (GCP) [207]. We selected virtual machines (VMs) from GCP's offering that might be representative of the hardware of an average practitioner, while being reasonably affordable for running large-scale experiments. For datasets with fewer than 250k samples, we ran MLPs (TabM, RealMLP) on a2-highgpu-1g [367] VMs with an NVIDIA A100 GPU with 40GB VRAM, 12 vCPUs, and 85 GB RAM. For MLPs on larger datasets, and for any TFM (TabDPT, TabPFN-2.6, TabICLv2), we use a2-ultragpu-1g [368] VMs with an NVIDIA A100 GPU with 80 GB VRAM, 12 vCPUs, and 170 GB RAM. For all other models, we use n4-standard-16 [369] VMs with 16 vCPUs and 64 GB RAM.

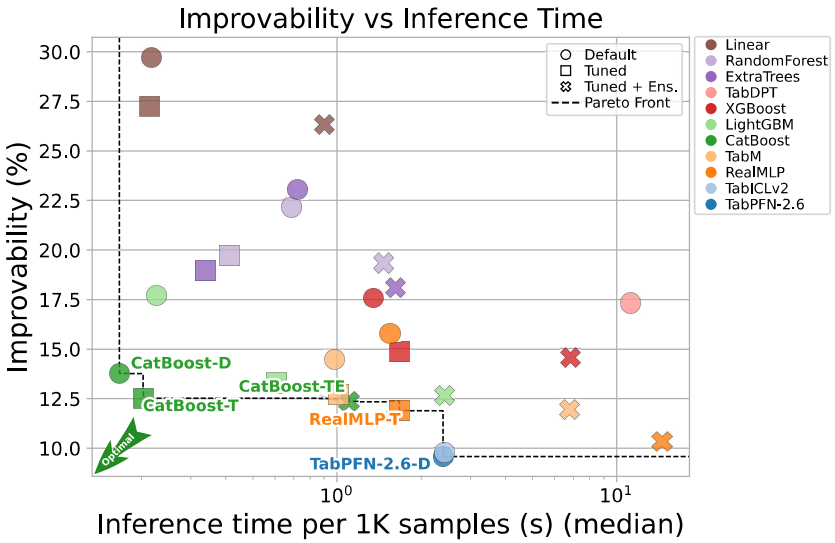

Figure D.1: **Improvability vs. Inference Time Pareto Front.** We show the Improvability and inference time of models with the default hyperparameters, with tuning, and with tuning and post-hoc ensembling. The default performance of TFMs equals their in-context learning performance.

D Results

D.1 Inference Time Overview

We show the Pareto front of Improvability vs. inference time in Figure D.1. We observe that CatBoost dominates the inference time Pareto front, while RealMLP is competitive by default. TabICLv2 and TabPFN-2.6 both require significantly longer inference time while achieving 2.5% Improvement over a tuned CatBoost. A tuned RealMLP incurs an even larger inference time.

D.2 BeyondArena Leaderboard as a Table

Table D.1 shows the per-model performance across all datasets.

D.3 TabArena-v0.1 Datasets vs. BeyondArena Datasets in the Same Scope

To assess whether the newly added datasets that were in the same scope as the TabArena datasets pose a bigger challenge, we compare ELO on two dataset subsets: (1) the 49 accepted TabArena datasets, (2) 20 datasets which fall in the selection criteria of TabArena (IID, $500 - 250,000$ samples). In Figure D.2, we see that while tabular foundation models (TFMs) dominate on the TabArena datasets, they are outperformed by RealMLP on the new datasets. While Elo performance is stable for non-foundation models, all three evaluated TFMs degrade substantially on the new datasets. This can be explained by the fact that several of the new dataset additions are at the upper end of TabArena's sample-size limits, are often high-dimensional, and frequently contain high-cardinality categorical features. Overall, this confirms that even datasets within the scope of previous benchmarks can help measure future improvements of TFM on more challenging datasets.

Table D.1: **BeyondArena Leaderboard.** We show default (D), tuned (T), and tuned + ensembled (T+E) performances. Results for TabPFN-2.6 and TabDPT are imputed for datasets with more than 100k samples. The best three values in columns are highlighted with gold, silver, and bronze colors. For Elo values, we also indicate their approximate 95% confidence intervals obtained through bootstrapping.

Model	Elo (↑)	Improvability (↓)	Avg. rank (↓)	Harm. mean rank (↓)	#wins (↑)	Train time per 1K [s]	Predict time per 1K [s]
RealMLP (T+E)	$1282_{-26,+30}$	10.3%	7.2	3.8	9.9	1316.51	14.54
TabPFN-2.6 (D)	$1224_{-35,+40}$	9.6%	8.9	3.0	24.4	25.26	2.39
TabM (T+E)	$1210_{-24,+28}$	11.9%	9.3	5.1	4.9	1072.09	6.78
CatBoost (T+E)	$1208_{-19,+27}$	12.3%	9.4	5.4	6.6	425.71	1.10
TabICLv2 (D)	$1205_{-42,+39}$	9.8%	9.5	2.7	33.3	11.41	2.42
RealMLP (T)	$1202_{-23,+31}$	11.9%	9.6	5.1	6.9	1316.51	1.68
CatBoost (T)	$1189_{-21,+27}$	12.5%	10.0	5.7	5.8	425.71	0.20
LightGBM (T+E)	$1187_{-22,+26}$	12.7%	10.0	5.9	5.7	163.08	2.42
TabM (T)	$1168_{-27,+25}$	12.7%	10.7	5.8	5.1	1072.09	1.01
LightGBM (T)	$1149_{-23,+23}$	13.3%	11.3	6.9	3.0	163.08	0.61
CatBoost (D)	$1139_{-20,+29}$	13.8%	11.6	7.3	2.8	14.38	0.17
XGBoost (T+E)	$1138_{-22,+29}$	14.6%	11.7	7.2	2.9	271.48	6.83
XGBoost (T)	$1112_{-22,+26}$	14.9%	12.5	8.1	2.6	271.48	1.68
TabM (D)	$1107_{-24,+24}$	14.5%	12.7	8.0	2.5	29.89	0.98
RealMLP (D)	$1056_{-25,+23}$	15.8%	14.4	8.8	2.5	43.31	1.55
TabDPT (D)	$1052_{-44,+34}$	17.3%	14.5	5.7	9.2	107.79	11.22
ExtraTrees (T+E)	$1008_{-27,+23}$	18.1%	16.0	11.0	1.1	113.01	1.62
XGBoost (D)	$1000_{-25,+29}$	17.6%	16.3	11.8	1.0	6.59	1.35
LightGBM (D)	$991_{-18,+23}$	17.7%	16.6	12.2	1.4	3.47	0.23
ExtraTrees (T)	$971_{-29,+26}$	19.0%	17.2	11.8	1.1	113.01	0.34
RandomForest (T+E)	$963_{-30,+23}$	19.3%	17.5	12.9	0.9	143.88	1.47
RandomForest (T)	$946_{-26,+24}$	19.7%	18.0	13.3	0.8	143.88	0.41
Linear (T+E)	$890_{-38,+28}$	26.3%	19.7	11.5	2.0	204.03	0.90
RandomForest (D)	$876_{-32,+25}$	22.2%	20.1	15.5	0.6	9.76	0.69
ExtraTrees (D)	$864_{-40,+34}$	23.1%	20.4	14.1	1.2	6.31	0.72
Linear (T)	$855_{-41,+32}$	27.2%	20.7	12.6	1.8	204.03	0.21
Linear (D)	$793_{-50,+38}$	29.7%	22.2	13.6	2.0	6.44	0.22

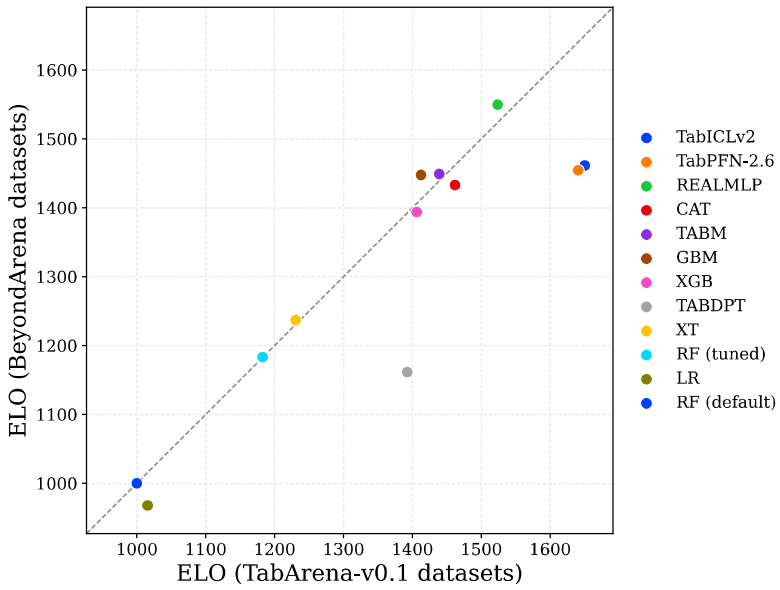

Figure D.2: **Elo parity scatterplot comparing the 49 accepted TabArena datasets with 20 BeyondArena datasets within the same scope.** The dashed diagonal indicates equal performance relative to a default random forest across the two dataset groups.

E Statistical Analysis

We follow the standard practice for comparing multiple learning algorithms across many datasets [370]. All tests are based on *normalized error* $e_{\text{norm}} \in [0, 1]$: on each dataset, methods are scored from best (0) to worst (1) among the 27 methods (11 default, 8 tuned, 8 tuned + ensembled), then averaged across cross-validation folds. Normalized errors let us work on a common scale, even though the benchmark spans heterogeneous tasks with their respective metrics: ROC AUC, log-loss, and RMSE.

Whenever a test produces multiple comparisons, we control the family-wise error rate with the **Holm–Šidák correction** [371, 372]. Holm makes no assumption that the individual tests are independent, which matters here because comparisons across the same 142 datasets yield p-values that are inherently correlated rather than independent. We report effect sizes alongside corrected p-values.

E.1 Test 1: Are the methods really different from each other?

The first question to settle is whether the leaderboard ranking is genuinely informative or whether it could plausibly arise from random fluctuations across the 142 datasets. We answer this with a **Friedman test** [373]. The Friedman test ranks the 27 methods within each dataset and asks whether the resulting rank distributions across datasets are consistent with all methods being equivalent. Because ranks are computed within each dataset, the test naturally accounts for the fact that the same 142 datasets are scored under every method.

The Friedman test rejects the null hypothesis: the observed test statistic is $\chi^2_{26} = 1374.9$, giving $p < 10^{-273}$.

To go beyond this global statement and identify which specific pairs of methods differ, we follow up with **pairwise Wilcoxon signed-rank** tests [374] over all $\binom{27}{2} = 351$ method pairs, with Holm correction. For each pair of methods (A, B) with normalized errors $\{e_i^A\}_{i=1}^N$ and $\{e_i^B\}_{i=1}^N$ on the $N = 142$ datasets, let $d_i = e_i^A - e_i^B$ be the per-dataset paired difference. The Wilcoxon signed-rank test drops the magnitudes of the d_i and works only with the ranks of their absolute values $|d_i|$, keeping their signs. The hypotheses are:

$$H_0 : \text{the differences } d_i \text{ are symmetrically distributed around zero;}$$
$$H_1 : \text{the differences are not symmetric around zero (two-sided).}$$

We reject H_0 at level $\alpha = 0.05$ when the corresponding two-sided p-value falls below 0.05. Of those 351 pairs, **260 (74.6%)** show performance differences that survive Holm correction. This sharpens the global Friedman result: not only is the leaderboard ranking real, but most individual pairs of methods can be reliably distinguished from one another.

E.2 Test 2: At what training-set size does using a TFM stop being a reasonable choice?

We split the 142 datasets into the four scale categories used throughout the paper — tiny (100-1k rows), small (1k-10k), medium (10k-100k), and large ($\geq 10^5$) — and compare the distribution of TFM normalized error across them with a **Kruskal-Wallis test** [375]. In Test E.1, every dataset is scored under all 27 methods, so the methods can be ranked *within each dataset* and Friedman aggregates those within-dataset rankings across the 142 rows. Here, by contrast, each dataset belongs to exactly one scale category: the four scale groups are disjoint sets of datasets so Wilcoxon and Friedman are not suitable.

Formally, let $g_i \in \{1, 2, 3, 4\}$ denote the scale category of dataset $i \in \{1, \ldots, N\}$ with $N = 142$, and let e_i be the TFM normalized error on dataset i. We rank the N datasets globally on e_i from best (1) to worst (N) across all four groups combined, and let $\bar{R}_g$ denote the mean rank within scale group g, with n_g the number of datasets in that group. The hypotheses are:

H_0 : all four scale groups have the same distribution of TFM normalized error;
i.e., dataset size has no systematic effect on TFM error.

H_1 : at least one group is stochastically different from the others;
i.e., TFM error tends to be larger (or smaller) on datasets of some sizes than others.

The Kruskal-Wallis test statistic is

$$H \;=\; \frac{12}{N(N+1)} \sum_{g=1}^{4} n_g \left(\bar{R}_g - \frac{N+1}{2} \right)^2 ,$$

which under H_0 is approximately χ^2-distributed with $k-1=3$ degrees of freedom. We reject H_0 at significance level $\alpha = 0.05$ when the corresponding p-value falls below 0.05.
The Kruskal-Wallis statistic is $H = 30.6$ on 3 degrees of freedom ($p = 1.0 \times 10^{-6}$), so the four scale categories do *not* share the same distribution of TFM errors.

Kruskal-Wallis tells us that the four scale groups are not identically distributed, but it does not say *which* pairs differ or in which direction. To localize where the change actually happens, we follow up with the two-sample analogue of Kruskal-Wallis: **p airwise Mann-Whitney U tests** [376] between every pair of scale categories, with Holm correction over the $\binom{4}{2} = 6$ comparisons (Table E.1).
The pattern is clear. Tiny, small, and medium scales are statistically indistinguishable from each other (every pairwise $p_{\text{adj}} > 0.5$): a TFM does not perform worse on a 50k-row dataset than on a 500-row one. This pattern breaks at the large-scale boundary: every comparison involving large datasets is highly significant ($p_{\text{adj}} < 0.001$) and the rank-biserial effect sizes are large (r between 0.68 and 0.79). The takeaway is that TFM degradation happens at $\sim 100,000$ *rows*.

Table E.1: Pairwise Mann-Whitney U tests for TFM normalized error across dataset scales, Holm-corrected over six comparisons. $\bar{e}_A$ and $\bar{e}_B$ are the mean TFM normalized errors in scales A and B. The rank-biserial correlation r measures the size of the gap on a $[-1,+1]$ scale: a positive r means scale B has the higher (worse) error. **The three rows comparing *large* against the other scales are statistically significant after correction (in bold)**; the other three are not. *For example:* the third row says that comparing tiny ($n_A = 52$) against large ($n_B = 22$), the mean TFM error rises from 0.183 to 0.528, with a large rank-biserial effect size of $r = +0.71$ (TFM errors on tiny datasets are systematically smaller than on large datasets), and the corresponding Holm-corrected p-value is below 0.001.

Scale A	Scale B	n_A	$\bar{e}_A$	$\bar{e}_B$	r	p_{adj}
tiny ($n = 52$)	small ($n = 38$)		0.183	0.135	-0.11	0.769
tiny	medium ($n = 30$)		0.183	0.212	$+0.06$	0.769
tiny	large ($n = 22$)		0.183	0.528	$+0.71$	< 0.001 ***
small	medium		0.135	0.212	$+0.20$	0.505
small	large		0.135	0.528	$+0.79$	< 0.001 ***
medium	large		0.212	0.528	$+0.68$	< 0.001 ***

E.3 Test 3: How much do tuning and ensembling help non-TFM models?

Traditional models seem to require tuning and ensembling to reach their full potential, while TFMs rely on in-context learning by default. We now test each of the eight traditional algorithms in BeyondArena to determine whether tuning and post hoc ensembling are actually required.
For each algorithm, across all 142 datasets, we have three matched observations: default error, tuned error, and tuned-plus-ensembled error. The matched structure is the right setting for a **Wilcoxon signed-rank test**. We run two such tests per algorithm — (default $\rightarrow$ tuned) and (tuned $\rightarrow$ tuned+ensemble) — for a total of 16 tests jointly Holm-corrected.

Formally, fix a traditional algorithm $a \in \{$CatBoost, LightGBM, Linear, RandomForest, RealMLP, TabM, XGBoost, ExtraTrees$\}$, and let $e_i^{\text{def}}, e_i^{\text{tun}}, e_i^{\text{ens}}$ denote its normalized errors on dataset $i \in \{1, \ldots, 142\}$ when run with default hyperparameters, with tuned hyperparameters, and with tuned-plus-ensembled hyperparameters respectively. We test two paired hypotheses per algorithm:

$$H_0^{(\text{tune})} : \text{the per-dataset differences } d_i^{(\text{tune})} = e_i^{\text{def}} - e_i^{\text{tun}} \text{ are symmetric around } 0;$$
$$H_0^{(\text{ens})} : \text{the per-dataset differences } d_i^{(\text{ens})} = e_i^{\text{tun}} - e_i^{\text{ens}} \text{ are symmetric around } 0.$$

Under each H_0, the matched pairs convey no systematic improvement; under the corresponding H_1, one pipeline variant is consistently better than the other. As in Test E.1, we compute the signed-rank statistic on the differences. We reject H_0 at level $\alpha = 0.05$ when the corresponding two-sided p-value

falls below 0.05 *after Holm correction across all* $2 \times 8 = 16$ *tests jointly*.
Every single one of the 16 tests is significant at $p_{\text{adj}} < 0.001$ (Table E.2). **Both HPO and ensembling consistently improve traditional models across the entire benchmark**. RealMLP shows the largest jump from tuning to ensembling ($r = 0.87$), while LightGBM shows the largest jump from the default to its tuned version ($r = 0.89$).

Table E.2: Wilcoxon signed-rank tests for the impact of tuning and post-hoc ensembling, computed independently for each of the eight non-TFM algorithms ($n = 142$ datasets per algorithm). The first three columns show the mean normalized error of each pipeline variant. The last two columns give the rank-biserial effect size r for the two transitions; a positive r indicates improvement. Every cell marked *** is significant at $p_{\text{adj}} < 0.001$ after Holm correction over all 16 tests. The entry shown in bold per column is the largest effect.

	Mean normalized error			Effect size r	
Algorithm	Default	Tuned	+Ensemble	Default→Tuned	Tuned→Ens.
CatBoost	0.204	0.174	0.167	0.48 ***	0.51 ***
LightGBM	0.343	0.195	0.176	**0.89 *** **	0.63 ***
Linear	0.713	0.614	0.571	0.53 ***	0.77 ***
RandomForest	0.497	0.393	0.379	0.57 ***	0.40 ***
RealMLP	0.271	0.166	0.121	0.80 ***	**0.87 *** **
TabM	0.237	0.189	0.173	0.46 ***	0.79 ***
XGBoost	0.346	0.236	0.224	0.84 ***	0.58 ***
ExtraTrees	0.517	0.370	0.341	0.65 ***	0.65 ***

E.4 Test 4: What dataset properties predict that GBDTs will beat TFMs?

For each of the 142 datasets, we summarize the head-to-head outcome with a single signed quantity, $\Delta = \bar{e}^{\star}_{\text{TFM}} - \bar{e}^{\star}_{\text{GBDT}}$, where $\bar{e}^{\star}$ is the minimum normalized error achieved by any member of that family on the dataset. Negative Δ means the best TFM beat the best GBDT; positive Δ means the opposite. We then correlate Δ with a set of meta-features of our datasets using **Spearman's rank correlation** ρ [377]. The eight meta-feature tests are jointly Holm-corrected.
Two meta-features stand out (Table E.3): dataset size and the number of high-cardinality categorical columns. The maximum cardinality of any single categorical column also predicts a GBDT advantage. Missing values, while a weaker signal, also push the balance toward GBDTs.

Table E.3: Spearman rank correlation between dataset meta-features and the per-dataset TFM-vs-GBDT advantage $\Delta = e_{\text{TFM}} - e_{\text{GBDT}}$. **A positive ρ means the meta-feature predicts a *larger* GBDT advantage**. The sample size n varies because a few features (e.g. minority-class share, max categorical cardinality) are undefined on regression tasks or on datasets with no categorical columns. We used Holm correction over all features.

Meta-feature	n	ρ	p_{raw}	p_{adj}	Sig.
`num_rows`	142	$+0.603$	$< 10^{-14}$	$< 10^{-13}$	***
`num_high_cardinality_cats`	142	$+0.466$	$< 10^{-8}$	$< 10^{-7}$	***
`max_categorical_cardinality`	105	$+0.413$	$< 10^{-4}$	$< 10^{-3}$	***
`missing_value_fraction`	142	$+0.286$	0.0006	0.003	**
`num_cols_after_preproc`	142	$+0.212$	0.011	0.045	*
`age`	142	-0.242	0.004	0.019	*
`minority_class_pct`	98	-0.211	0.038	0.075	ns
`num_cols`	142	$+0.197$	0.019	0.056	ns
`num_text_cols`	142	$+0.051$	0.545	0.545	ns

E.5 Test 5: Which method should a practitioner pick, given their data regime?

Test E.1 established that there is a real global ranking across all 27 methods that is not noise, and Tests E.2–E.4 answered specific sub-questions about how dataset size, tuning, and dataset meta-features affect performance. None of these directly answers the question a practitioner actually faces: *given a dataset with a specific combination of properties, which method should I deploy?* A global

ranking averaged across all 142 datasets may not match the ranking inside any one regime — TFMs lead on average but lose to neural networks on large or non-IID data, for instance.
To turn the benchmark into actionable model-selection guidance, we run Test E.1's procedure separately within each of the 11 sub-benchmarks. The subsets correspond to one Friedman test per axis level along four orthogonal axes: task type (IID, temporal, grouped — 3 subsets), dataset scale (tiny, small, medium, large — 4 subsets), dimensionality (low-dim, high-dim — 2 subsets), and special feature types (text, high-cardinality — 2 subsets), for 11 sub-benchmarks in total. Each subset gets its own **Friedman test** on the methods restricted to its datasets, followed by Holm-corrected **pairwise Wilcoxon tests** within that subset.

For each subset s with N_s datasets, the Friedman test asks whether the 27 methods are equivalent within that subset:

$$H_0^{(s)} : \text{all 27 methods have the same expected within-subset rank;}$$
$$H_1^{(s)} : \text{at least one method has a different expected rank within subset } s.$$

When $H_0^{(s)}$ is rejected, we localize which method pairs differ by running pairwise Wilcoxon signed-rank tests within the subset. For each pair of methods (A, B) with paired per-dataset normalized-error differences $d_i = e_i^A - e_i^B$ on the N_s datasets in subset s:

$$H_0^{(s,A,B)} : \text{the differences } d_i \text{ are symmetrically distributed around zero;}$$
$$H_1^{(s,A,B)} : \text{the differences are not symmetric around zero (two-sided).}$$

Holm correction is applied jointly over all $\binom{27}{2} = 351$ pairs within each subset.

A coherent picture emerges across the sub-benchmarks (Table E.4): **TFMs** are the top-1 method on IID, tiny/small, and low-dimensional data. **Tuned and ensembled neural networks** (RealMLP) lead on temporal, grouped, medium, large, high-dimensional, and text subsets. **Tuned GBDTs** (CatBoost) lead on high-cardinality categorical data. The grouped subset is the one place where this narrative deserves a caveat: although RealMLP is the top-ranked method, only 5.1% of pairwise comparisons within the grouped subset survive Holm correction, meaning the ranking among the remaining methods is uncertain.

Table E.4: The Friedman test is significant ($p < 0.001$) in every one of the 11 subsets: the per-subset rankings are real. The fraction of pairwise comparisons that survive Holm correction within a subset (column "% sig. pairs") varies considerably, from 75.5% down to 5.1%. A high fraction means the subset is large and homogeneous enough that most methods can be reliably ordered against each other; a low fraction (the grouped subset, with only 18 datasets) means that even though some method is clearly best, the ranking among the others is too noisy. We report the top method and its family in each subset; for subsets with low sig. pairs, only the top method itself should be read as a robust recommendation. "Top-1 method" is the method with the smallest mean Friedman rank in the subset, with [D] denoting default-mode and [T+E] tuned-plus-ensembled.

Subset	n	χ^2	p	Sig. pairs	% sig.	Top-1 method	Family
Task type							
task=random	103	1086.8	$< 10^{-3}$	265/351	75.5	TabICLv2 (default)	TFM
task=temporal	21	345.4	$< 10^{-3}$	154/351	43.9	RealMLP (t+ens)	Neural
task=grouped	18	147.4	$< 10^{-3}$	18/351	5.1	RealMLP (t+ens)	Neural
Dataset scale							
scale=tiny	52	408.9	$< 10^{-3}$	163/351	46.4	TabICLv2 (default)	TFM
scale=small	38	500.2	$< 10^{-3}$	214/351	61.0	TabPFN-2.6 (default)	TFM
scale=medium	30	481.9	$< 10^{-3}$	196/351	55.8	RealMLP (t+ens)	Neural
scale=large	22	398.5	$< 10^{-3}$	196/351	55.8	RealMLP (t+ens)	Neural
Dimensionality							
dim=low-dim	91	927.5	$< 10^{-3}$	254/351	72.4	TabICLv2 (default)	TFM
dim=high-dim	51	568.4	$< 10^{-3}$	207/351	59.0	RealMLP (t+ens)	Neural
Feature type							
feature=text	16	229.2	$< 10^{-3}$	85/351	24.2	RealMLP (t+ens)	Neural
feature=high-card.	30	502.5	$< 10^{-3}$	208/351	59.3	CAT (t+ens)	Tree

F Ablations

We ablate `BeyondArena` to determine the validity of our results. We summarize the results in Table 1 in the main paper. Here, we provide more details on the experimental setup for each ablation.

1. In Appendix F.1, we introduce `BeyondArena-Core` and show that one can use fewer splits while obtaining similar representative results per dataset.
2. In Appendix F.2, we show that using IID outer splits for non-IID grouped data significantly distorts the results. Benchmarks should use the correct splits for non-IID data.
3. In Appendix F.3, we show that using 5×5 cross-validation instead of 8-fold cross-validation for smaller data did lead to better performance for all models, making it a valid pipeline choice.
4. In Appendix F.4, we show that using non-IID inner splits for grouped and temporal data leads to better performance, solidifying the importance of using the appropriate splits per task.
5. In Appendix F.5, we show that our grouped preprocessing leads to big gains, but not for all models.
6. In Appendix F.6, we show that our default text encoding method (Qwen3) is better than a classical TF-IDF method for short text. Yet, it is consistently beaten on large text. Our results remain representative, as the model rankings are highly correlated.
7. In Appendix F.7, we show that probability calibration can significantly boost the performance, mostly for tree-based models. While `BeyondArena` did not use calibration, future benchmarks should.

F.1 (Outer Splits, A.1) Using Fewer Splits with BeyondArena-Core

We use up to 60 splits per dataset. Inspired by TabArena's Lite subset, which established that benchmark compute costs can be substantially reduced while preserving the aggregate ranking by using only the first split of each dataset, we introduce `BeyondArena-Core`. We refine the concept of TabArena-Lite to not only preserve aggregate rankings but also to preserve per-dataset rankings, by adjusting the split budget on a per-dataset basis to achieve a per-dataset stability of $\Phi = 0.8$.

Formally, for each dataset d with F_d available splits, we define ρ_d as the mean off-diagonal Spearman correlation of dataset d's $F_d \times F_d$ split-similarity matrix, i.e., averaged pairwise rank agreement across errors of methods between splits. For each stability target $\Phi \in \{0, 0.6, 0.8, 0.9, 0.95, 1\}$ and each dataset d, we determine a per-dataset split budget via Spearman-Brown extrapolation:

$$k_d(\Phi) \;=\; \min\Big(F_d,\; \Big\lceil \tfrac{\Phi\,(1-\rho_d)}{\rho_d\,(1-\Phi)} \Big\rceil\Big), \qquad k_d(1) := F_d. \tag{1}$$

We then recompute the leaderboard restricted to those tasks and overlay two views on a shared percentage-points axis: (i) aggregate per-method $|\Delta\mathrm{Winrate}|$ against the $\Phi = 1$ reference (Mean and Max curves), and (ii) per-dataset bootstrap winrate jitter at $k_d(\Phi)$ as a strip plot with cross-dataset medians. The bootstrap is taken *with replacement* so the $\Phi = 1$ column reflects the irreducible variance of the all-splits estimate rather than collapsing to zero, providing a precision floor against which fidelity loss at smaller Φ can be judged. The dashed reference line marks the per-split mean absolute deviation (MAD) under uniform-random rankings (an upper bound for the $k > 1$ regime).

Figure F.1 shows that the aggregate mean winrate delta is well below 1% regardless of Φ, meaning that even a single split per dataset is sufficient to faithfully represent the overall leaderboard and model rankings, aligning with TabArena's insights. The per-dataset jitter median, however, grows substantially from Full through Lite: individual datasets' orderings become noticeably less stable as compute is reduced, even when the benchmark-wide ranking is preserved. In the worst case ($\Phi = 0$), several datasets have a winrate delta of over 20%, which is approaching random noise and thus insufficient for effective dataset-level analysis. By adding splits to satisfy Φ, we efficiently increase compute for the most unstable datasets, resulting in Core ($\Phi = 0.8$) having all datasets with a mean winrate delta below 10% while costing only 60% more than Lite.
Benchmark users and creators should use a Core-like subset ($\Phi = 0.8$) when reporting per-dataset results (rank tables, per-dataset wins/losses, dataset-conditioned analyses), which preserves per-dataset rank stability while being 5x cheaper than Full; those reporting only aggregate metrics (overall

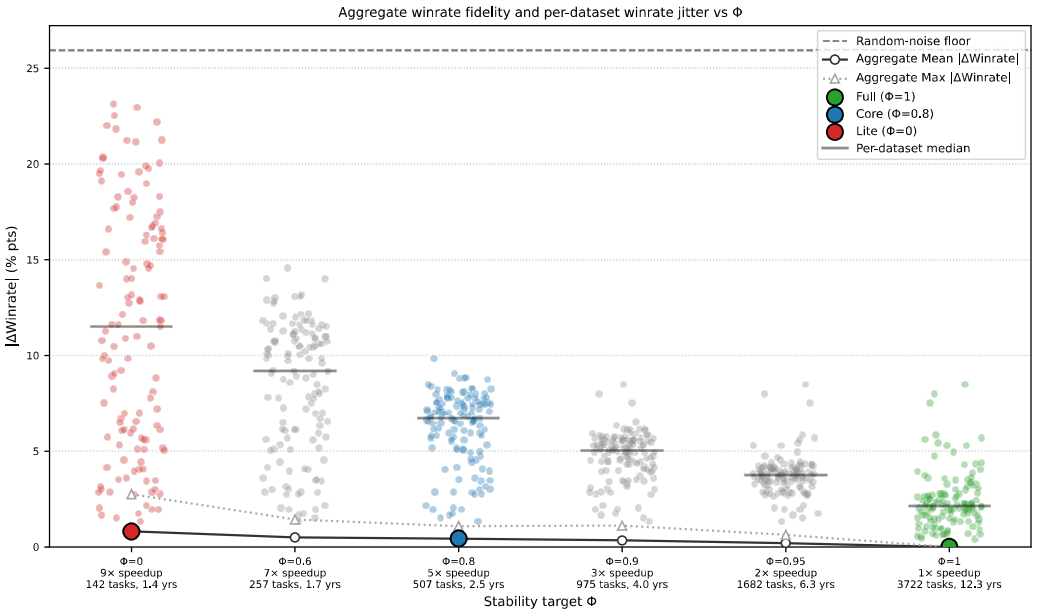

Figure F.1: **Compute–fidelity trade-off across stability targets Φ.** Aggregate per-method $|\Delta\text{Winrate}|$ vs. the $\Phi = 1$ reference (Mean: solid, Max: dotted) overlaid on per-dataset bootstrap winrate jitter at the per-dataset split budget $k_d(\Phi)$ (Eq. 1); one translucent dot per dataset, black bars mark the cross-dataset median. The dashed line is the per-split MAD under uniform-random rankings (upper bound for $k > 1$).

mean rank, headline winrate) can drop all the way to Lite ($\Phi = 0$) and recover the full speedup (9x faster than Full) with no meaningful loss in fidelity.

F.2 (Outer Splits, A.2) IID Splits for non-IID Grouped Data

We investigate the impact on benchmark results when using IID outer train-test splits for two grouped datasets (`musk` and `sat11_hand_algo_runtime`). We use the same experimental setup as described in Section 5, but reduce the costs by using a representative subset of models: Linear, ExtraTrees, LightGBM, RealMLP, TabPFN-2.6, and TabICLv2. Figure F.2 shows the results of running all models on the Grouped vs. IID outer splits. For `musk`, the task becomes trivial under IID splits, with almost all models having an error of 0. At the same time, the variance across folds and the model rankings are heavily distorted. `sat11_hand_algo_runtime` it looks similiar. While the task does not become trivial, it becomes much simpler. Ranks are distorted again.
We conclude, as with temporal data, that the appropriate outer split is crucial for comparing models.

F.3 (Inner Splits, B.1) Using 8-fold Cross-Validation for Tiny Data

We investigate the impact of changing the inner splits from 5-repeated 5-fold (5×5) CV to 8-fold CV This ablation only uses datasets with ≤ 500 training samples. To reduce the cost of the ablation, we (a) compare on IID datasets with ≤ 100 features, and (b) only run the first three outer folds. We then run the full tuning + ensembling for Linear, ExtraTrees, LightGBM, and RealMLP. We ignore TFMs because they are not affected by the validation protocol; they refit on the full data after cross-validation. Figure F.3 shows that 5×5-fold CV performance is better in all cases and gives the largest boost for RealMLP. The rankings are comparable across both protocols.

F.4 (Inner Splits, B.2) Using IID Splits for non-IID Data

We used non-IID validation splits for all non-IID datasets in `BeyondArena`. Here, we investigate how performance changes when using IID splits for validation. We follow the experimental setup from Section 5, using only Linear, ExtraTrees, and LightGBM, and running only the first outer split.

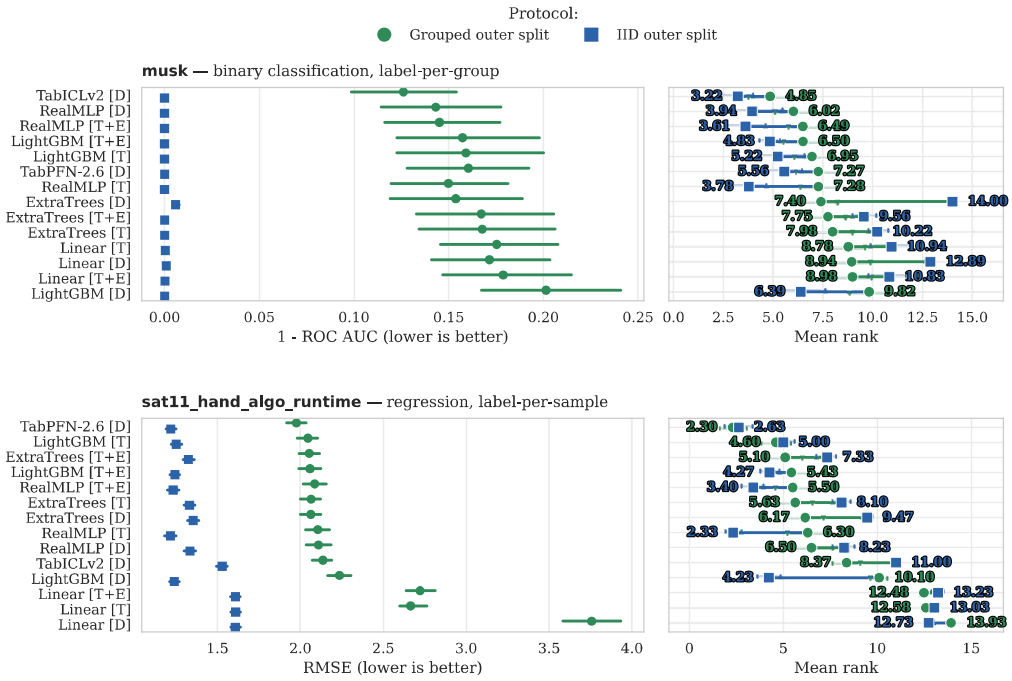

Figure F.2: **Grouped vs. IID Outer Splits.** We compare changing the outer splits for two grouped datasets. As a result, **(left)** the raw metric values and their variances change in a drastic manner, while **(right)** model ranks are heavily distorted (`musk` $\tau = 0.60$, `sat11_hand_algo_runtime` $\tau = 0.49$).

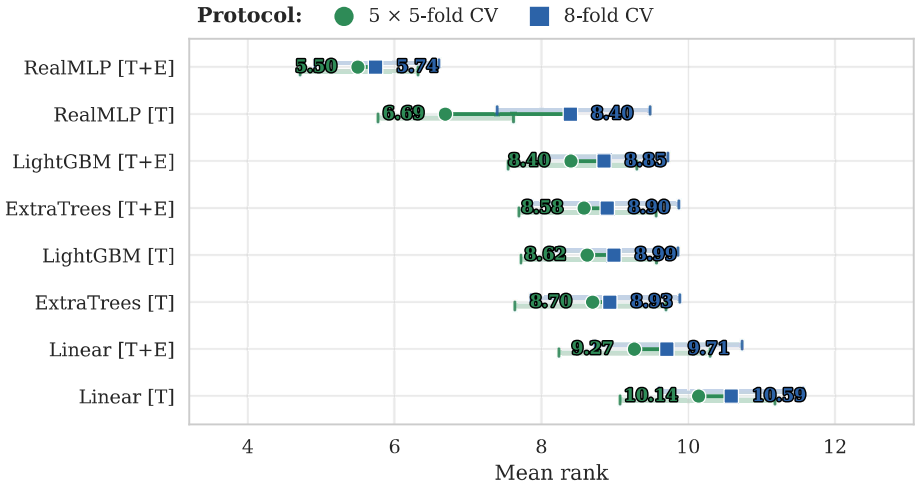

Figure F.3: **Performance of Cross-validation (CV) Strategies for Tiny Data.** We compare Linear, ExtraTrees, LightGBM, and RealMLP with tuning and with post-hoc ensembling on datasets with fewer than 500 samples, when changing the inner splits from 5-repeated 5-fold (5×5) CV to 8-fold CV; all other benchmark settings are identical. 5×5 CV consistently performs better than 8-fold. Moreover, tuned RealMLP gains a drastic boost, which aligns with it being the most overtuned model in TabArena [11].

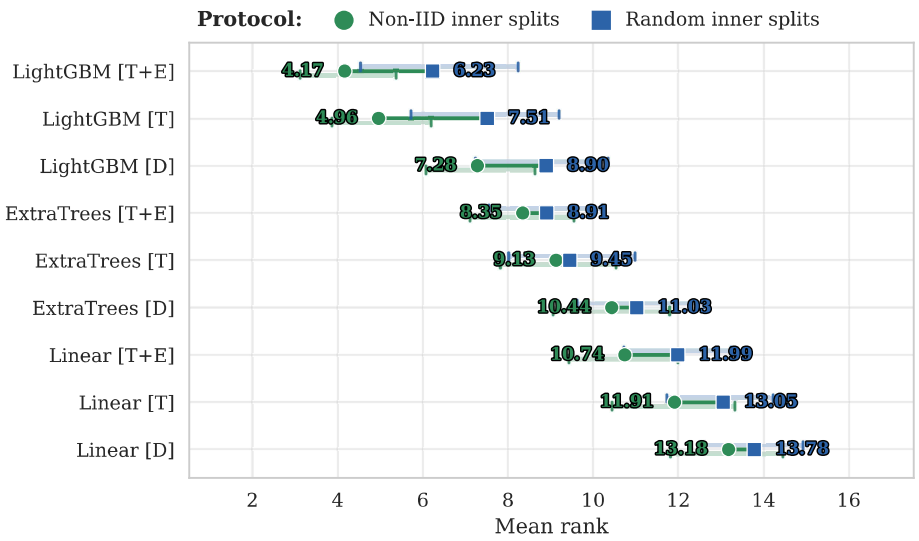

Figure F.4: **Performance of Non-IID vs IID Inner Splits for Non-IID Data.** We compare Linear, ExtraTrees, and LightGBM when using non-IID inner splits (grouped or temporal) as a validation protocol against always using random inner splits. We observe that Linear and LightGBM benefit significantly from non-IID splits, whereas ExtraTrees remains largely unaffected. Moreover, the ranking order is getting mixed: a tuned plus ensembled ExtraTrees model would beat a default LightGBM, only when using random inner splits.

In Figure F.4, we observe that non-IID splits led to better performance in almost all cases, and in such cases, providing a significant boost in performance.

F.5 (Grouped Data Preprocessing, C.1a/b) Disabling Preprocessing for Grouped Data.

In `BeyondArena`, we introduce a default model-agnostic preprocessing step for grouped data that follows industry practice. Here, we examine how the performance would change if we disable this preprocessing. We use the same experimental setup as in Section 5, but only compare the default performance of Linear, ExtraTrees, LightGBM, RealMLP, TabM, TabPFN-2.6, and TabICLv2. Moreover, we restrict the ablation to only using datasets with $\leq 100k$ training samples. In Figure F.5, we split the ablation by `label-per-group` (`L-P-G`) and `label-per-sample` (`L-P-S`) datasets, showing that for both, the rankings can change a lot while performance gains with grouped preprocessing are higher on average. For TabPFN-2.6, no preprocessing would have been better.

F.6 (Text Data Preprocessing, C.2a/b) TF-IDF for Text Encoding.

`BeyondArena` contains tabular data with text, which requires encoding for tabular machine learning models. By default, we used Qwen3-Embedding-8B with Matryoshka representation learning (MRL) for text encoding. Here, we ablate using TF-IDF from Skrub with SVD. We run the ablation across all datasets with text, following the experimental setup in Section 5. We use only the first split of each dataset and restrict the runs to the cheaper CPU models: Linear, ExtraTrees, and LightGBM. Figure F.6 shows mixed results. For short text, Qwen3 was the better choice. For long text, TF-IDF was the better choice. In the future, one would need to tune the preprocessing method for each dataset to achieve peak performance.

F.7 (Post-processing, D) Using Probability Calibration for Log-loss.

`BeyondArena` uses log-loss as a metric for multiclass classification. Post hoc calibration can improve the model's performance when measured by log-loss. Since `BeyondArena` does not use post hoc

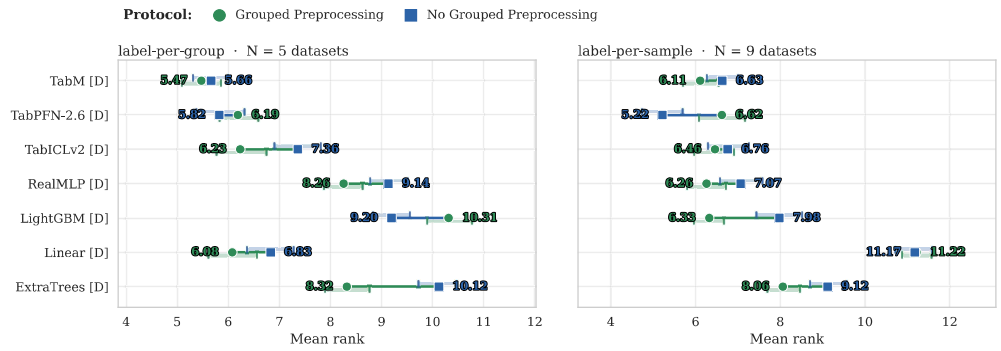

Figure F.5: **Impact of Preprocessing for Grouped Data.** We compare the default performance of Linear, ExtraTrees, LightGBM, RealMLP, TabICLv2, TabPFN-2.6, and TabM with and without the group preprocessing introduced by `BeyondArena`, split by `label-per-group` (`L-P-G`) and `label-per-sample` (`L-P-S`) data. We observe that the performance gains depend on the model, while, on average, enabling grouped preprocessing led to better performance. TabPFN-2.6 is a distinguishable outlier compared to all other models, as it benefited the most from not using grouped preprocessing.

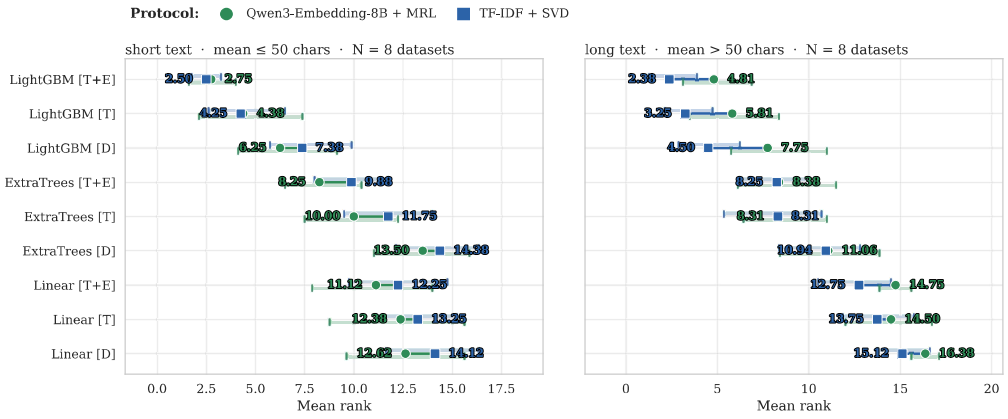

Figure F.6: **LM vs. TF-IDF for Encoding Text Features.** We compare the performance of Linear, ExtraTrees, and LightGBM when changing the preprocessing of text features from a language model (LM), in the form of Qwen3-Embedding-8B with Matryoshka representation learning (MRL), to a classical encoding method, in the form of TF-IDF with SVD. Both approaches create a 32-dimensional output vector for each text feature. We split the results across short-text ($\leq$ 50 characters on average across all text cells) and long-text datasets. For short texts, the LM consistently performs better, while for long texts, TF-IDF dominates. In both cases, the model rankings are not affected by the preprocessing choice.

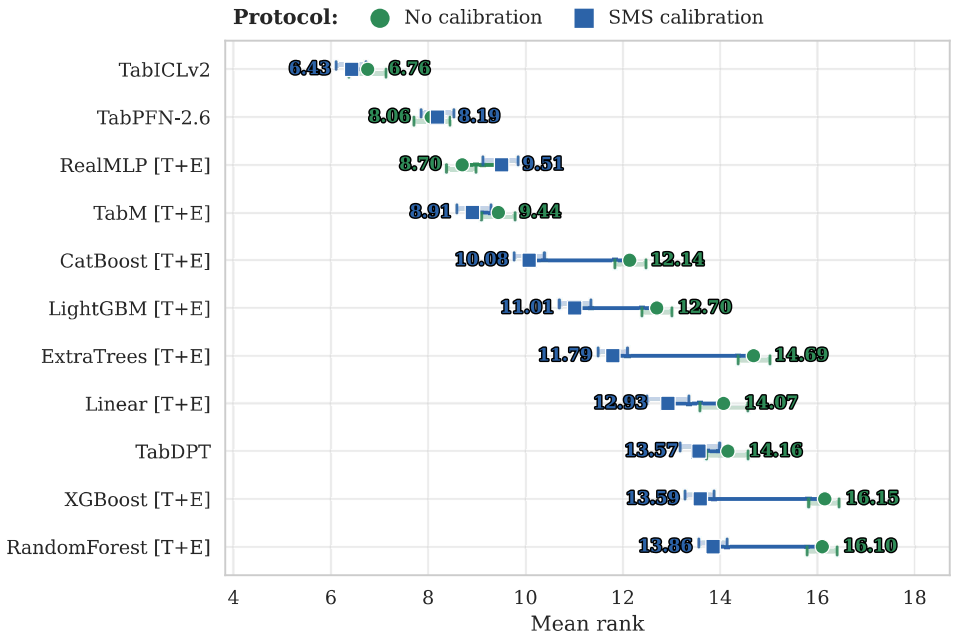

Figure F.7: **No Calibration vs. Probability Calibration for Multiclass Classification.** We compare the performance of all models when adding post hoc probability calibration (via SMS calibration [378]) after tuning and ensembling for multiclass classification. SMS calibration yields significant gains in average rank for several models and alters the model rank order.

calibration, we ablate using structured matrix scaling (SMS) for calibration [378], the default of the `probmetrics` framework [379]. We recompute the log-loss of all methods after SMS using the saved predictions from the main benchmark. In Figure F.7, we show that SMS calibration improved the performance for most methods. In particular, tree-based models improved significantly, while only TabPFN-2.6 and RealMLP performed worse in calibration.

G Per Dataset Results

Performance Per Dataset. We show the average predictive performance per dataset (metric error) with the standard deviation over folds. We show the performance for the default hyperparameter configuration (`Default`), for the model after tuning (`Tuned`), and for the ensemble after tuning (`Tuned + Ens.`). We highlight the best-performing methods with significance on three levels: (1) Green: The best performing method on average; (2) **Bold**: Methods that are not significantly worse than the best method on average, based on a Wilcoxon Signed-Rank test for paired samples with Holm-Bonferroni correction and $\alpha = 0.05$. (3) Underlined: Methods that are not significantly worse than the best method in the same pipeline regime (`Default`, `Tuned`, or `Tuned + Ens.`), based on a Wilcoxon Signed-Rank test for paired samples with Holm-Bonferroni correction and $\alpha = 0.05$. Datasets with only a single split do not receive bold/underline annotations because the significance test degenerates without paired samples.

HPO Pareto Trajectory. Beside each per-dataset table we plot the metric error (y-axis) of each method's tuned and ensembled configuration against cumulative training time in seconds (x-axis), as we increase the number of HPO trials. Each curve traces the Pareto frontier of validation error versus training-time budget for a single method, so points further down and to the left dominate. Reading the plot together with the table makes it possible to compare the cost (compute budget needed to reach a given error) and the achievable error of each method on that dataset, beyond the single endpoint summary in the table. Each point in the trajectory corresponds from left to right to an ensemble of {1, 2, 5, 10, 15, 20, 26} configurations. The default configuration is always present, with the remaining configurations being randomly selected and the results averaged across 10 seeds.

	Default	Tuned	Tuned + Ens.
Linear	**0.372 ± 0.038**	**0.376 ± 0.042**	**0.375 ± 0.041**
RandomForest	**0.365 ± 0.031**	**0.352 ± 0.045**	**0.353 ± 0.047**
ExtraTrees	**0.362 ± 0.035**	**0.357 ± 0.045**	**0.356 ± 0.044**
CatBoost	**0.355 ± 0.041**	**0.356 ± 0.042**	**0.356 ± 0.043**
LightGBM	**0.354 ± 0.048**	**0.352 ± 0.045**	**0.352 ± 0.044**
XGBoost	**0.354 ± 0.048**	**0.353 ± 0.046**	**0.353 ± 0.047**
RealMLP	**0.353 ± 0.049**	**0.351 ± 0.042**	**0.352 ± 0.045**
TabM	**0.353 ± 0.043**	**0.349 ± 0.042**	**0.348 ± 0.041**
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	**0.368 ± 0.041**	-	-

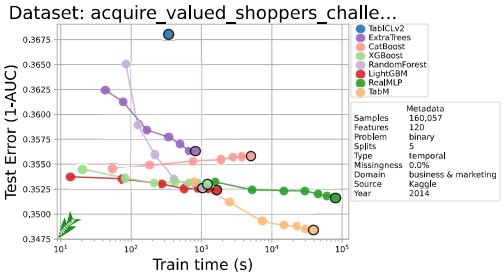

Figure G.1: **acquire_valued_shoppers_challenge**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	5.337 ± 0.152	4.745 ± 0.148	4.729 ± 0.135
RandomForest	1.945 ± 0.115	1.928 ± 0.113	1.874 ± 0.119
ExtraTrees	1.898 ± 0.114	1.677 ± 0.121	1.677 ± 0.123
CatBoost	1.602 ± 0.102	1.356 ± 0.122	1.361 ± 0.122
LightGBM	1.575 ± 0.130	1.530 ± 0.147	1.528 ± 0.135
XGBoost	1.574 ± 0.137	1.450 ± 0.126	1.456 ± 0.127
RealMLP	1.218 ± 0.101	1.185 ± 0.113	1.170 ± 0.110
TabM	1.283 ± 0.129	1.216 ± 0.107	1.195 ± 0.105
TabDPT	1.203 ± 0.092	-	-
TabPFN-2.6	**1.117 ± 0.096**	-	-
TabICLv2	**1.112 ± 0.092**	-	-

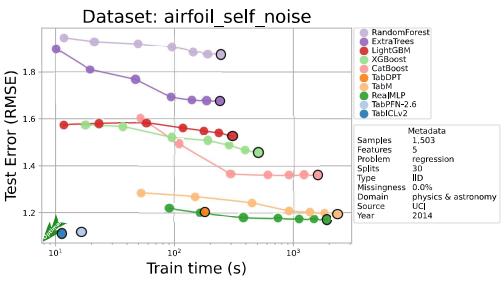

Figure G.2: **airfoil_self_noise**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.565 ± 0.001	0.565 ± 0.001	0.564 ± 0.001
RandomForest	0.560 ± 0.001	0.558 ± 0.001	0.557 ± 0.001
ExtraTrees	0.558 ± 0.001	0.558 ± 0.001	0.557 ± 0.001
CatBoost	0.534 ± 0.001	0.534 ± 0.001	0.534 ± 0.001
LightGBM	0.540 ± 0.001	0.536 ± 0.001	0.535 ± 0.001
XGBoost	0.538 ± 0.001	0.538 ± 0.001	0.536 ± 0.001
RealMLP	0.536 ± 0.001	0.533 ± 0.001	**0.532 ± 0.001**
TabM	0.533 ± 0.001	0.532 ± 0.001	0.532 ± 0.001
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.542 ± 0.001	-	-

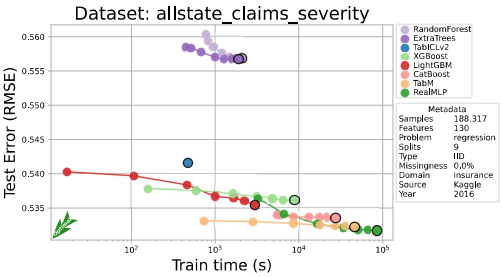

Figure G.3: **allstate_claims_severity**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.149 ± 0.009	0.147 ± 0.009	0.146 ± 0.010
RandomForest	0.172 ± 0.006	0.172 ± 0.006	0.172 ± 0.006
ExtraTrees	0.168 ± 0.007	0.160 ± 0.007	0.159 ± 0.007
CatBoost	0.120 ± 0.006	**0.118 ± 0.006**	**0.118 ± 0.006**
LightGBM	0.154 ± 0.007	0.145 ± 0.009	0.141 ± 0.008
XGBoost	0.166 ± 0.008	0.152 ± 0.011	0.143 ± 0.008
RealMLP	0.153 ± 0.009	0.132 ± 0.005	0.130 ± 0.007
TabM	0.163 ± 0.007	0.159 ± 0.011	0.151 ± 0.008
TabDPT	0.189 ± 0.008	-	-
TabPFN-2.6	0.152 ± 0.007	-	-
TabICLv2	0.145 ± 0.008	-	-

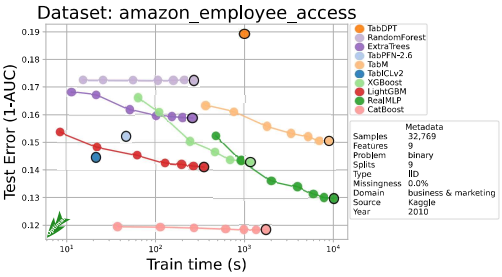

Figure G.4: **amazon_employee_access**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.061	0.061	0.058
RandomForest	0.057	0.057	0.058
ExtraTrees	0.060	0.061	0.060
CatBoost	0.057	0.056	0.056
LightGBM	0.057	0.055	0.055
XGBoost	0.056	0.055	0.055
RealMLP	0.058	0.055	0.055
TabM	0.055	0.054	0.055
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.063	-	-

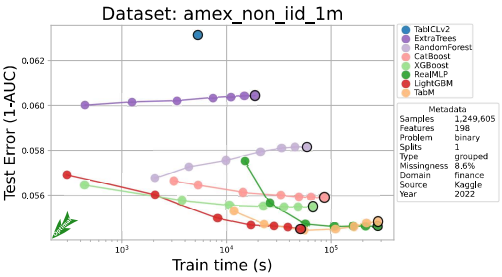

Figure G.5: **amex_non_iid_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.086 ± 0.044	0.077 ± 0.043	0.077 ± 0.043
RandomForest	0.081 ± 0.035	0.085 ± 0.058	0.077 ± 0.046
ExtraTrees	0.096 ± 0.028	0.082 ± 0.053	0.078 ± 0.048
CatBoost	0.068 ± 0.041	**0.067 ± 0.039**	**0.067 ± 0.039**
LightGBM	0.070 ± 0.039	**0.068 ± 0.040**	**0.067 ± 0.038**
XGBoost	**0.067 ± 0.038**	**0.067 ± 0.038**	**0.067 ± 0.038**
RealMLP	0.069 ± 0.041	**0.067 ± 0.040**	**0.067 ± 0.040**
TabM	**0.067 ± 0.039**	**0.067 ± 0.040**	**0.067 ± 0.039**
TabDPT	0.184 ± 0.063	-	-
TabPFN-2.6	**0.067 ± 0.040**	-	-
TabICLv2	0.069 ± 0.040	-	-

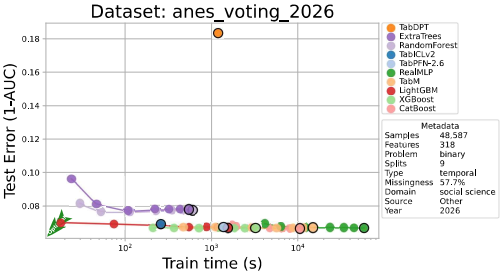

Figure G.6: **anes_voting_2026**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.012 ± 0.003	0.012 ± 0.003	0.010 ± 0.002
RandomForest	0.009 ± 0.002	0.009 ± 0.002	0.009 ± 0.002
ExtraTrees	0.010 ± 0.002	0.010 ± 0.003	0.009 ± 0.002
CatBoost	0.008 ± 0.002	0.008 ± 0.002	0.008 ± 0.002
LightGBM	0.008 ± 0.002	0.008 ± 0.002	0.008 ± 0.002
XGBoost	0.008 ± 0.002	0.008 ± 0.002	0.008 ± 0.002
RealMLP	0.010 ± 0.003	0.010 ± 0.002	0.008 ± 0.002
TabM	0.008 ± 0.002	0.008 ± 0.002	0.008 ± 0.002
TabDPT	0.010 ± 0.003	-	-
TabPFN-2.6	0.008 ± 0.002	-	-
TabICLv2	**0.006 ± 0.002**	-	-

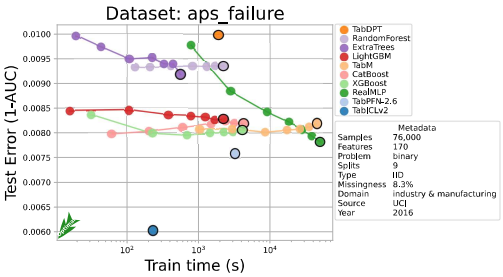

Figure G.7: **aps_failure**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	3.815 ± 0.655	3.063 ± 0.444	2.805 ± 0.401
RandomForest	2.286 ± 0.149	2.287 ± 0.149	2.279 ± 0.163
ExtraTrees	2.306 ± 0.144	2.265 ± 0.146	2.259 ± 0.139
CatBoost	2.231 ± 0.113	2.245 ± 0.110	2.242 ± 0.110
LightGBM	2.289 ± 0.078	2.267 ± 0.095	2.265 ± 0.091
XGBoost	2.282 ± 0.066	2.251 ± 0.079	2.249 ± 0.078
RealMLP	2.546 ± 0.317	2.257 ± 0.133	2.250 ± 0.117
TabM	2.307 ± 0.183	2.251 ± 0.129	2.242 ± 0.119
TabDPT	2.295 ± 0.114	-	-
TabPFN-2.6	**2.187 ± 0.101**	-	-
TabICLv2	**2.175 ± 0.097**	-	-

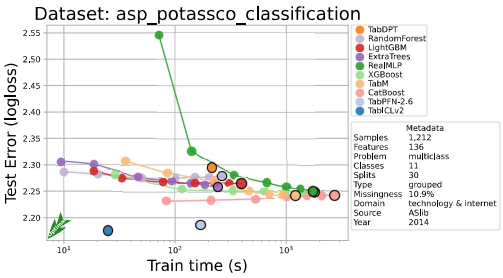

Figure G.8: **asp_potassco_classification**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.283 ± 0.086	0.293 ± 0.092	**0.276 ± 0.084**
RandomForest	0.361 ± 0.054	0.381 ± 0.086	0.365 ± 0.060
ExtraTrees	0.361 ± 0.052	0.380 ± 0.113	0.361 ± 0.066
CatBoost	0.427 ± 0.059	0.338 ± 0.065	0.336 ± 0.065
LightGBM	0.433 ± 0.053	0.359 ± 0.057	0.360 ± 0.056
XGBoost	0.368 ± 0.066	0.350 ± 0.057	0.349 ± 0.058
RealMLP	0.347 ± 0.052	0.310 ± 0.072	0.302 ± 0.066
TabM	0.342 ± 0.067	0.349 ± 0.070	0.347 ± 0.069
TabDPT	0.402 ± 0.053	-	-
TabPFN-2.6	0.309 ± 0.070	-	-
TabICLv2	**0.270 ± 0.093**	-	-

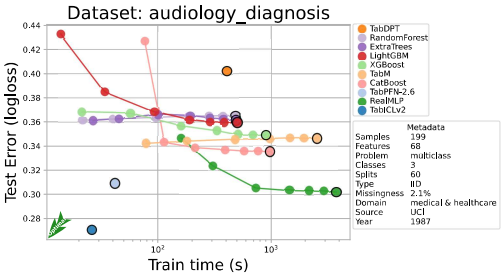

Figure G.9: **audiology_diagnosis**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.268 ± 0.017	0.269 ± 0.017	0.266 ± 0.018
RandomForest	0.269 ± 0.022	0.273 ± 0.026	0.269 ± 0.025
ExtraTrees	0.296 ± 0.022	0.279 ± 0.024	0.277 ± 0.025
CatBoost	**0.252 ± 0.020**	0.253 ± 0.022	**0.249 ± 0.020**
LightGBM	0.273 ± 0.023	0.264 ± 0.021	0.255 ± 0.019
XGBoost	0.269 ± 0.026	**0.253 ± 0.021**	**0.252 ± 0.021**
RealMLP	0.269 ± 0.025	0.264 ± 0.023	0.257 ± 0.020
TabM	0.261 ± 0.021	0.261 ± 0.023	0.261 ± 0.022
TabDPT	**0.254 ± 0.021**	-	-
TabPFN-2.6	**0.250 ± 0.021**	-	-
TabICLv2	**0.252 ± 0.020**	-	-

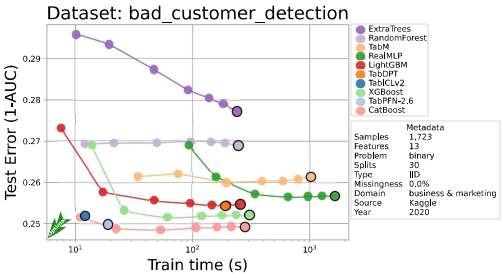

Figure G.10: **bad_customer_detection**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.229 ± 0.005	0.228 ± 0.005	0.228 ± 0.005
RandomForest	0.147 ± 0.007	0.143 ± 0.007	0.142 ± 0.007
ExtraTrees	0.146 ± 0.007	0.142 ± 0.006	0.140 ± 0.007
CatBoost	**0.130 ± 0.006**	**0.129 ± 0.005**	**0.129 ± 0.005**
LightGBM	0.135 ± 0.007	0.133 ± 0.006	0.132 ± 0.006
XGBoost	0.136 ± 0.007	0.135 ± 0.006	0.134 ± 0.006
RealMLP	0.134 ± 0.006	0.131 ± 0.005	0.130 ± 0.006
TabM	**0.130 ± 0.007**	**0.130 ± 0.007**	**0.129 ± 0.006**
TabDPT	0.134 ± 0.007	-	-
TabPFN-2.6	**0.129 ± 0.006**	-	-
TabICLv2	**0.129 ± 0.006**	-	-

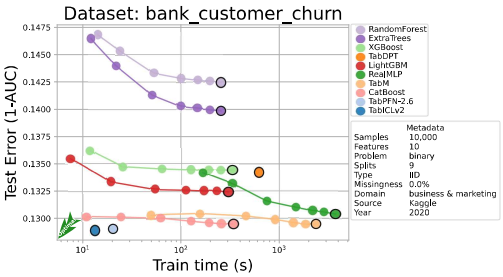

Figure G.11: **bank_customer_churn**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.260 ± 0.008	0.259 ± 0.008	0.259 ± 0.008
RandomForest	0.270 ± 0.006	0.241 ± 0.006	0.240 ± 0.005
ExtraTrees	0.275 ± 0.006	0.244 ± 0.006	0.243 ± 0.006
CatBoost	**0.235 ± 0.006**	**0.235 ± 0.006**	**0.235 ± 0.006**
LightGBM	0.237 ± 0.006	0.237 ± 0.006	**0.236 ± 0.006**
XGBoost	0.238 ± 0.007	**0.236 ± 0.006**	**0.235 ± 0.006**
RealMLP	0.240 ± 0.007	0.237 ± 0.006	**0.236 ± 0.006**
TabM	0.237 ± 0.006	0.237 ± 0.006	0.236 ± 0.006
TabDPT	0.238 ± 0.006	-	-
TabPFN-2.6	0.236 ± 0.006	-	-
TabICLv2	0.239 ± 0.007	-	-

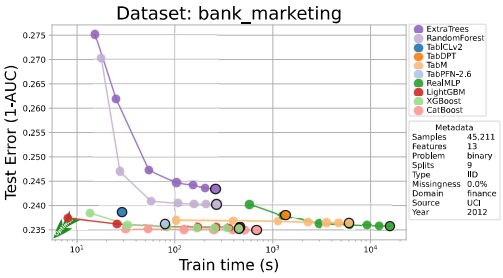

Figure G.12: **bank_marketing**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.842 ± 0.075	1.647 ± 0.048	1.647 ± 0.048
RandomForest	1.670 ± 0.013	1.573 ± 0.037	1.563 ± 0.032
ExtraTrees	1.684 ± 0.011	1.535 ± 0.036	1.535 ± 0.034
CatBoost	1.556 ± 0.028	1.527 ± 0.033	1.529 ± 0.032
LightGBM	1.694 ± 0.034	1.555 ± 0.034	1.552 ± 0.033
XGBoost	1.634 ± 0.040	1.571 ± 0.033	1.572 ± 0.034
RealMLP	1.576 ± 0.039	1.427 ± 0.037	1.420 ± 0.032
TabM	1.427 ± 0.048	1.418 ± 0.046	1.417 ± 0.045
TabDPT	1.511 ± 0.028	-	-
TabPFN-2.6	**1.373 ± 0.045**	-	-
TabICLv2	**1.369 ± 0.050**	-	-

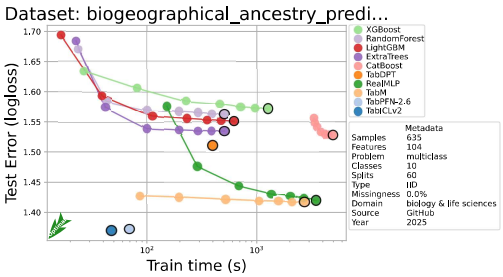

Figure G.13: **biogeographical_ancestry_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.373 ± 0.044	0.341 ± 0.057	0.340 ± 0.053
RandomForest	0.370 ± 0.029	0.368 ± 0.037	0.367 ± 0.037
ExtraTrees	0.411 ± 0.023	0.364 ± 0.030	0.364 ± 0.030
CatBoost	0.367 ± 0.043	0.349 ± 0.044	0.349 ± 0.044
LightGBM	0.377 ± 0.044	0.367 ± 0.044	0.364 ± 0.041
XGBoost	0.403 ± 0.050	0.376 ± 0.042	0.377 ± 0.041
RealMLP	0.355 ± 0.080	**0.336 ± 0.065**	**0.332 ± 0.054**
TabM	0.348 ± 0.036	0.341 ± 0.049	0.341 ± 0.048
TabDPT	0.340 ± 0.054	-	-
TabPFN-2.6	0.341 ± 0.056	-	-
TabICLv2	**0.332 ± 0.059**	-	-

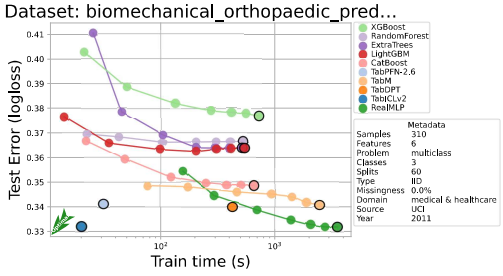

Figure G.14: **biomechanical_orthopaedic_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.211 ± 0.012	0.195 ± 0.011	0.186 ± 0.011
RandomForest	0.130 ± 0.012	0.130 ± 0.012	0.129 ± 0.012
ExtraTrees	0.133 ± 0.012	0.135 ± 0.012	0.132 ± 0.011
CatBoost	0.130 ± 0.009	**0.128 ± 0.009**	**0.128 ± 0.009**
LightGBM	0.132 ± 0.010	0.130 ± 0.011	0.129 ± 0.011
XGBoost	0.130 ± 0.009	**0.127 ± 0.011**	**0.127 ± 0.011**
RealMLP	0.142 ± 0.011	0.133 ± 0.009	**0.126 ± 0.010**
TabM	0.137 ± 0.010	**0.127 ± 0.010**	**0.127 ± 0.011**
TabDPT	0.135 ± 0.011	-	-
TabPFN-2.6	**0.125 ± 0.010**	-	-
TabICLv2	0.141 ± 0.011	-	-

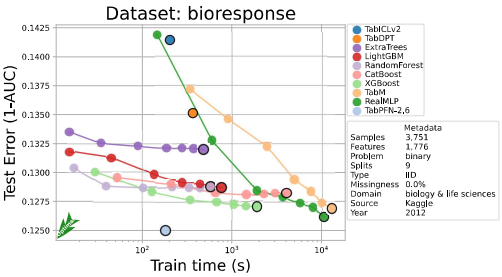

Figure G.15: **bioresponse**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.681 ± 0.030	0.683 ± 0.033	0.683 ± 0.030
RandomForest	0.699 ± 0.032	0.685 ± 0.031	0.686 ± 0.031
ExtraTrees	0.688 ± 0.029	0.683 ± 0.030	0.683 ± 0.029
CatBoost	0.684 ± 0.029	0.686 ± 0.028	0.685 ± 0.029
LightGBM	0.689 ± 0.029	0.686 ± 0.030	0.686 ± 0.030
XGBoost	0.701 ± 0.029	0.687 ± 0.030	0.687 ± 0.030
RealMLP	0.681 ± 0.031	0.681 ± 0.030	0.681 ± 0.030
TabM	0.685 ± 0.030	0.684 ± 0.031	0.684 ± 0.031
TabDPT	**0.673 ± 0.031**	-	-
TabPFN-2.6	0.678 ± 0.031	-	-
TabICLv2	0.678 ± 0.031	-	-

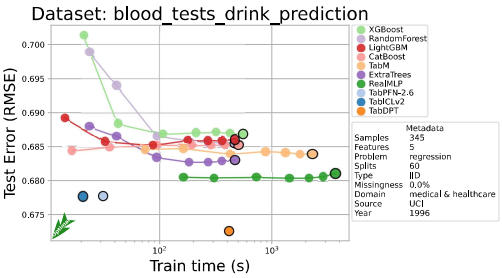

Figure G.16: **blood_tests_drink_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.269 ± 0.032	0.252 ± 0.034	**0.247 ± 0.032**
RandomForest	0.307 ± 0.029	0.289 ± 0.032	0.289 ± 0.032
ExtraTrees	0.304 ± 0.028	0.271 ± 0.032	0.272 ± 0.032
CatBoost	0.257 ± 0.032	0.256 ± 0.033	0.257 ± 0.032
LightGBM	0.269 ± 0.035	0.251 ± 0.033	0.252 ± 0.034
XGBoost	0.285 ± 0.035	0.264 ± 0.034	0.264 ± 0.034
RealMLP	**0.248 ± 0.032**	0.250 ± 0.030	0.249 ± 0.031
TabM	0.259 ± 0.031	0.262 ± 0.033	0.260 ± 0.032
TabDPT	0.251 ± 0.032	-	-
TabPFN-2.6	0.251 ± 0.032	-	-
TabICLv2	**0.247 ± 0.033**	-	-

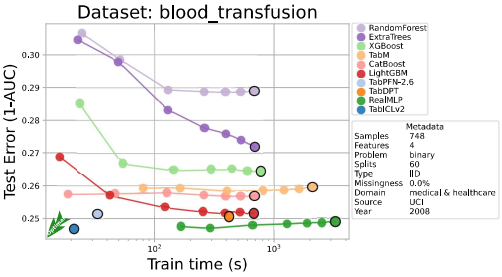

Figure G.17: **blood_transfusion**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.011 ± 0.001	0.011 ± 0.001	0.011 ± 0.001
RandomForest	0.011 ± 0.001	0.011 ± 0.001	0.011 ± 0.001
ExtraTrees	0.011 ± 0.001	0.011 ± 0.001	0.011 ± 0.001
CatBoost	0.011 ± 0.001	0.011 ± 0.001	0.011 ± 0.001
LightGBM	0.011 ± 0.001	0.011 ± 0.001	0.011 ± 0.001
XGBoost	0.012 ± 0.001	0.012 ± 0.001	0.012 ± 0.001
RealMLP	0.010 ± 0.001	0.010 ± 0.001	0.010 ± 0.001
TabM	0.011 ± 0.001	0.011 ± 0.001	0.011 ± 0.001
TabDPT	**0.010 ± 0.001**	-	-
TabPFN-2.6	0.010 ± 0.001	-	-
TabICLv2	**0.010 ± 0.001**	-	-

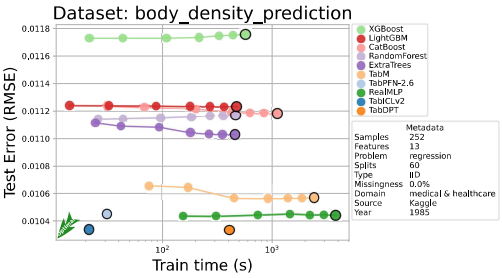

Figure G.18: **body_density_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.245 ± 0.012**	**0.242 ± 0.009**	**0.240 ± 0.009**
RandomForest	**0.211 ± 0.009**	**0.202 ± 0.009**	**0.202 ± 0.009**
ExtraTrees	**0.202 ± 0.011**	**0.201 ± 0.011**	**0.200 ± 0.011**
CatBoost	**0.196 ± 0.008**	**0.195 ± 0.008**	**0.195 ± 0.008**
LightGBM	**0.196 ± 0.010**	**0.195 ± 0.010**	**0.193 ± 0.010**
XGBoost	**0.198 ± 0.009**	**0.195 ± 0.010**	**0.194 ± 0.010**
RealMLP	**0.197 ± 0.009**	**0.193 ± 0.009**	**0.193 ± 0.009**
TabM	**0.199 ± 0.010**	**0.198 ± 0.012**	**0.197 ± 0.011**
TabDPT	**0.322 ± 0.018**	-	-
TabPFN-2.6	**0.189 ± 0.010**	-	-
TabICLv2	**0.197 ± 0.011**	-	-

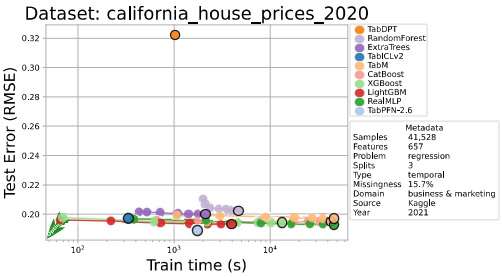

Figure G.19: **california_house_prices_2020**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.341 ± 0.065	0.345 ± 0.073	0.305 ± 0.062
RandomForest	0.253 ± 0.045	0.246 ± 0.052	0.241 ± 0.049
ExtraTrees	0.281 ± 0.046	0.245 ± 0.047	0.243 ± 0.046
CatBoost	0.225 ± 0.052	**0.222 ± 0.052**	**0.223 ± 0.052**
LightGBM	0.230 ± 0.051	0.232 ± 0.057	**0.225 ± 0.050**
XGBoost	0.231 ± 0.051	0.231 ± 0.056	**0.227 ± 0.052**
RealMLP	0.278 ± 0.062	0.263 ± 0.054	0.245 ± 0.051
TabM	0.270 ± 0.066	0.259 ± 0.060	0.252 ± 0.058
TabDPT	0.258 ± 0.059	-	-
TabPFN-2.6	0.254 ± 0.070	-	-
TabICLv2	**0.230 ± 0.064**	-	-

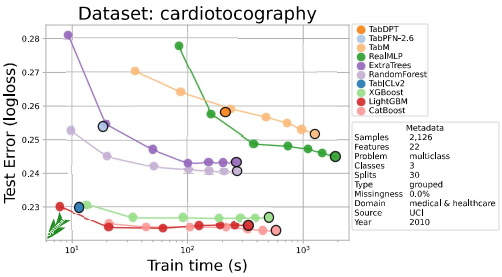

Figure G.20: **cardiotocography**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.223 ± 0.014	0.175 ± 0.006	0.175 ± 0.006
RandomForest	0.083 ± 0.012	0.082 ± 0.012	0.081 ± 0.012
ExtraTrees	0.085 ± 0.010	0.082 ± 0.011	0.080 ± 0.011
CatBoost	0.076 ± 0.005	0.076 ± 0.005	0.076 ± 0.005
LightGBM	0.083 ± 0.011	0.080 ± 0.011	0.081 ± 0.010
XGBoost	0.077 ± 0.008	0.077 ± 0.008	0.079 ± 0.010
RealMLP	0.081 ± 0.009	0.077 ± 0.010	0.077 ± 0.009
TabM	0.078 ± 0.010	0.079 ± 0.011	0.080 ± 0.011
TabDPT	0.074 ± 0.012	-	-
TabPFN-2.6	0.074 ± 0.010	-	-
TabICLv2	**0.067 ± 0.010**	-	-

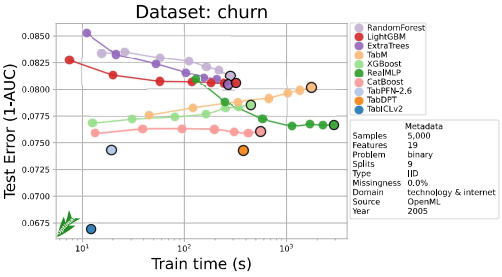

Figure G.21: **churn**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.905 ± 0.089	0.878 ± 0.084	0.870 ± 0.083
RandomForest	0.879 ± 0.082	0.867 ± 0.077	0.867 ± 0.078
ExtraTrees	0.877 ± 0.078	0.883 ± 0.079	0.882 ± 0.079
CatBoost	0.875 ± 0.080	0.863 ± 0.078	0.863 ± 0.078
LightGBM	0.884 ± 0.080	0.869 ± 0.081	0.868 ± 0.082
XGBoost	0.909 ± 0.085	0.873 ± 0.081	0.872 ± 0.080
RealMLP	0.840 ± 0.078	0.839 ± 0.076	**0.833 ± 0.075**
TabM	**0.826 ± 0.079**	0.834 ± 0.079	0.833 ± 0.078
TabDPT	0.848 ± 0.083	-	-
TabPFN-2.6	0.849 ± 0.082	-	-
TabICLv2	0.855 ± 0.086	-	-

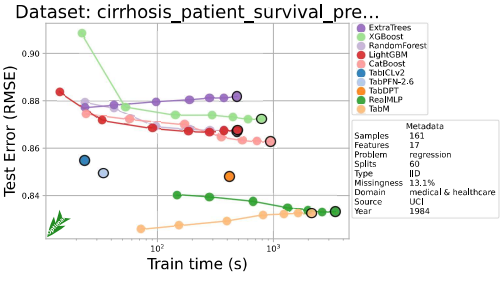

Figure G.22: **cirrhosis_patient_survival_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.538	1.545	1.537
RandomForest	1.496	1.496	1.494
ExtraTrees	1.526	1.519	1.519
CatBoost	1.339	1.330	1.325
LightGBM	1.353	1.329	1.321
XGBoost	1.341	1.341	1.327
RealMLP	1.332	1.300	1.293
TabM	1.340	1.312	1.312
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	1.392	-	-

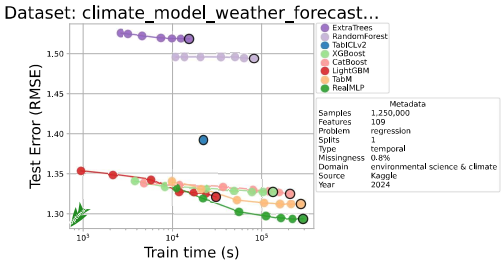

Figure G.23: **climate_model_weather_forecasting_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.528 ± 0.075	0.541 ± 0.069	0.542 ± 0.069
RandomForest	0.518 ± 0.069	0.511 ± 0.080	0.510 ± 0.079
ExtraTrees	0.509 ± 0.069	0.500 ± 0.071	0.498 ± 0.071
CatBoost	0.482 ± 0.065	0.477 ± 0.067	0.474 ± 0.067
LightGBM	**0.461 ± 0.062**	0.472 ± 0.058	0.470 ± 0.059
XGBoost	0.477 ± 0.066	**0.461 ± 0.066**	**0.461 ± 0.063**
RealMLP	0.503 ± 0.069	0.479 ± 0.066	**0.465 ± 0.063**
TabM	0.477 ± 0.063	**0.468 ± 0.062**	**0.469 ± 0.065**
TabDPT	0.544 ± 0.067	-	-
TabPFN-2.6	0.478 ± 0.068	-	-
TabICLv2	0.508 ± 0.068	-	-

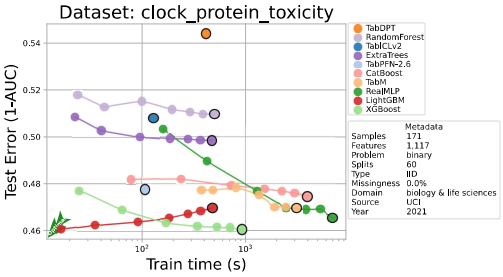

Figure G.24: **clock_protein_toxicity**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**1.205 ± 0.175**	**1.145 ± 0.150**	**1.148 ± 0.187**
RandomForest	**1.212 ± 0.205**	**1.206 ± 0.194**	**1.204 ± 0.204**
ExtraTrees	**1.247 ± 0.257**	**1.230 ± 0.250**	**1.229 ± 0.251**
CatBoost	**1.105 ± 0.183**	**1.093 ± 0.165**	**1.093 ± 0.168**
LightGBM	**1.136 ± 0.185**	**1.085 ± 0.186**	**1.085 ± 0.186**
XGBoost	**1.133 ± 0.196**	**1.091 ± 0.182**	**1.091 ± 0.182**
RealMLP	**1.079 ± 0.175**	**1.079 ± 0.184**	**1.070 ± 0.190**
TabM	**1.104 ± 0.179**	**1.104 ± 0.158**	**1.100 ± 0.168**
TabDPT	**1.111 ± 0.140**	-	-
TabPFN-2.6	**1.042 ± 0.183**	-	-
TabICLv2	**1.080 ± 0.172**	-	-

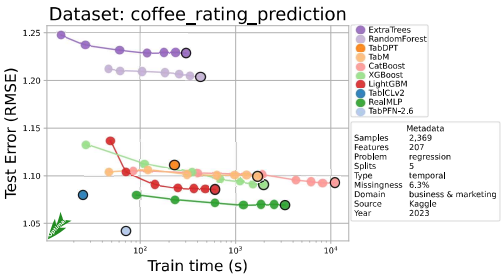

Figure G.25: **coffee_rating_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.267 ± 0.020	0.264 ± 0.022	0.264 ± 0.022
RandomForest	0.296 ± 0.019	0.257 ± 0.021	0.257 ± 0.020
ExtraTrees	0.296 ± 0.020	0.261 ± 0.023	0.255 ± 0.023
CatBoost	0.241 ± 0.021	0.240 ± 0.018	0.240 ± 0.021
LightGBM	0.251 ± 0.019	0.239 ± 0.019	0.238 ± 0.020
XGBoost	0.242 ± 0.016	0.247 ± 0.021	0.243 ± 0.018
RealMLP	0.263 ± 0.021	0.244 ± 0.019	0.242 ± 0.017
TabM	0.246 ± 0.019	0.239 ± 0.022	0.239 ± 0.021
TabDPT	0.276 ± 0.020	-	-
TabPFN-2.6	**0.225 ± 0.021**	-	-
TabICLv2	**0.229 ± 0.022**	-	-

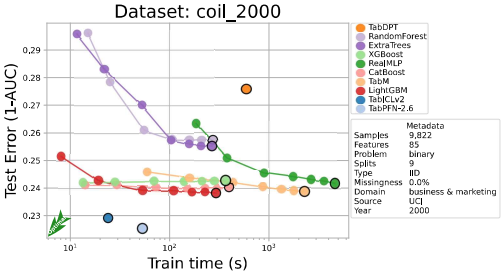

Figure G.26: **coil_2000**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	8.2 ± 0.4	8.2 ± 0.4	8.2 ± 0.4
RandomForest	5.2 ± 0.4	5.0 ± 0.4	5.0 ± 0.4
ExtraTrees	5.2 ± 0.4	4.9 ± 0.4	4.9 ± 0.4
CatBoost	4.1 ± 0.4	4.1 ± 0.4	4.1 ± 0.4
LightGBM	4.4 ± 0.3	4.2 ± 0.4	4.2 ± 0.3
XGBoost	4.6 ± 0.4	4.2 ± 0.3	4.2 ± 0.3
RealMLP	4.7 ± 0.3	4.1 ± 0.3	4.0 ± 0.3
TabM	4.2 ± 0.3	4.2 ± 0.3	131.7 ± 583.8
TabDPT	4.2 ± 0.4	-	-
TabPFN-2.6	4.0 ± 0.3	-	-
TabICLv2	**3.9 ± 0.4**	-	-

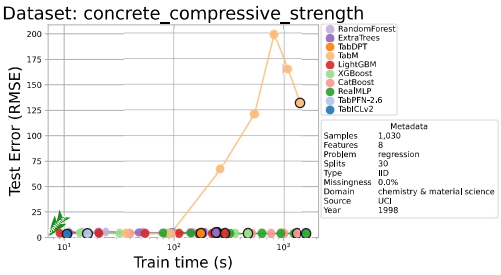

Figure G.27: **concrete_compressive_strength**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.561	0.601	0.565
RandomForest	0.437	0.437	0.432
ExtraTrees	0.479	0.427	0.430
CatBoost	0.431	0.423	0.425
LightGBM	0.429	0.430	0.428
XGBoost	0.432	0.433	0.431
RealMLP	0.446	0.446	0.435
TabM	0.443	0.426	0.424
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.445	-	-

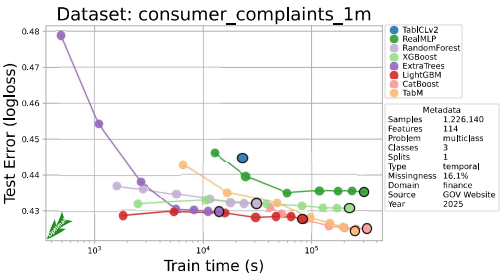

Figure G.28: **consumer_complaints_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.487	0.485	0.483
RandomForest	0.483	0.483	0.483
ExtraTrees	0.485	0.485	0.485
CatBoost	0.476	0.476	0.476
LightGBM	0.481	0.477	0.476
XGBoost	0.478	0.478	0.477
RealMLP	0.480	0.477	0.476
TabM	0.482	0.479	0.478
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.488	-	-

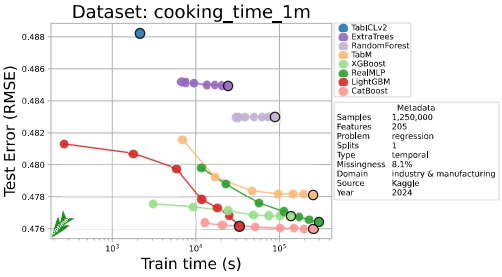

Figure G.29: **cooking_time_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.819 ± 0.128**	**0.805 ± 0.143**	**0.805 ± 0.140**
RandomForest	**0.769 ± 0.056**	**0.754 ± 0.049**	**0.752 ± 0.053**
ExtraTrees	**0.783 ± 0.034**	**0.754 ± 0.025**	**0.762 ± 0.019**
CatBoost	**0.739 ± 0.026**	**0.715 ± 0.046**	**0.717 ± 0.034**
LightGBM	**0.807 ± 0.016**	**0.720 ± 0.058**	**0.724 ± 0.051**
XGBoost	**0.812 ± 0.038**	**0.743 ± 0.046**	**0.744 ± 0.045**
RealMLP	**0.776 ± 0.088**	**0.733 ± 0.055**	**0.712 ± 0.061**
TabM	**0.850 ± 0.102**	**0.696 ± 0.109**	**0.707 ± 0.098**
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	**1.025 ± 0.256**	-	-

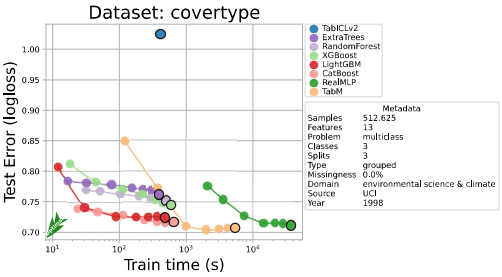

Figure G.30: **covertype**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.079 ± 0.016	0.076 ± 0.017	0.075 ± 0.017
RandomForest	0.066 ± 0.014	0.065 ± 0.015	0.065 ± 0.015
ExtraTrees	0.074 ± 0.016	0.076 ± 0.016	0.075 ± 0.016
CatBoost	0.062 ± 0.014	0.065 ± 0.015	0.064 ± 0.014
LightGBM	0.063 ± 0.014	0.064 ± 0.015	0.062 ± 0.014
XGBoost	0.067 ± 0.014	0.063 ± 0.014	0.062 ± 0.014
RealMLP	0.075 ± 0.016	0.069 ± 0.015	0.068 ± 0.015
TabM	0.064 ± 0.015	0.065 ± 0.015	0.064 ± 0.015
TabDPT	0.066 ± 0.016	-	-
TabPFN-2.6	**0.060 ± 0.014**	-	-
TabICLv2	0.061 ± 0.013	-	-

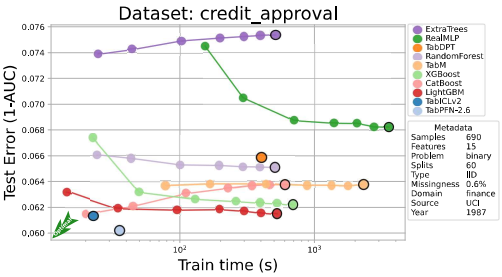

Figure G.31: **credit_approval**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.254 ± 0.005	0.254 ± 0.005	0.255 ± 0.005
RandomForest	0.231 ± 0.004	0.219 ± 0.004	0.219 ± 0.004
ExtraTrees	0.229 ± 0.003	0.220 ± 0.002	0.218 ± 0.003
CatBoost	0.215 ± 0.003	0.215 ± 0.003	0.215 ± 0.003
LightGBM	0.216 ± 0.003	0.215 ± 0.003	0.214 ± 0.003
XGBoost	0.217 ± 0.003	0.215 ± 0.003	0.214 ± 0.003
RealMLP	0.215 ± 0.003	0.214 ± 0.003	0.214 ± 0.003
TabM	0.216 ± 0.004	0.211 ± 0.003	0.212 ± 0.003
TabDPT	0.216 ± 0.002	-	-
TabPFN-2.6	0.210 ± 0.003	-	-
TabICLv2	**0.209 ± 0.003**	-	-

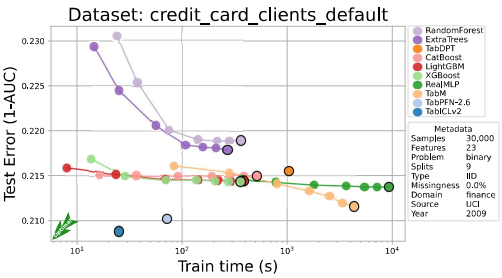

Figure G.32: **credit_card_clients_default**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.217 ± 0.026	0.211 ± 0.026	0.211 ± 0.027
RandomForest	0.215 ± 0.029	0.218 ± 0.028	0.218 ± 0.028
ExtraTrees	0.219 ± 0.028	0.221 ± 0.027	0.220 ± 0.027
CatBoost	0.210 ± 0.028	0.207 ± 0.028	**0.205 ± 0.028**
LightGBM	0.234 ± 0.025	**0.208 ± 0.029**	**0.206 ± 0.028**
XGBoost	0.218 ± 0.030	0.208 ± 0.030	0.208 ± 0.030
RealMLP	0.222 ± 0.025	0.214 ± 0.029	0.208 ± 0.028
TabM	**0.207 ± 0.029**	**0.209 ± 0.030**	**0.205 ± 0.029**
TabDPT	0.215 ± 0.028	-	-
TabPFN-2.6	**0.207 ± 0.026**	-	-
TabICLv2	**0.205 ± 0.026**	-	-

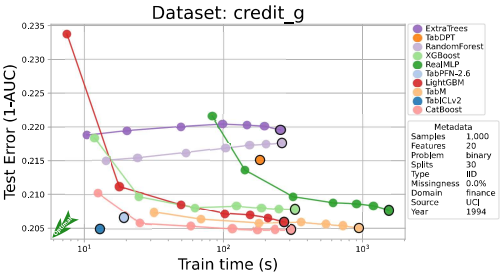

Figure G.33: **credit_g**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.038 ± 0.002	0.038 ± 0.002	0.038 ± 0.002
RandomForest	0.008 ± 0.000	0.008 ± 0.000	0.007 ± 0.000
ExtraTrees	0.008 ± 0.000	0.006 ± 0.000	0.006 ± 0.000
CatBoost	0.005 ± 0.000	0.005 ± 0.000	0.005 ± 0.000
LightGBM	0.006 ± 0.000	0.006 ± 0.000	0.006 ± 0.000
XGBoost	0.006 ± 0.000	0.006 ± 0.000	0.006 ± 0.000
RealMLP	0.005 ± 0.000	0.005 ± 0.000	0.005 ± 0.000
TabM	0.005 ± 0.000	**0.005 ± 0.000**	**0.005 ± 0.000**
TabDPT	0.006 ± 0.000	-	-
TabPFN-2.6	**0.005 ± 0.000**	-	-
TabICLv2	0.005 ± 0.000	-	-

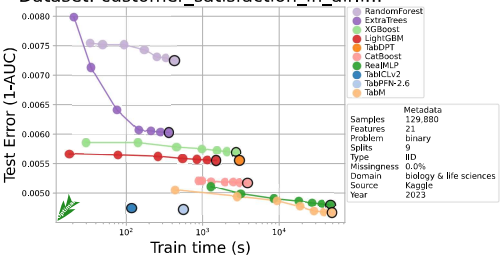

Figure G.34: **customer_satisfaction_in_airline**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.004	0.999	0.874
RandomForest	0.530	0.530	0.531
ExtraTrees	0.533	0.533	0.533
CatBoost	0.516	0.515	0.515
LightGBM	0.517	0.515	0.515
XGBoost	0.516	0.516	0.514
RealMLP	0.517	0.514	0.514
TabM	0.517	0.517	0.515
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.524	-	-

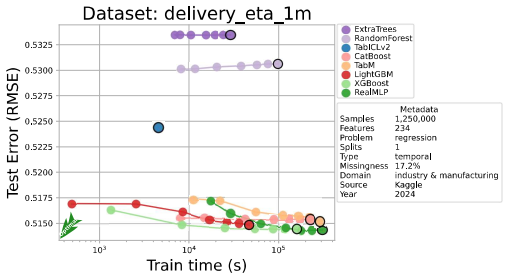

Figure G.35: **delivery_eta_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.719 ± 0.071	**0.625 ± 0.062**	**0.624 ± 0.062**
RandomForest	0.687 ± 0.048	0.663 ± 0.058	0.662 ± 0.053
ExtraTrees	0.714 ± 0.044	0.645 ± 0.051	0.645 ± 0.051
CatBoost	0.668 ± 0.052	0.645 ± 0.051	0.646 ± 0.051
LightGBM	0.663 ± 0.053	0.636 ± 0.061	0.635 ± 0.057
XGBoost	0.714 ± 0.052	0.685 ± 0.058	0.683 ± 0.058
RealMLP	0.649 ± 0.083	**0.624 ± 0.071**	**0.619 ± 0.066**
TabM	0.653 ± 0.066	0.647 ± 0.072	0.644 ± 0.068
TabDPT	0.706 ± 0.079	-	-
TabPFN-2.6	0.671 ± 0.074	-	-
TabICLv2	0.682 ± 0.077	-	-

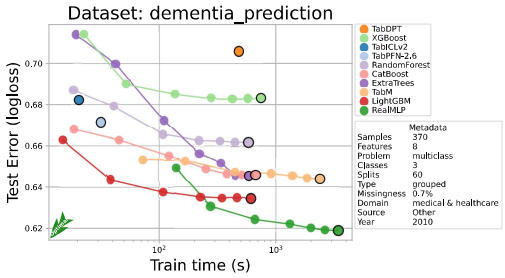

Figure G.36: **dementia_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.352 ± 0.002	0.342 ± 0.002	0.342 ± 0.002
RandomForest	0.364 ± 0.002	0.346 ± 0.002	0.344 ± 0.003
ExtraTrees	0.369 ± 0.003	0.350 ± 0.003	0.347 ± 0.003
CatBoost	0.328 ± 0.003	**0.327 ± 0.003**	**0.327 ± 0.003**
LightGBM	0.350 ± 0.003	0.338 ± 0.002	0.333 ± 0.002
XGBoost	0.335 ± 0.003	0.330 ± 0.003	0.329 ± 0.003
RealMLP	0.341 ± 0.004	0.330 ± 0.004	0.330 ± 0.002
TabM	0.339 ± 0.003	0.338 ± 0.003	0.337 ± 0.003
TabDPT	0.377 ± 0.005	-	-
TabPFN-2.6	0.332 ± 0.002	-	-
TabICLv2	0.335 ± 0.002	-	-

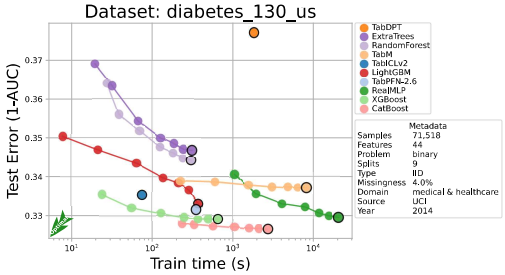

Figure G.37: **diabetes_130_us**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.195 ± 0.036	0.202 ± 0.063	0.198 ± 0.056
RandomForest	0.092 ± 0.002	0.092 ± 0.002	0.092 ± 0.002
ExtraTrees	0.091 ± 0.001	0.090 ± 0.001	0.090 ± 0.001
CatBoost	0.084 ± 0.001	0.084 ± 0.001	0.084 ± 0.001
LightGBM	0.085 ± 0.001	0.085 ± 0.001	0.084 ± 0.001
XGBoost	0.085 ± 0.000	0.085 ± 0.001	0.085 ± 0.001
RealMLP	0.086 ± 0.001	0.082 ± 0.001	0.081 ± 0.001
TabM	0.082 ± 0.001	0.081 ± 0.001	0.081 ± 0.000
TabDPT	0.085 ± 0.002	-	-
TabPFN-2.6	0.082 ± 0.001	-	-
TabICLv2	**0.080 ± 0.001**	-	-

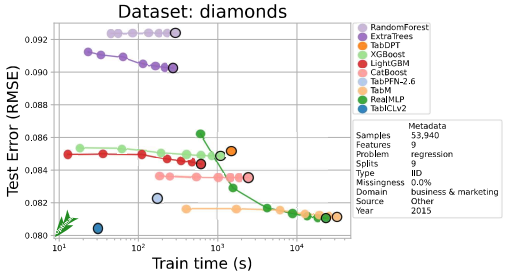

Figure G.38: **diamonds**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.245 ± 0.030	0.247 ± 0.034	0.234 ± 0.031
RandomForest	0.174 ± 0.032	0.174 ± 0.032	0.173 ± 0.033
ExtraTrees	0.154 ± 0.032	0.153 ± 0.032	0.154 ± 0.032
CatBoost	0.164 ± 0.031	0.158 ± 0.031	0.156 ± 0.031
LightGBM	0.187 ± 0.035	0.169 ± 0.034	0.170 ± 0.034
XGBoost	0.181 ± 0.032	0.181 ± 0.035	0.176 ± 0.033
RealMLP	0.188 ± 0.033	0.170 ± 0.033	0.159 ± 0.028
TabM	0.177 ± 0.033	0.170 ± 0.029	0.167 ± 0.029
TabDPT	0.182 ± 0.035	-	-
TabPFN-2.6	0.150 ± 0.027	-	-
TabICLv2	**0.140 ± 0.028**	-	-

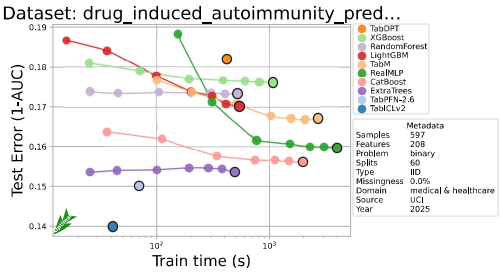

Figure G.39: **drug_induced_autoimmunity_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	17.0 ± 4.0	14.0 ± 1.0	13.0 ± 0.0
RandomForest	13.0 ± 0.0	13.0 ± 0.0	13.0 ± 0.0
ExtraTrees	13.0 ± 0.0	13.0 ± 0.0	13.0 ± 0.0
CatBoost	13.0 ± 0.0	13.0 ± 0.0	**13.0 ± 0.0**
LightGBM	13.0 ± 0.0	13.0 ± 0.0	13.0 ± 0.0
XGBoost	13.0 ± 0.0	13.0 ± 0.0	13.0 ± 0.0
RealMLP	13.0 ± 0.0	13.0 ± 0.0	13.0 ± 0.0
TabM	13415.0 ± 31489.0	29.0 ± 46.0	138.0 ± 373.(
TabDPT	13.0 ± 0.0	-	-
TabPFN-2.6	13.0 ± 0.0	-	-
TabICLv2	13.0 ± 0.0	-	-

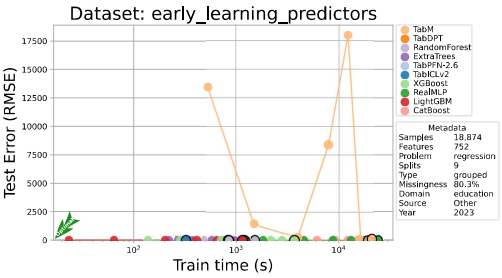

Figure G.40: **early_learning_predictors**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.053 ± 0.018	0.050 ± 0.017	0.050 ± 0.017
RandomForest	0.028 ± 0.014	0.029 ± 0.014	0.029 ± 0.014
ExtraTrees	0.027 ± 0.014	0.029 ± 0.015	0.029 ± 0.014
CatBoost	0.028 ± 0.014	0.028 ± 0.013	0.028 ± 0.013
LightGBM	0.041 ± 0.017	0.038 ± 0.017	0.038 ± 0.016
XGBoost	0.041 ± 0.016	0.037 ± 0.016	0.036 ± 0.016
RealMLP	0.037 ± 0.017	0.032 ± 0.015	0.033 ± 0.015
TabM	0.040 ± 0.017	0.037 ± 0.018	0.038 ± 0.017
TabDPT	0.033 ± 0.015	-	-
TabPFN-2.6	0.031 ± 0.013	-	-
TabICLv2	**0.025 ± 0.012**	-	-

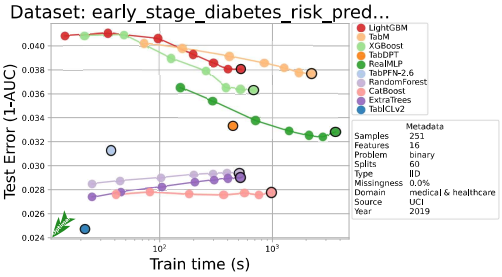

Figure G.41: **early_stage_diabetes_risk_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.386 ± 0.057	0.388 ± 0.063	0.384 ± 0.062
RandomForest	0.396 ± 0.054	0.404 ± 0.064	0.401 ± 0.061
ExtraTrees	0.420 ± 0.038	0.394 ± 0.055	0.392 ± 0.055
CatBoost	0.385 ± 0.060	**0.369 ± 0.056**	**0.368 ± 0.055**
LightGBM	0.408 ± 0.059	0.384 ± 0.058	0.382 ± 0.054
XGBoost	0.443 ± 0.059	0.405 ± 0.053	0.406 ± 0.054
RealMLP	0.405 ± 0.097	0.382 ± 0.071	0.371 ± 0.067
TabM	0.386 ± 0.059	0.395 ± 0.062	0.391 ± 0.063
TabDPT	0.372 ± 0.065	-	-
TabPFN-2.6	0.393 ± 0.077	-	-
TabICLv2	**0.362 ± 0.076**	-	-

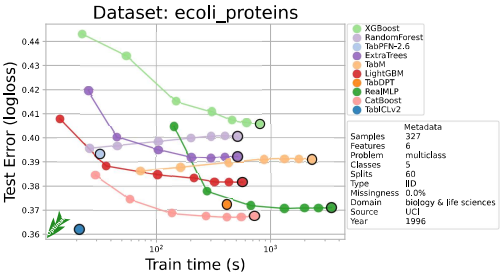

Figure G.42: **ecoli_proteins**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.295 ± 0.010	0.278 ± 0.007	0.276 ± 0.008
RandomForest	**0.256 ± 0.007**	**0.253 ± 0.008**	**0.254 ± 0.006**
ExtraTrees	0.258 ± 0.008	**0.256 ± 0.009**	**0.255 ± 0.008**
CatBoost	**0.253 ± 0.006**	**0.255 ± 0.007**	**0.255 ± 0.006**
LightGBM	**0.255 ± 0.006**	**0.256 ± 0.004**	**0.256 ± 0.005**
XGBoost	**0.255 ± 0.005**	**0.255 ± 0.004**	**0.255 ± 0.005**
RealMLP	**0.258 ± 0.006**	**0.256 ± 0.006**	**0.253 ± 0.007**
TabM	**0.254 ± 0.008**	**0.255 ± 0.008**	**0.254 ± 0.006**
TabDPT	0.259 ± 0.007	-	-
TabPFN-2.6	**0.254 ± 0.008**	-	-
TabICLv2	**0.253 ± 0.008**	-	-

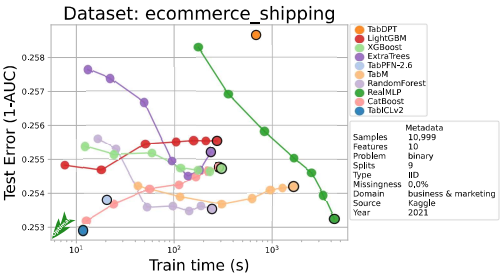

Figure G.43: **ecommerce_shipping**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	3.053	2.221	2.208
RandomForest	3.390	2.920	2.947
ExtraTrees	2.464	2.460	2.475
CatBoost	2.493	2.548	2.480
LightGBM	2.694	2.428	2.358
XGBoost	2.835	2.520	2.523
RealMLP	1.858	1.702	1.772
TabM	2.151	2.059	2.039
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	2.454	-	-

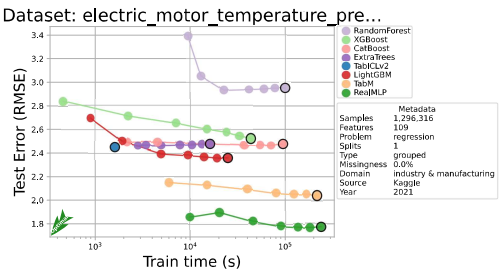

Figure G.44: **electric_motor_temperature_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.035 ± 0.003	0.036 ± 0.004	0.034 ± 0.003
RandomForest	0.013 ± 0.001	0.013 ± 0.001	0.013 ± 0.001
ExtraTrees	0.013 ± 0.001	0.013 ± 0.002	0.011 ± 0.002
CatBoost	0.012 ± 0.001	0.012 ± 0.002	0.012 ± 0.002
LightGBM	0.011 ± 0.002	0.012 ± 0.002	0.010 ± 0.002
XGBoost	0.012 ± 0.002	0.011 ± 0.002	0.011 ± 0.002
RealMLP	0.012 ± 0.004	0.010 ± 0.003	0.007 ± 0.002
TabM	0.012 ± 0.002	0.010 ± 0.002	0.010 ± 0.002
TabDPT	0.011 ± 0.002	-	-
TabPFN-2.6	**0.005 ± 0.001**	-	-
TabICLv2	0.006 ± 0.001	-	-

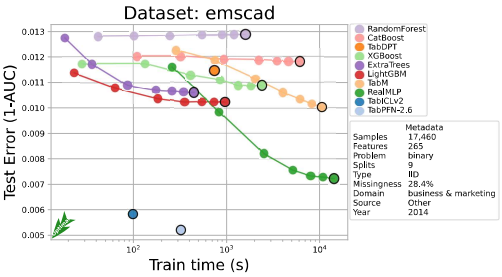

Figure G.45: **emscad**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.109 ± 0.016	0.066 ± 0.025	0.065 ± 0.025
RandomForest	0.158 ± 0.014	0.107 ± 0.027	0.107 ± 0.027
ExtraTrees	0.172 ± 0.013	0.095 ± 0.020	0.096 ± 0.019
CatBoost	0.084 ± 0.031	0.069 ± 0.025	0.069 ± 0.025
LightGBM	0.849 ± 0.066	0.069 ± 0.019	0.070 ± 0.019
XGBoost	0.121 ± 0.034	0.092 ± 0.023	0.092 ± 0.024
RealMLP	0.071 ± 0.018	0.066 ± 0.019	0.065 ± 0.019
TabM	0.074 ± 0.014	0.075 ± 0.015	0.075 ± 0.014
TabDPT	0.083 ± 0.021	-	-
TabPFN-2.6	0.066 ± 0.029	-	-
TabICLv2	**0.054 ± 0.020**	-	-

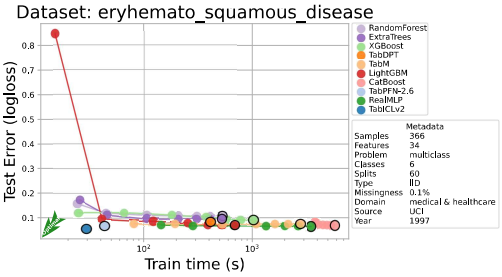

Figure G.46: **eryhemato_squamous_disease**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	791.7 ± 27.5	763.9 ± 24.5	764.1 ± 24.5
RandomForest	745.0 ± 28.0	733.0 ± 26.9	731.5 ± 26.9
ExtraTrees	738.3 ± 29.6	735.9 ± 29.7	734.3 ± 28.7
CatBoost	737.3 ± 27.0	733.7 ± 25.3	730.2 ± 26.2
LightGBM	742.5 ± 26.7	740.9 ± 25.0	731.2 ± 26.3
XGBoost	751.9 ± 29.7	736.8 ± 26.1	733.6 ± 26.6
RealMLP	755.3 ± 27.0	730.6 ± 29.1	723.5 ± 28.3
TabM	750.9 ± 28.6	753.0 ± 28.1	748.1 ± 29.1
TabDPT	720.1 ± 30.1	-	-
TabPFN-2.6	726.8 ± 27.9	-	-
TabICLv2	**714.3 ± 30.7**	-	-

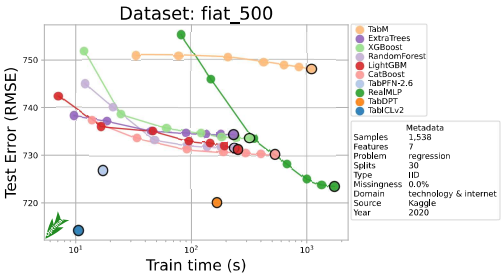

Figure G.47: **fiat_500**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.181 ± 0.018**	**0.181 ± 0.018**	**0.181 ± 0.018**
RandomForest	0.219 ± 0.018	0.206 ± 0.018	0.205 ± 0.018
ExtraTrees	0.224 ± 0.016	0.185 ± 0.019	0.187 ± 0.018
CatBoost	0.185 ± 0.018	0.187 ± 0.018	0.187 ± 0.018
LightGBM	0.206 ± 0.018	0.183 ± 0.019	0.183 ± 0.019
XGBoost	0.202 ± 0.016	0.191 ± 0.018	0.192 ± 0.018
RealMLP	0.188 ± 0.017	0.184 ± 0.018	0.182 ± 0.017
TabM	0.183 ± 0.018	0.183 ± 0.018	0.182 ± 0.018
TabDPT	**0.180 ± 0.018**	-	-
TabPFN-2.6	**0.179 ± 0.018**	-	-
TabICLv2	0.180 ± 0.017	-	-

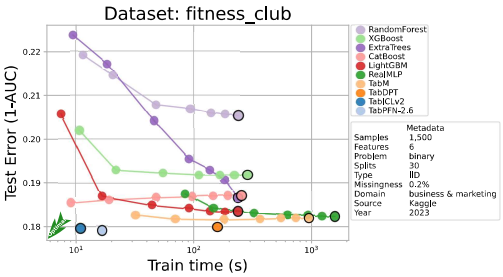

Figure G.48: **fitness_club**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	8.557 ± 0.046	8.333 ± 0.068	8.295 ± 0.066
RandomForest	7.774 ± 0.042	7.586 ± 0.049	7.589 ± 0.047
ExtraTrees	8.108 ± 0.044	7.778 ± 0.058	7.762 ± 0.057
CatBoost	7.376 ± 0.055	**7.366 ± 0.055**	**7.366 ± 0.054**
LightGBM	7.618 ± 0.056	7.513 ± 0.056	7.461 ± 0.054
XGBoost	7.392 ± 0.057	7.392 ± 0.057	7.396 ± 0.056
RealMLP	7.925 ± 0.062	7.452 ± 0.057	7.448 ± 0.056
TabM	7.784 ± 0.045	7.675 ± 0.054	7.675 ± 0.054
TabDPT	7.461 ± 0.070	-	-
TabPFN-2.6	7.381 ± 0.057	-	-
TabICLv2	7.593 ± 0.053	-	-

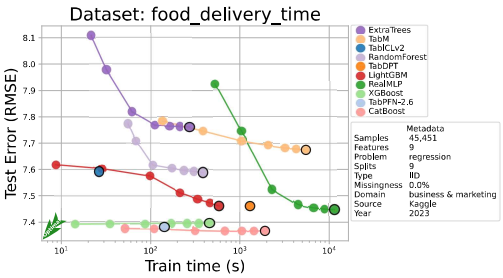

Figure G.49: **food_delivery_time**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.964 ± 0.129	0.988 ± 0.119	0.888 ± 0.101
RandomForest	0.669 ± 0.063	0.662 ± 0.071	0.655 ± 0.073
ExtraTrees	0.690 ± 0.055	0.639 ± 0.079	0.635 ± 0.072
CatBoost	0.636 ± 0.083	0.626 ± 0.089	0.623 ± 0.084
LightGBM	0.725 ± 0.070	0.676 ± 0.085	0.677 ± 0.084
XGBoost	0.746 ± 0.092	0.732 ± 0.082	0.726 ± 0.083
RealMLP	0.788 ± 0.124	0.645 ± 0.105	0.631 ± 0.084
TabM	0.680 ± 0.091	0.675 ± 0.096	0.662 ± 0.080
TabDPT	0.591 ± 0.088	-	-
TabPFN-2.6	**0.560 ± 0.098**	-	-
TabICLv2	**0.552 ± 0.093**	-	-

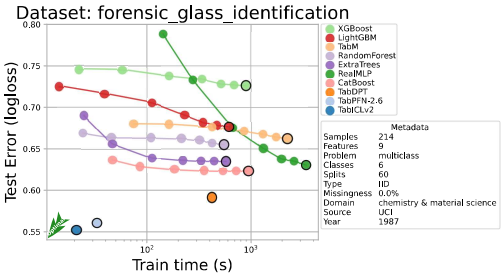

Figure G.50: **forensic_glass_identification**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.428 ± 0.075	1.415 ± 0.088	1.404 ± 0.079
RandomForest	1.463 ± 0.070	1.440 ± 0.074	1.441 ± 0.073
ExtraTrees	1.444 ± 0.072	1.405 ± 0.080	1.407 ± 0.080
CatBoost	1.397 ± 0.080	1.400 ± 0.079	1.399 ± 0.080
LightGBM	1.399 ± 0.080	1.402 ± 0.080	1.401 ± 0.080
XGBoost	1.398 ± 0.079	1.399 ± 0.080	1.399 ± 0.080
RealMLP	1.404 ± 0.078	1.402 ± 0.080	1.400 ± 0.079
TabM	1.399 ± 0.080	1.400 ± 0.081	1.400 ± 0.081
TabDPT	**1.392 ± 0.079**	-	-
TabPFN-2.6	1.403 ± 0.082	-	-
TabICLv2	1.399 ± 0.079	-	-

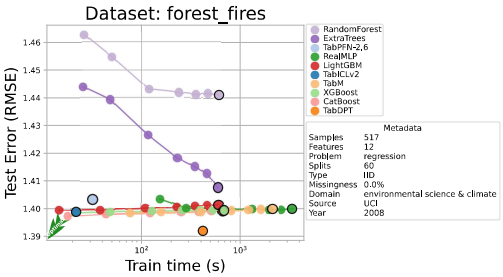

Figure G.51: **forest_fires**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.142 ± 0.032	0.146 ± 0.030	0.130 ± 0.027
RandomForest	0.152 ± 0.032	0.154 ± 0.032	0.153 ± 0.031
ExtraTrees	0.162 ± 0.041	0.147 ± 0.031	0.147 ± 0.032
CatBoost	0.147 ± 0.030	0.142 ± 0.029	0.142 ± 0.030
LightGBM	0.152 ± 0.031	0.140 ± 0.028	0.136 ± 0.028
XGBoost	0.158 ± 0.030	0.149 ± 0.031	0.147 ± 0.030
RealMLP	0.122 ± 0.025	0.125 ± 0.027	0.113 ± 0.027
TabM	0.135 ± 0.028	0.134 ± 0.027	0.135 ± 0.027
TabDPT	0.107 ± 0.026	-	-
TabPFN-2.6	0.104 ± 0.026	-	-
TabICLv2	**0.100 ± 0.029**	-	-

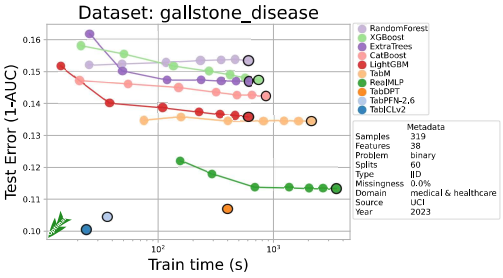

Figure G.52: **gallstone_disease**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.27 ± 0.66	0.19 ± 0.12	0.22 ± 0.35
RandomForest	0.13 ± 0.03	0.13 ± 0.03	0.13 ± 0.03
ExtraTrees	0.13 ± 0.03	0.13 ± 0.03	0.13 ± 0.03
CatBoost	0.13 ± 0.03	0.13 ± 0.03	0.13 ± 0.03
LightGBM	0.13 ± 0.03	0.13 ± 0.03	0.13 ± 0.03
XGBoost	0.13 ± 0.03	0.13 ± 0.03	0.13 ± 0.03
RealMLP	0.14 ± 0.03	0.13 ± 0.04	0.13 ± 0.03
TabM	73.83 ± 284.20	**0.22 ± 0.38**	**0.13 ± 0.04**
TabDPT	**0.13 ± 0.03**	-	-
TabPFN-2.6	**0.13 ± 0.03**	-	-
TabICLv2	**0.13 ± 0.04**	-	-

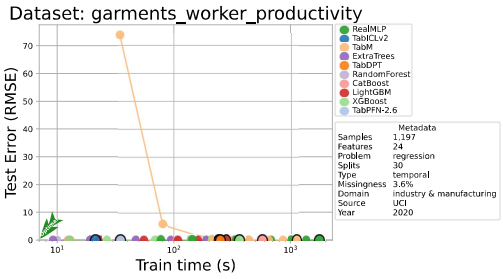

Figure G.53: **garments_worker_productivity**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.231 ± 0.138**	**0.209 ± 0.110**	**0.206 ± 0.114**
RandomForest	0.239 ± 0.125	0.251 ± 0.144	0.242 ± 0.136
ExtraTrees	0.218 ± 0.108	0.232 ± 0.134	0.236 ± 0.141
CatBoost	0.204 ± 0.123	**0.184 ± 0.108**	**0.189 ± 0.106**
LightGBM	**0.225 ± 0.158**	**0.191 ± 0.102**	**0.182 ± 0.102**
XGBoost	0.236 ± 0.119	0.216 ± 0.136	**0.214 ± 0.131**
RealMLP	**0.171 ± 0.118**	**0.174 ± 0.091**	**0.174 ± 0.092**
TabM	**0.183 ± 0.106**	**0.202 ± 0.126**	**0.186 ± 0.107**
TabDPT	0.244 ± 0.091	-	-
TabPFN-2.6	**0.193 ± 0.092**	-	-
TabICLv2	0.243 ± 0.082	-	-

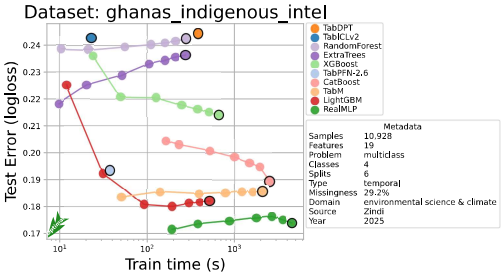

Figure G.54: **ghanas_indigenous_intel**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.176 ± 0.003	0.175 ± 0.003	0.173 ± 0.003
RandomForest	0.225 ± 0.020	0.178 ± 0.003	0.177 ± 0.002
ExtraTrees	0.212 ± 0.011	0.181 ± 0.004	0.181 ± 0.004
CatBoost	0.157 ± 0.003	0.156 ± 0.004	0.156 ± 0.003
LightGBM	0.153 ± 0.004	**0.153 ± 0.004**	**0.152 ± 0.004**
XGBoost	0.154 ± 0.004	0.154 ± 0.004	0.154 ± 0.003
RealMLP	0.156 ± 0.003	0.154 ± 0.004	0.154 ± 0.003
TabM	0.154 ± 0.004	0.153 ± 0.003	**0.153 ± 0.004**
TabDPT	0.222 ± 0.010	-	-
TabPFN-2.6	0.169 ± 0.006	-	-
TabICLv2	**0.153 ± 0.003**	-	-

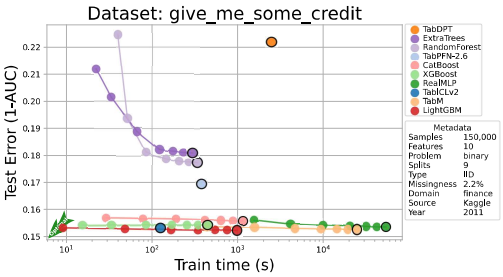

Figure G.55: **give_me_some_credit**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.051 ± 0.007	0.047 ± 0.006	0.047 ± 0.006
RandomForest	0.043 ± 0.007	0.042 ± 0.006	0.041 ± 0.006
ExtraTrees	0.047 ± 0.007	0.037 ± 0.006	0.037 ± 0.006
CatBoost	0.027 ± 0.006	0.026 ± 0.006	0.027 ± 0.006
LightGBM	0.028 ± 0.005	0.023 ± 0.006	0.023 ± 0.006
XGBoost	0.028 ± 0.006	0.026 ± 0.006	0.026 ± 0.006
RealMLP	0.016 ± 0.005	0.016 ± 0.004	0.015 ± 0.004
TabM	0.033 ± 0.006	0.018 ± 0.005	0.018 ± 0.005
TabDPT	0.007 ± 0.003	-	-
TabPFN-2.6	0.008 ± 0.003	-	-
TabICLv2	**0.005 ± 0.002**	-	-

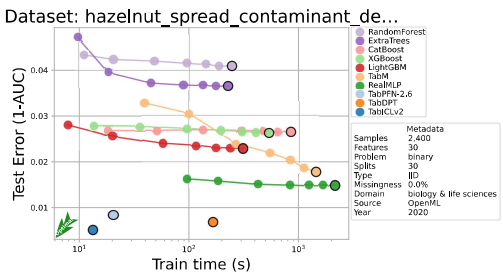

Figure G.56: **hazelnut_spread_contaminant_detection**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.446 ± 0.024	0.446 ± 0.024	0.446 ± 0.024
RandomForest	0.394 ± 0.026	0.379 ± 0.028	0.378 ± 0.028
ExtraTrees	0.390 ± 0.027	0.372 ± 0.031	0.372 ± 0.031
CatBoost	0.373 ± 0.031	0.372 ± 0.030	0.372 ± 0.030
LightGBM	0.377 ± 0.028	0.370 ± 0.030	0.368 ± 0.031
XGBoost	0.384 ± 0.026	0.370 ± 0.030	0.371 ± 0.030
RealMLP	0.372 ± 0.031	0.374 ± 0.029	0.371 ± 0.030
TabM	0.373 ± 0.030	0.371 ± 0.031	0.371 ± 0.031
TabDPT	0.364 ± 0.032	-	-
TabPFN-2.6	**0.360 ± 0.037**	-	-
TabICLv2	**0.360 ± 0.034**	-	-

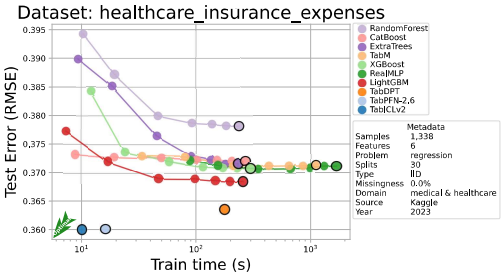

Figure G.57: **healthcare_insurance_expenses**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.092 ± 0.025**	**0.093 ± 0.025**	**0.090 ± 0.025**
RandomForest	0.097 ± 0.027	0.100 ± 0.028	0.100 ± 0.027
ExtraTrees	**0.090 ± 0.025**	0.092 ± 0.026	0.092 ± 0.026
CatBoost	0.098 ± 0.028	0.094 ± 0.027	0.094 ± 0.026
LightGBM	0.113 ± 0.028	0.094 ± 0.026	**0.092 ± 0.026**
XGBoost	0.104 ± 0.028	0.101 ± 0.028	0.100 ± 0.027
RealMLP	0.097 ± 0.025	**0.093 ± 0.026**	**0.092 ± 0.026**
TabM	0.093 ± 0.026	0.095 ± 0.026	0.094 ± 0.027
TabDPT	0.093 ± 0.026	-	-
TabPFN-2.6	**0.090 ± 0.027**	-	-
TabICLv2	**0.091 ± 0.026**	-	-

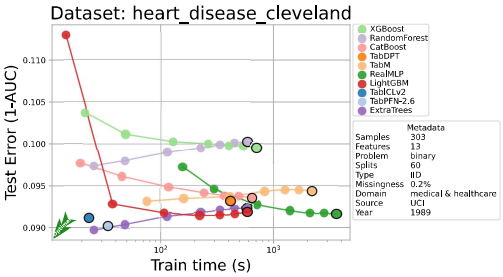

Figure G.58: **heart_disease_cleveland**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.097 ± 0.026	0.096 ± 0.028	**0.093 ± 0.026**
RandomForest	0.110 ± 0.023	0.109 ± 0.025	0.110 ± 0.024
ExtraTrees	0.102 ± 0.022	0.096 ± 0.023	0.096 ± 0.023
CatBoost	0.098 ± 0.021	0.100 ± 0.025	0.099 ± 0.024
LightGBM	0.154 ± 0.032	0.099 ± 0.026	0.099 ± 0.024
XGBoost	0.124 ± 0.025	0.101 ± 0.022	0.101 ± 0.022
RealMLP	**0.094 ± 0.023**	**0.091 ± 0.023**	**0.091 ± 0.023**
TabM	**0.094 ± 0.023**	**0.093 ± 0.023**	0.095 ± 0.022
TabDPT	0.095 ± 0.021	-	-
TabPFN-2.6	**0.093 ± 0.024**	-	-
TabICLv2	**0.094 ± 0.023**	-	-

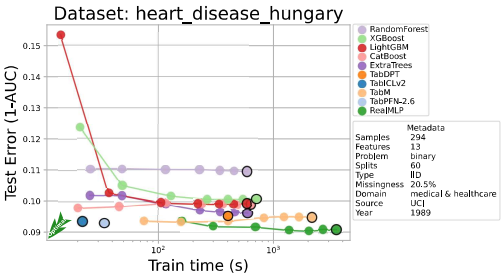

Figure G.59: **heart_disease_hungary**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.338 ± 0.056	0.334 ± 0.063	0.335 ± 0.061
RandomForest	0.322 ± 0.049	0.327 ± 0.051	0.327 ± 0.050
ExtraTrees	0.304 ± 0.047	0.311 ± 0.049	0.311 ± 0.050
CatBoost	0.305 ± 0.050	0.321 ± 0.048	0.318 ± 0.048
LightGBM	0.371 ± 0.064	0.333 ± 0.065	0.334 ± 0.064
XGBoost	0.368 ± 0.061	0.358 ± 0.063	0.349 ± 0.062
RealMLP	0.342 ± 0.057	0.348 ± 0.060	0.351 ± 0.055
TabM	0.324 ± 0.057	0.347 ± 0.065	0.344 ± 0.061
TabDPT	**0.294 ± 0.049**	-	-
TabPFN-2.6	0.319 ± 0.055	-	-
TabICLv2	0.327 ± 0.051	-	-

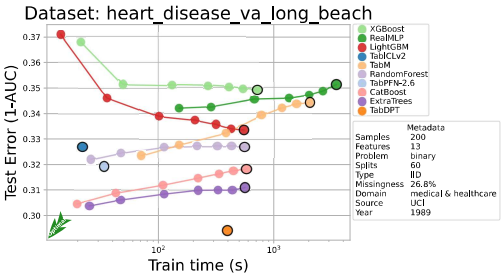

Figure G.60: **heart_disease_va_long_beach**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.129 ± 0.029	0.129 ± 0.028	0.128 ± 0.027
RandomForest	**0.086 ± 0.021**	**0.087 ± 0.023**	**0.087 ± 0.022**
ExtraTrees	0.106 ± 0.021	0.092 ± 0.023	0.091 ± 0.022
CatBoost	0.090 ± 0.021	0.092 ± 0.022	0.090 ± 0.022
LightGBM	0.090 ± 0.021	0.092 ± 0.023	0.091 ± 0.022
XGBoost	0.096 ± 0.025	0.090 ± 0.021	0.089 ± 0.021
RealMLP	0.121 ± 0.026	0.099 ± 0.024	0.095 ± 0.020
TabM	**0.088 ± 0.023**	0.090 ± 0.024	0.089 ± 0.024
TabDPT	0.094 ± 0.022	-	-
TabPFN-2.6	0.089 ± 0.020	-	-
TabICLv2	**0.084 ± 0.020**	-	-

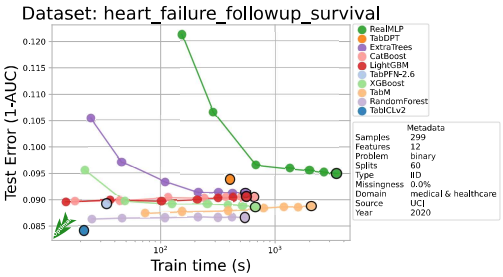

Figure G.61: **heart_failure_followup_survival**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.214 ± 0.005	0.214 ± 0.006	0.214 ± 0.005
RandomForest	0.207 ± 0.004	0.207 ± 0.004	0.207 ± 0.004
ExtraTrees	0.207 ± 0.005	0.207 ± 0.004	0.206 ± 0.005
CatBoost	0.202 ± 0.005	0.201 ± 0.005	0.201 ± 0.005
LightGBM	0.205 ± 0.004	0.201 ± 0.005	0.201 ± 0.005
XGBoost	0.206 ± 0.004	0.203 ± 0.004	0.202 ± 0.004
RealMLP	0.202 ± 0.005	0.200 ± 0.005	0.199 ± 0.004
TabM	0.203 ± 0.004	0.202 ± 0.004	0.201 ± 0.004
TabDPT	0.204 ± 0.005	-	-
TabPFN-2.6	**0.198 ± 0.004**	-	-
TabICLv2	**0.198 ± 0.004**	-	-

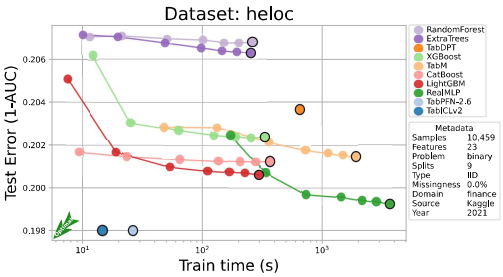

Figure G.62: **heloc**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.195 ± 0.044	0.196 ± 0.037	0.186 ± 0.034
RandomForest	0.162 ± 0.013	0.155 ± 0.023	0.154 ± 0.019
ExtraTrees	0.184 ± 0.010	0.147 ± 0.013	0.147 ± 0.013
CatBoost	0.150 ± 0.022	0.150 ± 0.024	0.149 ± 0.022
LightGBM	0.160 ± 0.025	0.144 ± 0.023	0.144 ± 0.022
XGBoost	0.164 ± 0.028	0.157 ± 0.027	0.156 ± 0.026
RealMLP	0.170 ± 0.039	0.144 ± 0.029	0.141 ± 0.026
TabM	0.150 ± 0.026	0.136 ± 0.027	0.134 ± 0.026
TabDPT	0.160 ± 0.030	-	-
TabPFN-2.6	**0.125 ± 0.026**	-	-
TabICLv2	0.131 ± 0.034	-	-

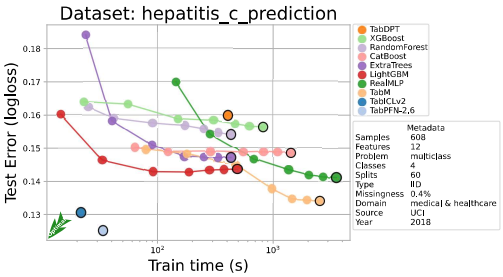

Figure G.63: **hepatitis_c_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.156 ± 0.058	0.143 ± 0.062	0.146 ± 0.059
RandomForest	0.130 ± 0.048	0.141 ± 0.055	0.142 ± 0.054
ExtraTrees	0.128 ± 0.044	0.138 ± 0.048	0.135 ± 0.046
CatBoost	0.137 ± 0.049	0.132 ± 0.050	0.132 ± 0.050
LightGBM	0.170 ± 0.057	0.150 ± 0.056	0.150 ± 0.055
XGBoost	0.168 ± 0.061	0.140 ± 0.058	0.143 ± 0.059
RealMLP	0.146 ± 0.060	0.150 ± 0.066	0.145 ± 0.064
TabM	0.131 ± 0.057	0.136 ± 0.060	0.131 ± 0.057
TabDPT	**0.121 ± 0.045**	-	-
TabPFN-2.6	0.135 ± 0.054	-	-
TabICLv2	0.127 ± 0.055	-	-

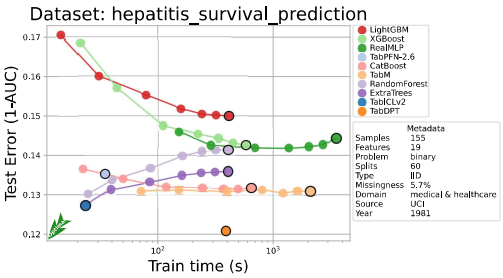

Figure G.64: **hepatitis_survival_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.262 ± 0.026	0.266 ± 0.031	0.266 ± 0.031
RandomForest	**0.212 ± 0.024**	**0.212 ± 0.024**	**0.213 ± 0.027**
ExtraTrees	**0.208 ± 0.025**	**0.215 ± 0.021**	**0.214 ± 0.022**
CatBoost	**0.210 ± 0.023**	**0.215 ± 0.021**	**0.214 ± 0.021**
LightGBM	**0.208 ± 0.033**	**0.210 ± 0.038**	**0.209 ± 0.034**
XGBoost	**0.208 ± 0.031**	**0.213 ± 0.029**	**0.209 ± 0.028**
RealMLP	0.233 ± 0.030	0.240 ± 0.043	**0.220 ± 0.030**
TabM	0.237 ± 0.032	0.232 ± 0.027	0.232 ± 0.029
TabDPT	**0.214 ± 0.025**	-	-
TabPFN-2.6	**0.212 ± 0.020**	-	-
TabICLv2	0.250 ± 0.028	-	-

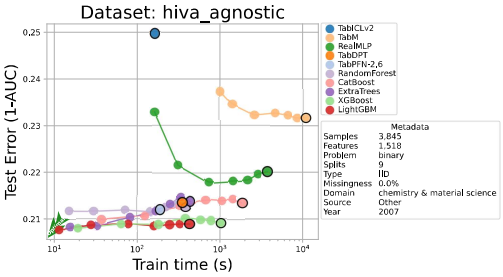

Figure G.65: **hiva_agnostic**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.226 ± 0.001	0.225 ± 0.001	0.222 ± 0.001
RandomForest	0.260 ± 0.002	0.252 ± 0.002	0.245 ± 0.002
ExtraTrees	0.257 ± 0.002	0.252 ± 0.002	0.244 ± 0.002
CatBoost	0.209 ± 0.001	0.208 ± 0.001	0.208 ± 0.001
LightGBM	0.213 ± 0.001	0.208 ± 0.001	0.208 ± 0.001
XGBoost	0.212 ± 0.001	**0.208 ± 0.001**	**0.208 ± 0.001**
RealMLP	0.216 ± 0.001	0.209 ± 0.001	**0.208 ± 0.001**
TabM	0.211 ± 0.001	0.209 ± 0.001	0.208 ± 0.001
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.224 ± 0.002	-	-

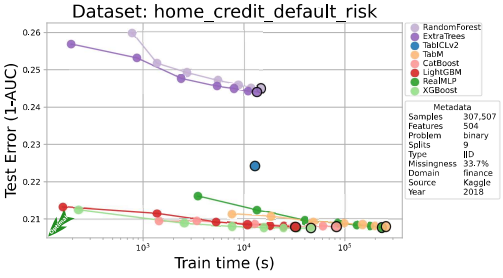

Figure G.66: **home_credit_default_risk**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.160	0.157	0.153
RandomForest	0.172	0.172	0.169
ExtraTrees	0.181	0.181	0.169
CatBoost	0.134	0.130	0.131
LightGBM	0.146	0.138	0.136
XGBoost	0.133	0.129	0.128
RealMLP	0.145	0.127	0.128
TabM	0.130	0.131	0.130
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.164	-	-

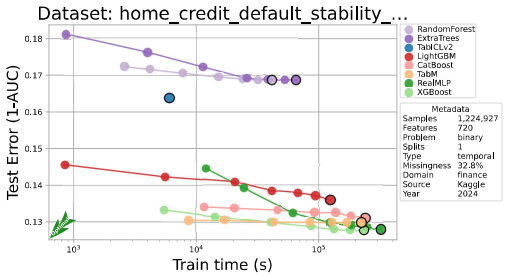

Figure G.67: **home_credit_default_stability_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.206 ± 0.010	0.200 ± 0.010	0.197 ± 0.009
RandomForest	0.042 ± 0.005	0.042 ± 0.005	0.042 ± 0.005
ExtraTrees	0.034 ± 0.004	0.034 ± 0.004	0.034 ± 0.004
CatBoost	0.032 ± 0.003	0.030 ± 0.004	0.029 ± 0.004
LightGBM	0.035 ± 0.004	0.032 ± 0.002	0.032 ± 0.003
XGBoost	0.039 ± 0.004	0.032 ± 0.003	0.032 ± 0.004
RealMLP	0.023 ± 0.005	0.013 ± 0.003	0.010 ± 0.002
TabM	0.033 ± 0.004	0.026 ± 0.004	0.025 ± 0.003
TabDPT	0.007 ± 0.002	-	-
TabPFN-2.6	**0.001 ± 0.000**	-	-
TabICLv2	0.003 ± 0.001	-	-

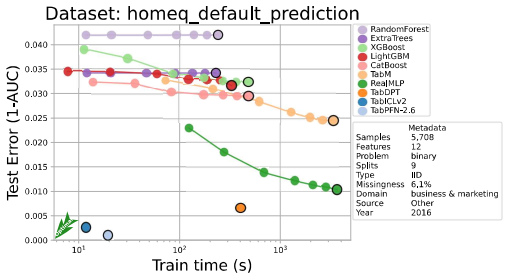

Figure G.68: **homeq_default_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.056 ± 0.000	0.050 ± 0.000	0.050 ± 0.000
RandomForest	0.043 ± 0.001	0.040 ± 0.001	0.039 ± 0.000
ExtraTrees	0.045 ± 0.001	0.039 ± 0.001	0.039 ± 0.001
CatBoost	**0.032 ± 0.000**	0.032 ± 0.001	**0.032 ± 0.001**
LightGBM	0.034 ± 0.001	0.033 ± 0.001	0.033 ± 0.001
XGBoost	0.033 ± 0.001	0.033 ± 0.001	0.033 ± 0.001
RealMLP	0.034 ± 0.001	0.033 ± 0.001	0.033 ± 0.001
TabM	0.033 ± 0.001	0.033 ± 0.001	0.033 ± 0.001
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.042 ± 0.001	-	-

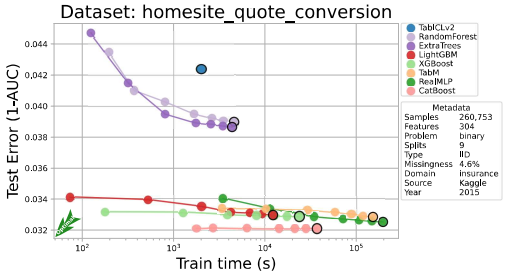

Figure G.69: **homesite_quote_conversion**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.854 ± 0.081	0.750 ± 0.046	0.749 ± 0.044
RandomForest	0.725 ± 0.035	0.722 ± 0.034	0.722 ± 0.036
ExtraTrees	0.727 ± 0.033	0.730 ± 0.050	0.725 ± 0.041
CatBoost	0.732 ± 0.032	0.735 ± 0.033	0.733 ± 0.032
LightGBM	0.731 ± 0.033	0.714 ± 0.039	0.712 ± 0.037
XGBoost	0.734 ± 0.034	0.713 ± 0.038	0.713 ± 0.038
RealMLP	0.820 ± 0.051	0.717 ± 0.042	0.707 ± 0.037
TabM	0.702 ± 0.033	**0.696 ± 0.037**	**0.696 ± 0.036**
TabDPT	0.726 ± 0.045	-	-
TabPFN-2.6	0.720 ± 0.049	-	-
TabICLv2	**0.701 ± 0.049**	-	-

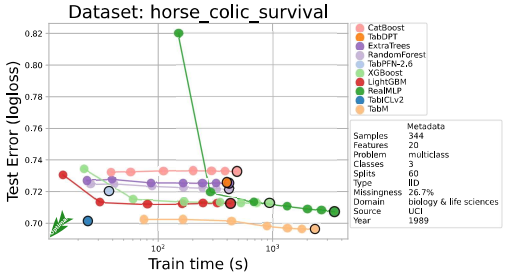

Figure G.70: **horse_colic_survival**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.207 ± 0.020	0.207 ± 0.022	0.207 ± 0.022
RandomForest	0.278 ± 0.035	0.269 ± 0.035	0.273 ± 0.033
ExtraTrees	0.304 ± 0.033	0.275 ± 0.029	0.275 ± 0.030
CatBoost	**0.179 ± 0.017**	**0.181 ± 0.021**	**0.182 ± 0.021**
LightGBM	0.203 ± 0.022	0.201 ± 0.022	0.199 ± 0.022
XGBoost	0.198 ± 0.026	0.195 ± 0.025	0.197 ± 0.025
RealMLP	**0.183 ± 0.023**	0.208 ± 0.027	0.193 ± 0.022
TabM	0.190 ± 0.025	0.200 ± 0.021	0.193 ± 0.024
TabDPT	0.324 ± 0.040	-	-
TabPFN-2.6	0.213 ± 0.026	-	-
TabICLv2	0.223 ± 0.024	-	-

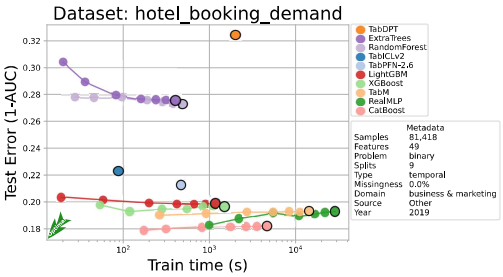

Figure G.71: **hotel_booking_demand**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.313 ± 0.005	0.313 ± 0.005	0.312 ± 0.005
RandomForest	0.224 ± 0.004	0.224 ± 0.004	0.224 ± 0.004
ExtraTrees	0.235 ± 0.004	0.231 ± 0.004	0.231 ± 0.004
CatBoost	0.208 ± 0.004	0.208 ± 0.004	0.207 ± 0.004
LightGBM	0.213 ± 0.004	0.210 ± 0.004	0.211 ± 0.004
XGBoost	0.211 ± 0.004	0.212 ± 0.004	0.211 ± 0.004
RealMLP	0.221 ± 0.005	0.199 ± 0.004	0.198 ± 0.004
TabM	0.209 ± 0.003	0.204 ± 0.003	0.202 ± 0.003
TabDPT	0.202 ± 0.003	-	-
TabPFN-2.6	0.194 ± 0.004	-	-
TabICLv2	**0.191 ± 0.004**	-	-

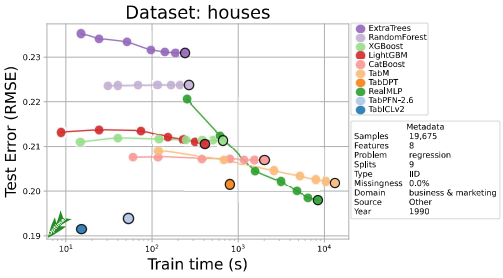

Figure G.72: **houses**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.204 ± 0.004	0.203 ± 0.003	0.203 ± 0.003
RandomForest	0.208 ± 0.003	0.198 ± 0.002	0.198 ± 0.002
ExtraTrees	0.212 ± 0.004	0.200 ± 0.003	0.199 ± 0.003
CatBoost	**0.195 ± 0.003**	**0.196 ± 0.003**	0.196 ± 0.003
LightGBM	0.199 ± 0.004	0.199 ± 0.004	0.197 ± 0.003
XGBoost	**0.195 ± 0.003**	0.196 ± 0.003	**0.195 ± 0.003**
RealMLP	0.199 ± 0.003	0.197 ± 0.004	0.197 ± 0.004
TabM	0.199 ± 0.003	0.199 ± 0.003	0.198 ± 0.003
TabDPT	0.197 ± 0.004	-	-
TabPFN-2.6	**0.194 ± 0.004**	-	-
TabICLv2	**0.193 ± 0.004**	-	-

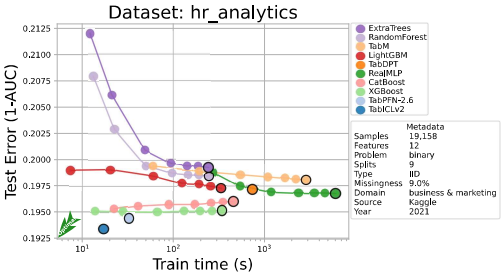

Figure G.73: **hr_analytics**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.141 ± 0.009**	**0.140 ± 0.009**	**0.136 ± 0.009**
RandomForest	**0.145 ± 0.038**	**0.140 ± 0.010**	**0.133 ± 0.012**
ExtraTrees	**0.123 ± 0.007**	**0.123 ± 0.007**	**0.121 ± 0.005**
CatBoost	**0.096 ± 0.010**	**0.092 ± 0.010**	**0.091 ± 0.006**
LightGBM	**0.120 ± 0.017**	**0.105 ± 0.015**	**0.093 ± 0.012**
XGBoost	**0.103 ± 0.010**	**0.091 ± 0.012**	**0.091 ± 0.016**
RealMLP	**0.110 ± 0.015**	**0.104 ± 0.014**	**0.093 ± 0.010**
TabM	**0.111 ± 0.010**	**0.110 ± 0.010**	**0.103 ± 0.011**
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	**0.121 ± 0.014**	-	-

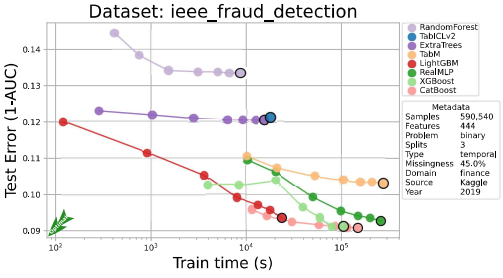

Figure G.74: **ieee_fraud_detection**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.473 ± 0.007	0.472 ± 0.007	0.470 ± 0.008
RandomForest	0.454 ± 0.007	0.434 ± 0.007	0.436 ± 0.006
ExtraTrees	0.462 ± 0.008	0.449 ± 0.008	0.448 ± 0.008
CatBoost	0.352 ± 0.006	0.353 ± 0.006	0.352 ± 0.006
LightGBM	0.364 ± 0.006	0.358 ± 0.007	0.356 ± 0.007
XGBoost	0.369 ± 0.006	0.359 ± 0.007	0.357 ± 0.007
RealMLP	0.367 ± 0.006	0.351 ± 0.007	**0.348 ± 0.007**
TabM	0.354 ± 0.007	0.353 ± 0.007	0.350 ± 0.007
TabDPT	0.440 ± 0.008	-	-
TabPFN-2.6	**0.346 ± 0.005**	-	-
TabICLv2	0.352 ± 0.005	-	-

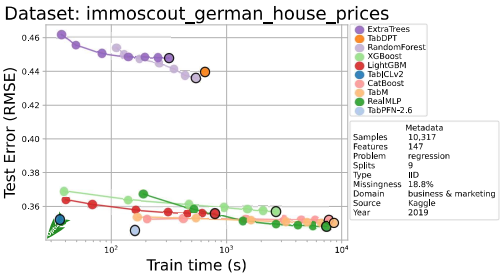

Figure G.75: **immoscout_german_house_prices**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.264 ± 0.006	0.264 ± 0.006	0.264 ± 0.006
RandomForest	0.186 ± 0.007	0.182 ± 0.009	0.177 ± 0.007
ExtraTrees	0.202 ± 0.008	0.195 ± 0.008	0.189 ± 0.008
CatBoost	0.160 ± 0.005	0.156 ± 0.006	0.154 ± 0.006
LightGBM	0.162 ± 0.005	0.160 ± 0.006	0.157 ± 0.005
XGBoost	0.166 ± 0.006	0.158 ± 0.006	0.157 ± 0.005
RealMLP	0.162 ± 0.004	0.153 ± 0.006	0.150 ± 0.005
TabM	0.151 ± 0.005	**0.149 ± 0.005**	**0.148 ± 0.005**
TabDPT	0.188 ± 0.008	-	-
TabPFN-2.6	0.150 ± 0.006	-	-
TabICLv2	0.154 ± 0.006	-	-

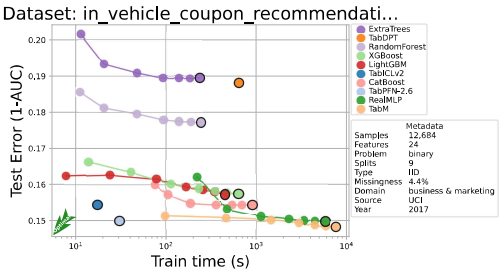

Figure G.76: **in_vehicle_coupon_recommendation**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.257 ± 0.030	0.263 ± 0.029	0.255 ± 0.030
RandomForest	0.256 ± 0.029	0.261 ± 0.029	0.261 ± 0.029
ExtraTrees	0.241 ± 0.030	0.247 ± 0.029	0.245 ± 0.031
CatBoost	0.255 ± 0.032	0.252 ± 0.031	0.251 ± 0.032
LightGBM	0.273 ± 0.031	0.266 ± 0.032	0.265 ± 0.030
XGBoost	0.273 ± 0.032	0.269 ± 0.028	0.269 ± 0.029
RealMLP	0.262 ± 0.031	0.259 ± 0.031	0.253 ± 0.031
TabM	0.255 ± 0.031	0.256 ± 0.031	0.252 ± 0.031
TabDPT	**0.236 ± 0.029**	-	-
TabPFN-2.6	0.251 ± 0.031	-	-
TabICLv2	**0.234 ± 0.028**	-	-

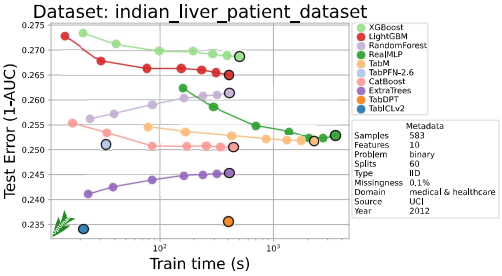

Figure G.77: **indian_liver_patient_dataset**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.074 ± 0.008	0.067 ± 0.007	0.066 ± 0.007
RandomForest	0.016 ± 0.003	0.017 ± 0.003	0.017 ± 0.003
ExtraTrees	0.015 ± 0.002	0.015 ± 0.003	0.014 ± 0.002
CatBoost	0.013 ± 0.002	0.011 ± 0.002	0.011 ± 0.002
LightGBM	0.014 ± 0.003	0.015 ± 0.003	0.013 ± 0.003
XGBoost	0.016 ± 0.003	0.014 ± 0.003	0.014 ± 0.003
RealMLP	0.010 ± 0.003	0.007 ± 0.002	0.007 ± 0.002
TabM	0.015 ± 0.004	0.009 ± 0.003	0.008 ± 0.002
TabDPT	0.005 ± 0.002	-	-
TabPFN-2.6	0.005 ± 0.002	-	-
TabICLv2	**0.004 ± 0.002**	-	-

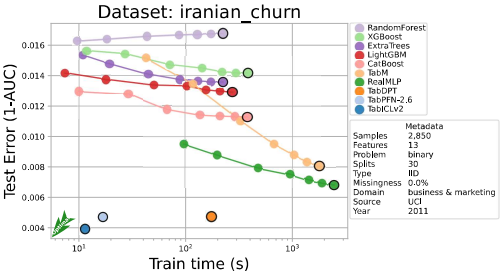

Figure G.78: **iranian_churn**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.277 ± 0.004	0.277 ± 0.004	0.277 ± 0.004
RandomForest	0.249 ± 0.007	0.251 ± 0.007	0.242 ± 0.005
ExtraTrees	0.243 ± 0.007	0.240 ± 0.005	0.236 ± 0.006
CatBoost	0.255 ± 0.003	0.251 ± 0.005	0.251 ± 0.003
LightGBM	0.252 ± 0.005	0.253 ± 0.005	0.251 ± 0.004
XGBoost	0.251 ± 0.004	0.251 ± 0.006	0.250 ± 0.005
RealMLP	0.269 ± 0.004	0.256 ± 0.007	0.242 ± 0.006
TabM	0.267 ± 0.005	0.266 ± 0.004	0.258 ± 0.004
TabDPT	**0.224 ± 0.006**	-	-
TabPFN-2.6	0.228 ± 0.004	-	-
TabICLv2	**0.223 ± 0.005**	-	-

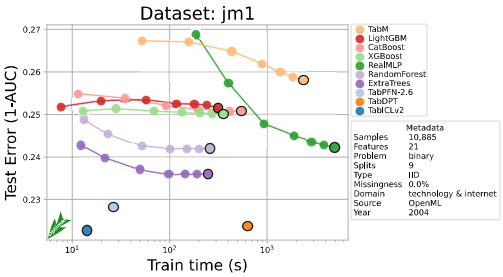

Figure G.79: **jm1**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.207 ± 0.011	0.184 ± 0.013	0.183 ± 0.012
RandomForest	0.288 ± 0.018	0.193 ± 0.005	0.193 ± 0.007
ExtraTrees	0.218 ± 0.014	0.188 ± 0.011	0.181 ± 0.010
CatBoost	0.161 ± 0.010	**0.159 ± 0.011**	**0.159 ± 0.011**
LightGBM	0.203 ± 0.009	0.179 ± 0.011	0.175 ± 0.009
XGBoost	0.174 ± 0.011	0.167 ± 0.012	0.165 ± 0.010
RealMLP	0.195 ± 0.011	0.183 ± 0.007	0.180 ± 0.008
TabM	0.198 ± 0.007	0.185 ± 0.009	0.183 ± 0.009
TabDPT	0.266 ± 0.019	-	-
TabPFN-2.6	0.173 ± 0.010	-	-
TabICLv2	0.172 ± 0.012	-	-

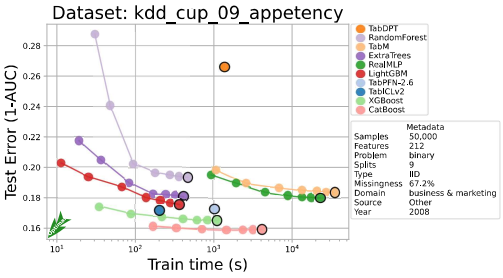

Figure G.80: **kdd_cup_09_appetency**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.254 ± 0.033	0.249 ± 0.031	0.249 ± 0.031
RandomForest	0.256 ± 0.018	0.249 ± 0.016	0.248 ± 0.014
ExtraTrees	0.249 ± 0.027	0.250 ± 0.025	0.247 ± 0.027
CatBoost	**0.232 ± 0.029**	**0.231 ± 0.027**	**0.230 ± 0.027**
LightGBM	0.247 ± 0.022	0.238 ± 0.031	0.235 ± 0.028
XGBoost	**0.231 ± 0.025**	**0.234 ± 0.022**	**0.231 ± 0.023**
RealMLP	0.237 ± 0.028	**0.231 ± 0.027**	**0.230 ± 0.027**
TabM	0.240 ± 0.030	0.239 ± 0.030	0.238 ± 0.031
TabDPT	0.248 ± 0.035	-	-
TabPFN-2.6	**0.228 ± 0.031**	-	-
TabICLv2	**0.232 ± 0.032**	-	-

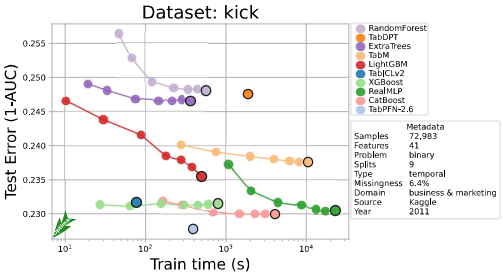

Figure G.81: **kick**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.115 ± 0.018**	**0.115 ± 0.019**	**0.115 ± 0.018**
RandomForest	**0.093 ± 0.005**	**0.089 ± 0.003**	**0.088 ± 0.003**
ExtraTrees	**0.121 ± 0.000**	**0.094 ± 0.006**	**0.093 ± 0.006**
CatBoost	**0.073 ± 0.006**	**0.072 ± 0.005**	**0.072 ± 0.005**
LightGBM	**0.077 ± 0.006**	**0.075 ± 0.007**	**0.074 ± 0.006**
XGBoost	**0.078 ± 0.008**	**0.072 ± 0.004**	**0.072 ± 0.005**
RealMLP	**0.076 ± 0.008**	**0.072 ± 0.005**	**0.072 ± 0.006**
TabM	**0.073 ± 0.004**	**0.073 ± 0.006**	**0.072 ± 0.004**
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	**0.079 ± 0.003**	-	-

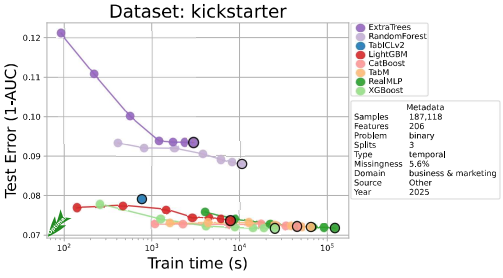

Figure G.82: **kickstarter**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.289 ± 0.003	0.288 ± 0.003	0.288 ± 0.003
RandomForest	0.254 ± 0.004	0.234 ± 0.004	0.233 ± 0.004
ExtraTrees	0.264 ± 0.004	0.233 ± 0.004	0.231 ± 0.004
CatBoost	0.221 ± 0.004	0.220 ± 0.004	0.220 ± 0.004
LightGBM	0.226 ± 0.005	0.225 ± 0.004	0.224 ± 0.004
XGBoost	0.225 ± 0.005	0.223 ± 0.004	0.222 ± 0.004
RealMLP	0.233 ± 0.005	0.218 ± 0.004	0.215 ± 0.004
TabM	0.223 ± 0.004	0.222 ± 0.004	0.222 ± 0.004
TabDPT	0.257 ± 0.003	-	-
TabPFN-2.6	0.226 ± 0.012	-	-
TabICLv2	**0.213 ± 0.004**	-	-

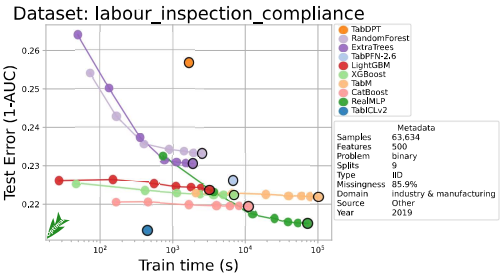

Figure G.83: **labour_inspection_compliance**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.306	0.305	0.303
RandomForest	0.308	0.317	0.313
ExtraTrees	0.319	0.312	0.310
CatBoost	0.287	0.286	0.285
LightGBM	0.288	0.287	0.286
XGBoost	0.288	0.285	0.284
RealMLP	0.287	0.287	0.285
TabM	0.290	0.291	0.289
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.315	-	-

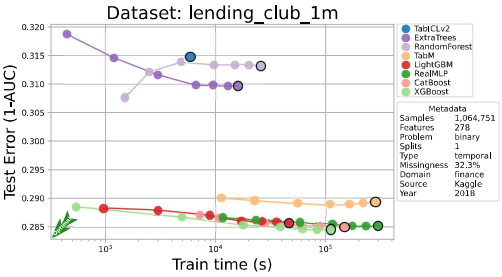

Figure G.84: **lending_club_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.317 ± 0.048	0.306 ± 0.048	0.308 ± 0.049
RandomForest	0.332 ± 0.044	0.328 ± 0.049	0.328 ± 0.048
ExtraTrees	0.333 ± 0.047	0.313 ± 0.053	0.315 ± 0.051
CatBoost	**0.287 ± 0.043**	0.304 ± 0.051	0.303 ± 0.048
LightGBM	0.319 ± 0.043	0.312 ± 0.048	0.318 ± 0.042
XGBoost	0.342 ± 0.044	0.307 ± 0.046	0.306 ± 0.045
RealMLP	0.323 ± 0.044	0.324 ± 0.046	0.313 ± 0.046
TabM	0.304 ± 0.044	0.313 ± 0.047	0.310 ± 0.044
TabDPT	0.288 ± 0.048	-	-
TabPFN-2.6	**0.284 ± 0.045**	-	-
TabICLv2	**0.284 ± 0.047**	-	-

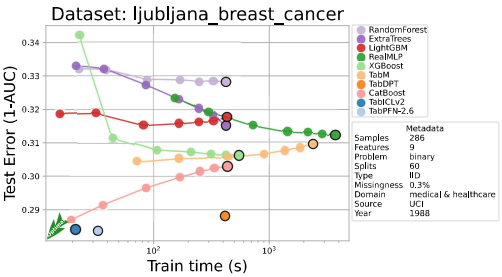

Figure G.85: **ljubljana_breast_cancer**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.423 ± 0.105	1.399 ± 0.078	1.396 ± 0.078
RandomForest	1.483 ± 0.108	1.404 ± 0.086	1.401 ± 0.082
ExtraTrees	1.493 ± 0.114	1.399 ± 0.081	1.401 ± 0.078
CatBoost	1.449 ± 0.063	1.387 ± 0.067	1.381 ± 0.068
LightGBM	1.421 ± 0.083	1.355 ± 0.081	1.348 ± 0.081
XGBoost	1.460 ± 0.088	1.400 ± 0.083	1.395 ± 0.076
RealMLP	1.454 ± 0.087	1.343 ± 0.087	1.329 ± 0.082
TabM	**1.313 ± 0.080**	1.320 ± 0.082	**1.315 ± 0.078**
TabDPT	1.391 ± 0.071	-	-
TabPFN-2.6	1.336 ± 0.084	-	-
TabICLv2	1.345 ± 0.094	-	-

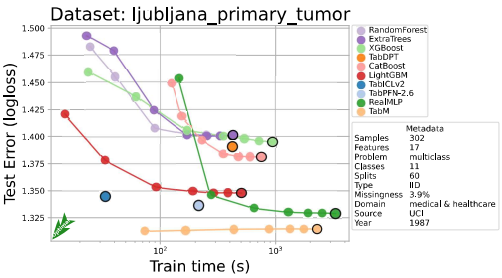

Figure G.86: **ljubljana_primary_tumor**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.143 ± 0.072	0.153 ± 0.078	0.140 ± 0.067
RandomForest	0.307 ± 0.031	0.239 ± 0.045	0.240 ± 0.045
ExtraTrees	0.326 ± 0.028	0.182 ± 0.030	0.182 ± 0.030
CatBoost	0.224 ± 0.086	0.219 ± 0.079	0.217 ± 0.078
LightGBM	0.191 ± 0.050	0.168 ± 0.054	0.168 ± 0.054
XGBoost	0.229 ± 0.057	0.213 ± 0.057	0.213 ± 0.057
RealMLP	0.150 ± 0.078	0.155 ± 0.087	0.148 ± 0.078
TabM	**0.133 ± 0.064**	0.143 ± 0.065	0.140 ± 0.062
TabDPT	0.144 ± 0.056	-	-
TabPFN-2.6	**0.127 ± 0.057**	-	-
TabICLv2	0.172 ± 0.080	-	-

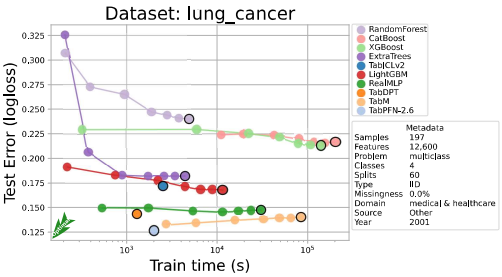

Figure G.87: **lung_cancer**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.223 ± 0.052**	**0.224 ± 0.052**	**0.224 ± 0.051**
RandomForest	0.266 ± 0.053	0.258 ± 0.057	0.256 ± 0.058
ExtraTrees	0.265 ± 0.051	0.265 ± 0.055	0.263 ± 0.055
CatBoost	0.249 ± 0.053	0.252 ± 0.050	0.251 ± 0.050
LightGBM	0.242 ± 0.059	0.241 ± 0.055	0.240 ± 0.054
XGBoost	0.250 ± 0.062	0.247 ± 0.060	0.244 ± 0.059
RealMLP	0.246 ± 0.053	0.245 ± 0.054	0.246 ± 0.050
TabM	0.246 ± 0.053	0.249 ± 0.053	0.249 ± 0.053
TabDPT	**0.223 ± 0.054**	-	-
TabPFN-2.6	0.241 ± 0.052	-	-
TabICLv2	0.273 ± 0.053	-	-

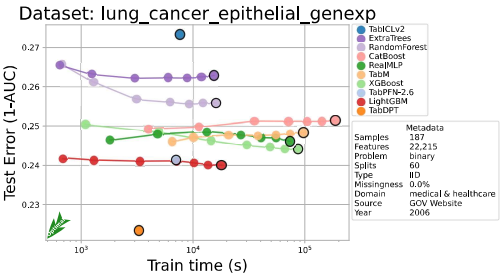

Figure G.88: **lung_cancer_epithelial_genexp**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.163	0.163	0.163
RandomForest	0.162	0.162	0.162
ExtraTrees	0.162	0.162	0.162
CatBoost	0.157	0.157	0.157
LightGBM	0.157	0.157	0.157
XGBoost	0.157	0.157	0.157
RealMLP	0.158	0.157	0.157
TabM	0.158	0.158	0.158
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.167	-	-

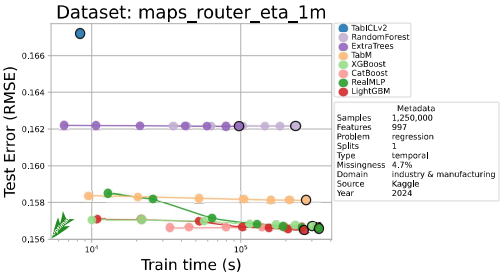

Figure G.89: **maps_router_eta_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.096 ± 0.012	0.097 ± 0.013	0.096 ± 0.012
RandomForest	0.115 ± 0.013	0.116 ± 0.013	0.117 ± 0.014
ExtraTrees	0.117 ± 0.013	0.118 ± 0.013	0.115 ± 0.012
CatBoost	0.095 ± 0.011	0.097 ± 0.012	0.096 ± 0.011
LightGBM	0.100 ± 0.012	0.090 ± 0.011	0.090 ± 0.011
XGBoost	0.103 ± 0.011	0.098 ± 0.010	0.097 ± 0.010
RealMLP	0.096 ± 0.012	0.093 ± 0.012	0.091 ± 0.012
TabM	0.097 ± 0.011	0.091 ± 0.009	0.087 ± 0.009
TabDPT	0.096 ± 0.014	-	-
TabPFN-2.6	0.079 ± 0.010	-	-
TabICLv2	**0.076 ± 0.010**	-	-

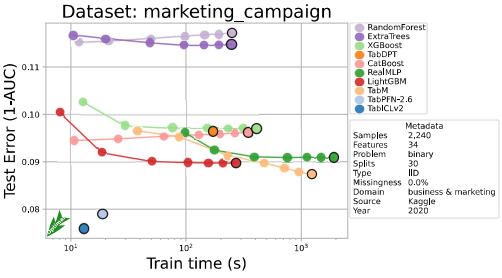

Figure G.90: **marketing_campaign**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.795 ± 0.022	0.789 ± 0.026	0.782 ± 0.023
RandomForest	0.452 ± 0.043	0.468 ± 0.038	0.436 ± 0.037
ExtraTrees	0.457 ± 0.038	0.455 ± 0.048	0.426 ± 0.034
CatBoost	0.462 ± 0.033	0.452 ± 0.033	0.449 ± 0.032
LightGBM	0.472 ± 0.037	0.458 ± 0.034	0.465 ± 0.029
XGBoost	0.456 ± 0.036	0.455 ± 0.032	0.453 ± 0.033
RealMLP	0.544 ± 0.041	0.436 ± 0.040	0.424 ± 0.034
TabM	0.510 ± 0.038	0.472 ± 0.043	0.466 ± 0.038
TabDPT	0.384 ± 0.041	-	-
TabPFN-2.6	0.398 ± 0.044	-	-
TabICLv2	**0.377 ± 0.040**	-	-

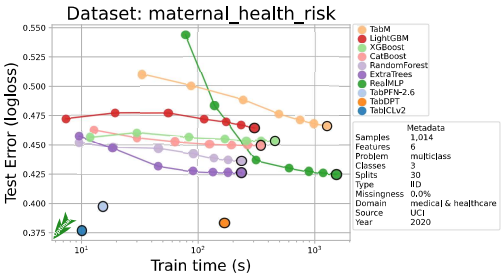

Figure G.91: **maternal_health_risk**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.648	0.648	0.648
RandomForest	0.512	0.512	0.512
ExtraTrees	0.518	0.516	0.516
CatBoost	0.490	0.467	0.468
LightGBM	0.481	0.466	0.470
XGBoost	0.472	0.472	0.469
RealMLP	0.473	0.451	0.452
TabM	0.473	0.469	0.467
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.464	-	-

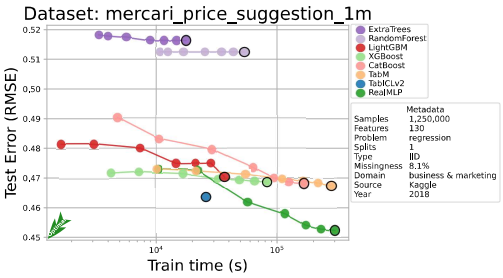

Figure G.92: **mercari_price_suggestion_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	8.220 ± 1.240	7.902 ± 1.162	7.871 ± 1.136
RandomForest	8.918 ± 1.497	8.541 ± 1.177	8.574 ± 1.217
ExtraTrees	8.548 ± 1.138	7.906 ± 1.147	7.903 ± 1.158
CatBoost	7.725 ± 1.215	7.743 ± 1.211	7.748 ± 1.226
LightGBM	7.855 ± 1.267	7.746 ± 1.235	7.724 ± 1.216
XGBoost	8.428 ± 1.583	7.797 ± 1.243	7.875 ± 1.290
RealMLP	7.813 ± 1.214	7.748 ± 1.171	7.742 ± 1.185
TabM	7.764 ± 1.187	7.765 ± 1.285	7.724 ± 1.253
TabDPT	7.889 ± 1.121	-	-
TabPFN-2.6	**7.614 ± 1.234**	-	-
TabICLv2	**7.671 ± 1.136**	-	-

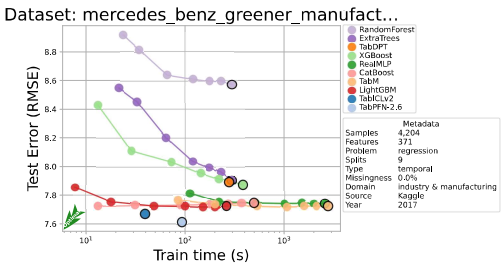

Figure G.93: **mercedes_benz_greener_manufacturing**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.269 ± 0.003	0.269 ± 0.003	0.264 ± 0.003
RandomForest	0.159 ± 0.004	0.157 ± 0.004	0.156 ± 0.003
ExtraTrees	0.156 ± 0.003	0.150 ± 0.003	0.150 ± 0.003
CatBoost	0.135 ± 0.002	0.135 ± 0.002	0.134 ± 0.002
LightGBM	0.141 ± 0.002	0.139 ± 0.003	0.137 ± 0.003
XGBoost	0.143 ± 0.002	0.139 ± 0.003	0.138 ± 0.002
RealMLP	0.139 ± 0.003	0.134 ± 0.003	0.133 ± 0.003
TabM	0.136 ± 0.002	0.135 ± 0.003	0.133 ± 0.003
TabDPT	0.131 ± 0.003	-	-
TabPFN-2.6	0.129 ± 0.003	-	-
TabICLv2	**0.128 ± 0.003**	-	-

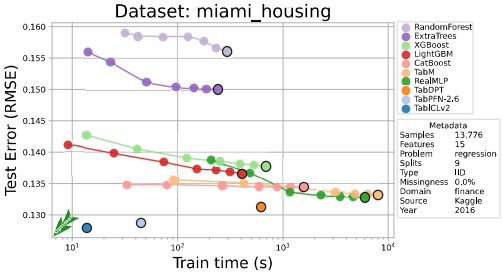

Figure G.94: **miami_housing**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.585 ± 0.038	0.480 ± 0.025	0.479 ± 0.025
RandomForest	0.477 ± 0.017	0.478 ± 0.017	0.469 ± 0.017
ExtraTrees	0.498 ± 0.012	0.491 ± 0.020	0.471 ± 0.017
CatBoost	0.453 ± 0.018	0.455 ± 0.017	0.451 ± 0.017
LightGBM	0.500 ± 0.016	0.460 ± 0.021	0.457 ± 0.018
XGBoost	0.469 ± 0.018	0.438 ± 0.019	0.438 ± 0.019
RealMLP	0.490 ± 0.036	**0.433 ± 0.020**	**0.434 ± 0.018**
TabM	0.438 ± 0.018	0.439 ± 0.019	**0.436 ± 0.018**
TabDPT	0.508 ± 0.014	-	-
TabPFN-2.6	**0.431 ± 0.021**	-	-
TabICLv2	0.438 ± 0.022	-	-

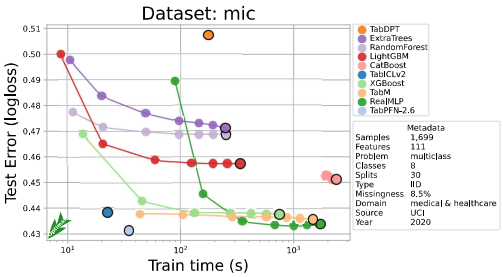

Figure G.95: **mic**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.265 ± 0.347	1.234 ± 0.304	1.213 ± 0.300
RandomForest	1.240 ± 0.102	1.259 ± 0.138	1.237 ± 0.137
ExtraTrees	1.300 ± 0.081	1.226 ± 0.115	1.225 ± 0.113
CatBoost	1.372 ± 0.120	1.323 ± 0.133	1.321 ± 0.128
LightGBM	1.683 ± 0.120	1.294 ± 0.134	1.295 ± 0.135
XGBoost	1.669 ± 0.114	1.504 ± 0.140	1.505 ± 0.140
RealMLP	1.343 ± 0.194	1.196 ± 0.161	1.183 ± 0.152
TabM	1.153 ± 0.176	1.121 ± 0.146	1.118 ± 0.141
TabDPT	1.352 ± 0.158	-	-
TabPFN-2.6	1.275 ± 0.101	-	-
TabICLv2	**0.910 ± 0.190**	-	-

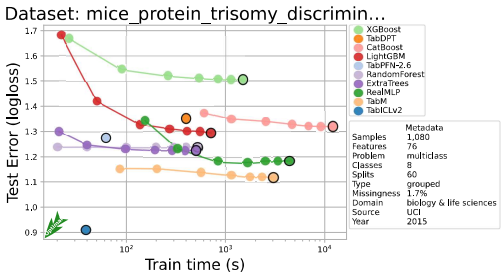

Figure G.96: **mice_protein_trisomy_discriminant**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.413 ± 0.104**	**0.415 ± 0.108**	**0.412 ± 0.108**
RandomForest	0.843 ± 0.039	0.702 ± 0.083	0.689 ± 0.078
ExtraTrees	0.990 ± 0.033	0.716 ± 0.058	0.716 ± 0.058
CatBoost	0.538 ± 0.087	0.483 ± 0.082	0.484 ± 0.082
LightGBM	0.698 ± 0.085	0.512 ± 0.087	0.518 ± 0.082
XGBoost	0.897 ± 0.090	0.787 ± 0.091	0.787 ± 0.090
RealMLP	0.690 ± 0.070	0.613 ± 0.069	0.611 ± 0.064
TabM	0.428 ± 0.076	0.429 ± 0.076	0.429 ± 0.074
TabDPT	1.184 ± 0.184	-	-
TabPFN-2.6	0.432 ± 0.087	-	-
TabICLv2	0.563 ± 0.107	-	-

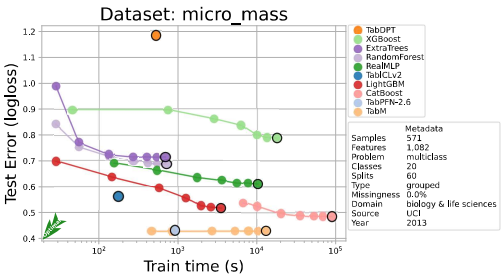

Figure G.97: **micro_mass**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.172 ± 0.123	0.175 ± 0.130	0.179 ± 0.130
RandomForest	0.144 ± 0.137	0.168 ± 0.158	0.166 ± 0.146
ExtraTrees	0.154 ± 0.131	0.168 ± 0.142	0.167 ± 0.142
CatBoost	0.234 ± 0.180	0.218 ± 0.174	0.220 ± 0.182
LightGBM	0.201 ± 0.151	0.159 ± 0.150	0.157 ± 0.149
XGBoost	0.193 ± 0.156	0.173 ± 0.157	0.168 ± 0.156
RealMLP	0.143 ± 0.128	0.150 ± 0.121	0.145 ± 0.116
TabM	0.129 ± 0.121	**0.119 ± 0.119**	**0.119 ± 0.119**
TabDPT	0.152 ± 0.114	-	-
TabPFN-2.6	0.160 ± 0.125	-	-
TabICLv2	**0.126 ± 0.111**	-	-

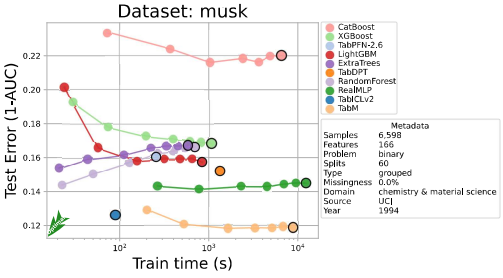

Figure G.98: **musk**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	5.457 ± 0.258	5.447 ± 0.274	5.312 ± 0.312
RandomForest	5.249 ± 0.379	4.967 ± 0.365	5.009 ± 0.380
ExtraTrees	5.210 ± 0.396	5.093 ± 0.388	5.076 ± 0.387
CatBoost	5.245 ± 0.396	4.967 ± 0.364	4.961 ± 0.365
LightGBM	5.143 ± 0.382	5.022 ± 0.374	4.919 ± 0.372
XGBoost	4.984 ± 0.400	4.986 ± 0.348	4.872 ± 0.354
RealMLP	4.914 ± 0.369	4.870 ± 0.354	4.812 ± 0.357
TabM	5.085 ± 0.328	4.872 ± 0.352	4.854 ± 0.345
TabDPT	5.377 ± 0.391	-	-
TabPFN-2.6	4.605 ± 0.379	-	-
TabICLv2	**4.559 ± 0.388**	-	-

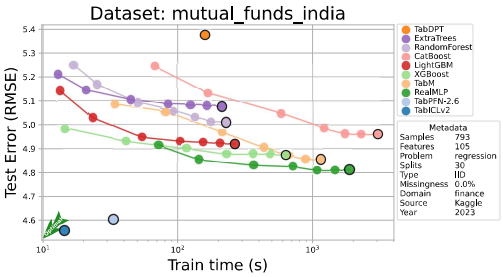

Figure G.99: **mutual_funds_india**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.019 ± 0.002	0.019 ± 0.002	0.019 ± 0.002
RandomForest	0.021 ± 0.002	0.020 ± 0.002	0.019 ± 0.002
ExtraTrees	0.021 ± 0.002	0.017 ± 0.002	0.017 ± 0.002
CatBoost	0.014 ± 0.002	0.014 ± 0.002	0.014 ± 0.002
LightGBM	0.015 ± 0.002	0.014 ± 0.002	0.014 ± 0.002
XGBoost	0.015 ± 0.002	0.016 ± 0.002	0.015 ± 0.002
RealMLP	0.015 ± 0.002	0.014 ± 0.002	0.014 ± 0.002
TabM	0.014 ± 0.002	0.014 ± 0.002	0.014 ± 0.002
TabDPT	0.016 ± 0.002	-	-
TabPFN-2.6	0.014 ± 0.002	-	-
TabICLv2	**0.012 ± 0.002**	-	-

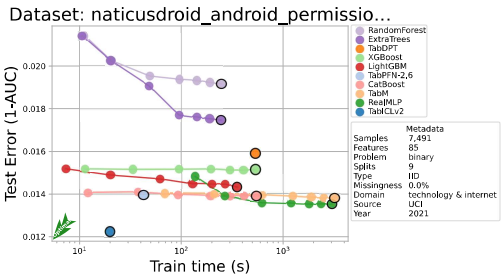

Figure G.100: **naticusdroid_android_permissions_dataset**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	4.440 ± 0.280	4.365 ± 0.280	4.364 ± 0.276
RandomForest	4.548 ± 0.253	4.406 ± 0.265	4.412 ± 0.266
ExtraTrees	4.489 ± 0.251	4.408 ± 0.261	4.406 ± 0.260
CatBoost	4.393 ± 0.276	4.397 ± 0.274	4.395 ± 0.273
LightGBM	4.454 ± 0.267	4.385 ± 0.288	4.374 ± 0.280
XGBoost	4.508 ± 0.271	4.422 ± 0.269	4.423 ± 0.270
RealMLP	4.358 ± 0.281	4.375 ± 0.283	4.362 ± 0.280
TabM	4.350 ± 0.276	4.371 ± 0.277	4.366 ± 0.277
TabDPT	**4.280 ± 0.267**	-	-
TabPFN-2.6	4.341 ± 0.284	-	-
TabICLv2	4.324 ± 0.279	-	-

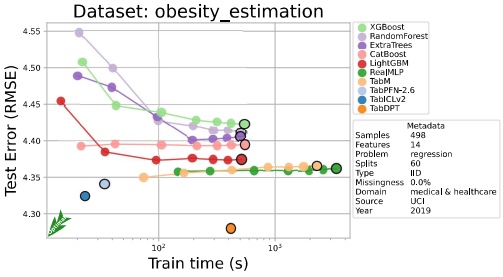

Figure G.101: **obesity_estimation**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.088 ± 0.006	0.088 ± 0.006	0.083 ± 0.006
RandomForest	0.073 ± 0.005	0.070 ± 0.004	0.069 ± 0.004
ExtraTrees	0.080 ± 0.006	0.070 ± 0.005	0.070 ± 0.005
CatBoost	0.066 ± 0.003	0.066 ± 0.004	0.065 ± 0.003
LightGBM	0.065 ± 0.004	0.064 ± 0.003	0.064 ± 0.003
XGBoost	0.064 ± 0.004	0.065 ± 0.004	0.064 ± 0.003
RealMLP	0.071 ± 0.004	0.067 ± 0.003	0.066 ± 0.004
TabM	0.065 ± 0.004	0.064 ± 0.004	0.064 ± 0.004
TabDPT	0.070 ± 0.005	-	-
TabPFN-2.6	**0.062 ± 0.004**	-	-
TabICLv2	0.063 ± 0.004	-	-

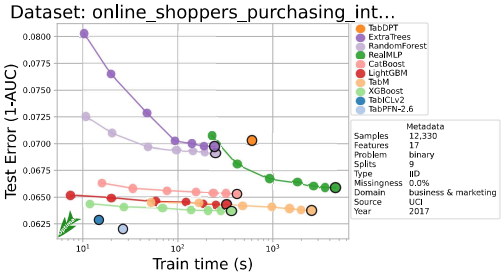

Figure G.102: **online_shoppers_purchasing_intention_dataset**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.754 ± 0.005	0.640 ± 0.004	0.625 ± 0.003
RandomForest	0.587 ± 0.003	0.543 ± 0.004	0.546 ± 0.004
ExtraTrees	0.628 ± 0.003	0.545 ± 0.004	0.543 ± 0.003
CatBoost	0.475 ± 0.005	0.469 ± 0.005	0.466 ± 0.005
LightGBM	0.476 ± 0.005	0.463 ± 0.005	0.460 ± 0.005
XGBoost	0.482 ± 0.004	0.461 ± 0.004	0.461 ± 0.005
RealMLP	0.474 ± 0.005	0.449 ± 0.005	0.440 ± 0.004
TabM	0.445 ± 0.005	0.445 ± 0.005	0.441 ± 0.004
TabDPT	0.494 ± 0.006	-	-
TabPFN-2.6	0.463 ± 0.003	-	-
TabICLv2	**0.421 ± 0.005**	-	-

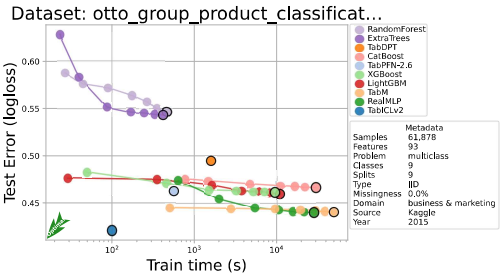

Figure G.103: **otto_group_product_classification_challenge**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.373 ± 0.090**	0.383 ± 0.090	**0.381 ± 0.089**
RandomForest	0.482 ± 0.061	0.507 ± 0.087	0.507 ± 0.087
ExtraTrees	0.465 ± 0.063	0.502 ± 0.090	0.506 ± 0.092
CatBoost	0.497 ± 0.079	0.491 ± 0.072	0.488 ± 0.074
LightGBM	0.512 ± 0.093	0.500 ± 0.085	0.503 ± 0.084
XGBoost	0.524 ± 0.110	0.513 ± 0.099	0.514 ± 0.096
RealMLP	0.452 ± 0.077	0.465 ± 0.082	0.458 ± 0.081
TabM	0.475 ± 0.075	0.483 ± 0.072	0.479 ± 0.072
TabDPT	**0.384 ± 0.099**	-	-
TabPFN-2.6	0.404 ± 0.091	-	-
TabICLv2	0.511 ± 0.072	-	-

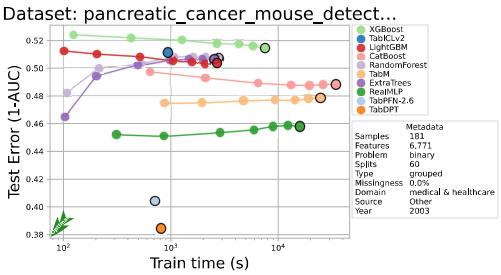

Figure G.104: **pancreatic_cancer_mouse_detection**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.200 ± 0.116	0.221 ± 0.131	0.227 ± 0.132
RandomForest	0.159 ± 0.113	0.179 ± 0.126	0.181 ± 0.125
ExtraTrees	0.144 ± 0.114	0.154 ± 0.115	0.155 ± 0.117
CatBoost	0.177 ± 0.125	0.166 ± 0.120	0.167 ± 0.119
LightGBM	0.213 ± 0.156	0.166 ± 0.128	0.170 ± 0.131
XGBoost	0.213 ± 0.135	0.209 ± 0.135	0.207 ± 0.135
RealMLP	0.211 ± 0.141	**0.126 ± 0.104**	**0.122 ± 0.100**
TabM	0.134 ± 0.089	0.157 ± 0.089	0.152 ± 0.090
TabDPT	0.232 ± 0.156	-	-
TabPFN-2.6	0.182 ± 0.126	-	-
TabICLv2	0.169 ± 0.138	-	-

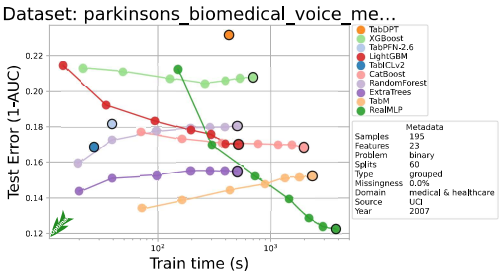

Figure G.105: **parkinsons_biomedical_voice_measurements**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	5.195 ± 0.028	5.185 ± 0.027	5.143 ± 0.027
RandomForest	3.589 ± 0.027	3.481 ± 0.027	3.476 ± 0.026
ExtraTrees	3.577 ± 0.023	3.469 ± 0.024	3.470 ± 0.024
CatBoost	3.525 ± 0.019	3.434 ± 0.026	3.405 ± 0.020
LightGBM	3.485 ± 0.022	3.448 ± 0.022	3.453 ± 0.022
XGBoost	3.520 ± 0.022	3.408 ± 0.021	3.409 ± 0.021
RealMLP	3.449 ± 0.038	3.185 ± 0.020	3.106 ± 0.024
TabM	3.419 ± 0.020	3.290 ± 0.028	3.247 ± 0.023
TabDPT	**2.898 ± 0.022**	-	-
TabPFN-2.6	3.011 ± 0.023	-	-
TabICLv2	2.944 ± 0.023	-	-

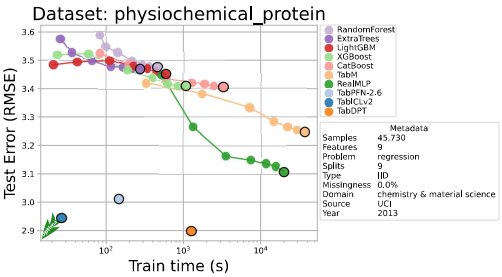

Figure G.106: **physiochemical_protein**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.134 ± 0.014	0.112 ± 0.013	0.105 ± 0.010
RandomForest	0.074 ± 0.009	0.069 ± 0.013	0.065 ± 0.011
ExtraTrees	0.121 ± 0.012	0.107 ± 0.010	0.107 ± 0.008
CatBoost	0.041 ± 0.008	0.042 ± 0.008	0.041 ± 0.008
LightGBM	0.048 ± 0.010	0.045 ± 0.010	0.045 ± 0.011
XGBoost	0.043 ± 0.010	0.043 ± 0.009	0.044 ± 0.009
RealMLP	0.043 ± 0.010	0.037 ± 0.008	0.037 ± 0.009
TabM	0.052 ± 0.009	0.031 ± 0.007	0.033 ± 0.008
TabDPT	0.034 ± 0.008	-	-
TabPFN-2.6	**0.014 ± 0.006**	-	-
TabICLv2	0.018 ± 0.007	-	-

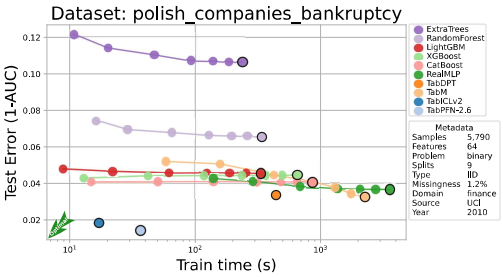

Figure G.107: **polish_companies_bankruptcy**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.369 ± 0.001**	**0.369 ± 0.001**	**0.370 ± 0.001**
RandomForest	**0.383 ± 0.002**	**0.379 ± 0.004**	**0.373 ± 0.004**
ExtraTrees	**0.379 ± 0.002**	**0.380 ± 0.003**	**0.374 ± 0.004**
CatBoost	**0.359 ± 0.001**	**0.357 ± 0.000**	**0.357 ± 0.000**
LightGBM	**0.369 ± 0.001**	**0.360 ± 0.001**	**0.360 ± 0.000**
XGBoost	**0.359 ± 0.001**	**0.360 ± 0.002**	**0.359 ± 0.000**
RealMLP	**0.361 ± 0.001**	**0.358 ± 0.001**	**0.357 ± 0.001**
TabM	**0.359 ± 0.001**	**0.358 ± 0.001**	**0.357 ± 0.001**
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	**0.366 ± 0.001**	-	-

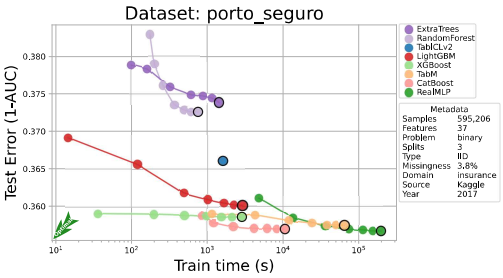

Figure G.108: **porto_seguro**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.567 ± 0.016	0.562 ± 0.015	0.559 ± 0.014
RandomForest	0.583 ± 0.014	0.578 ± 0.017	0.574 ± 0.016
ExtraTrees	0.589 ± 0.013	0.573 ± 0.016	0.567 ± 0.016
CatBoost	0.551 ± 0.016	0.541 ± 0.019	0.540 ± 0.018
LightGBM	0.554 ± 0.018	0.542 ± 0.020	0.541 ± 0.018
XGBoost	0.556 ± 0.021	0.552 ± 0.018	0.550 ± 0.017
RealMLP	0.553 ± 0.017	0.542 ± 0.019	0.537 ± 0.017
TabM	0.545 ± 0.017	0.545 ± 0.018	0.541 ± 0.017
TabDPT	0.555 ± 0.020	-	-
TabPFN-2.6	**0.533 ± 0.019**	-	-
TabICLv2	**0.531 ± 0.017**	-	-

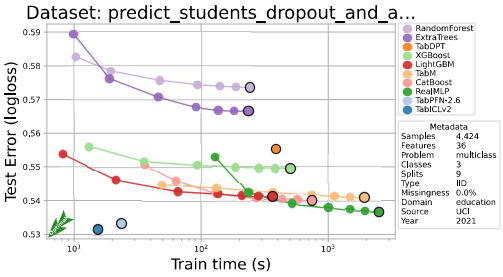

Figure G.109: **predict_students_dropout_and_academic_success**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.024 ± 0.016	0.027 ± 0.016	0.025 ± 0.016
RandomForest	0.026 ± 0.014	0.028 ± 0.015	0.022 ± 0.012
ExtraTrees	0.026 ± 0.015	0.020 ± 0.011	0.014 ± 0.008
CatBoost	0.020 ± 0.013	0.018 ± 0.011	0.018 ± 0.012
LightGBM	0.023 ± 0.014	0.017 ± 0.011	0.017 ± 0.011
XGBoost	0.032 ± 0.017	0.025 ± 0.015	0.025 ± 0.015
RealMLP	0.021 ± 0.016	0.018 ± 0.015	0.017 ± 0.015
TabM	0.025 ± 0.021	0.022 ± 0.019	0.022 ± 0.019
TabDPT	0.022 ± 0.011	-	-
TabPFN-2.6	**0.007 ± 0.005**	-	-
TabICLv2	0.056 ± 0.025	-	-

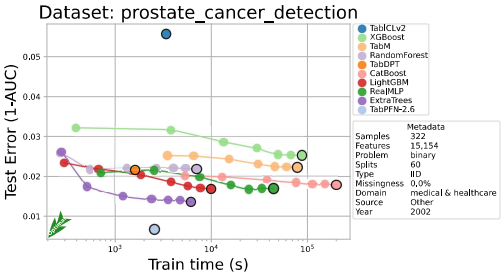

Figure G.110: **prostate_cancer_detection**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.409 ± 0.007	0.407 ± 0.007	0.404 ± 0.006
RandomForest	0.417 ± 0.004	0.390 ± 0.006	0.389 ± 0.007
ExtraTrees	0.403 ± 0.003	0.387 ± 0.005	0.385 ± 0.006
CatBoost	0.370 ± 0.006	0.369 ± 0.006	0.369 ± 0.006
LightGBM	0.383 ± 0.006	**0.368 ± 0.007**	**0.368 ± 0.006**
XGBoost	0.374 ± 0.007	0.369 ± 0.006	0.369 ± 0.006
RealMLP	0.373 ± 0.004	0.372 ± 0.004	0.372 ± 0.005
TabM	0.376 ± 0.004	0.374 ± 0.005	0.374 ± 0.005
TabDPT	0.400 ± 0.005	-	-
TabPFN-2.6	0.399 ± 0.006	-	-
TabICLv2	0.397 ± 0.003	-	-

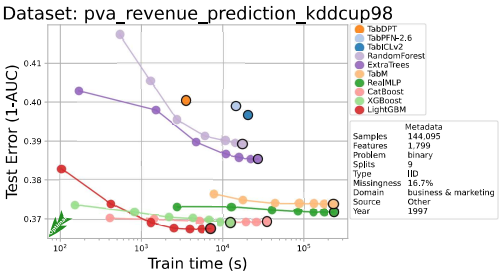

Figure G.111: **pva_revenue_prediction_kddcup98**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.290 ± 0.074	1.219 ± 0.071	1.221 ± 0.071
RandomForest	1.140 ± 0.072	1.124 ± 0.068	1.120 ± 0.067
ExtraTrees	1.108 ± 0.069	1.096 ± 0.066	1.095 ± 0.067
CatBoost	1.082 ± 0.065	1.083 ± 0.065	1.082 ± 0.065
LightGBM	1.134 ± 0.069	1.116 ± 0.073	1.117 ± 0.069
XGBoost	1.137 ± 0.075	1.115 ± 0.067	1.114 ± 0.067
RealMLP	1.068 ± 0.065	1.068 ± 0.065	1.067 ± 0.066
TabM	1.158 ± 0.066	1.137 ± 0.068	1.122 ± 0.065
TabDPT	**1.039 ± 0.069**	-	-
TabPFN-2.6	1.053 ± 0.071	-	-
TabICLv2	1.050 ± 0.069	-	-

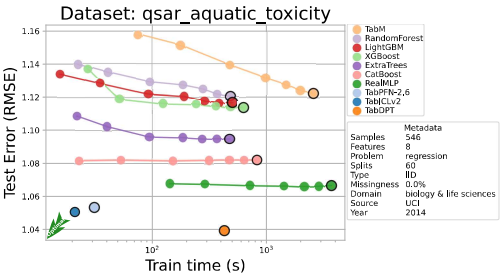

Figure G.112: **qsar_aquatic_toxicity**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.092 ± 0.012	0.078 ± 0.013	0.077 ± 0.012
RandomForest	0.069 ± 0.010	0.069 ± 0.011	0.071 ± 0.011
ExtraTrees	0.066 ± 0.009	0.067 ± 0.011	0.066 ± 0.010
CatBoost	0.069 ± 0.011	0.069 ± 0.011	0.068 ± 0.011
LightGBM	0.074 ± 0.011	0.068 ± 0.012	0.068 ± 0.011
XGBoost	0.074 ± 0.012	0.069 ± 0.010	0.069 ± 0.011
RealMLP	0.072 ± 0.011	0.069 ± 0.012	0.067 ± 0.011
TabM	0.069 ± 0.011	0.066 ± 0.012	0.064 ± 0.010
TabDPT	0.064 ± 0.010	-	-
TabPFN-2.6	0.063 ± 0.010	-	-
TabICLv2	**0.059 ± 0.009**	-	-

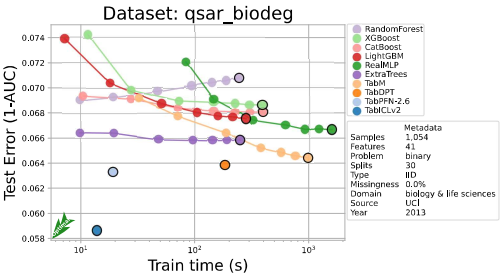

Figure G.113: **qsar_biodeg**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.947 ± 0.049	0.947 ± 0.049	0.947 ± 0.049
RandomForest	0.902 ± 0.053	0.887 ± 0.049	0.885 ± 0.050
ExtraTrees	0.874 ± 0.051	0.871 ± 0.051	0.867 ± 0.050
CatBoost	0.876 ± 0.052	0.874 ± 0.052	0.874 ± 0.050
LightGBM	0.898 ± 0.045	0.890 ± 0.048	0.886 ± 0.048
XGBoost	0.906 ± 0.050	0.881 ± 0.053	0.879 ± 0.053
RealMLP	0.874 ± 0.048	0.874 ± 0.050	0.863 ± 0.049
TabM	0.909 ± 0.050	0.897 ± 0.055	0.893 ± 0.052
TabDPT	**0.853 ± 0.047**	-	-
TabPFN-2.6	0.857 ± 0.050	-	-
TabICLv2	**0.855 ± 0.052**	-	-

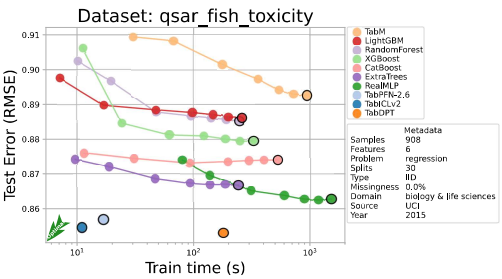

Figure G.114: **qsar_fish_toxicity**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.995 ± 0.019	0.923 ± 0.015	0.921 ± 0.013
RandomForest	0.789 ± 0.011	0.771 ± 0.014	0.772 ± 0.010
ExtraTrees	0.789 ± 0.011	0.774 ± 0.014	0.765 ± 0.013
CatBoost	0.749 ± 0.016	0.751 ± 0.013	0.748 ± 0.015
LightGBM	0.748 ± 0.016	**0.728 ± 0.014**	0.731 ± 0.013
XGBoost	0.763 ± 0.016	0.738 ± 0.012	0.735 ± 0.012
RealMLP	0.737 ± 0.019	0.736 ± 0.017	**0.730 ± 0.017**
TabM	0.739 ± 0.016	0.738 ± 0.018	**0.734 ± 0.018**
TabDPT	0.738 ± 0.020	-	-
TabPFN-2.6	**0.733 ± 0.017**	-	-
TabICLv2	0.792 ± 0.014	-	-

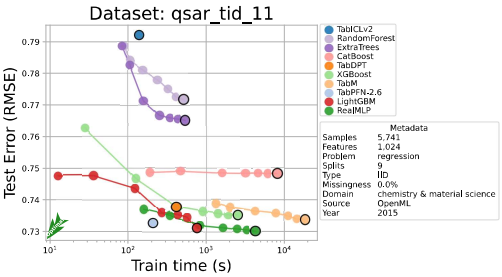

Figure G.115: **qsar_tid_11**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.094 ± 0.023	0.084 ± 0.020	0.085 ± 0.020
RandomForest	0.079 ± 0.016	0.081 ± 0.016	0.081 ± 0.018
ExtraTrees	0.082 ± 0.014	0.083 ± 0.015	0.082 ± 0.015
CatBoost	0.080 ± 0.018	0.081 ± 0.018	0.081 ± 0.017
LightGBM	0.092 ± 0.020	0.076 ± 0.015	0.076 ± 0.016
XGBoost	0.100 ± 0.020	0.084 ± 0.018	0.084 ± 0.018
RealMLP	0.091 ± 0.019	0.086 ± 0.021	0.086 ± 0.021
TabM	0.088 ± 0.020	0.087 ± 0.020	0.086 ± 0.020
TabDPT	0.094 ± 0.019	-	-
TabPFN-2.6	**0.071 ± 0.015**	-	-
TabICLv2	**0.069 ± 0.017**	-	-

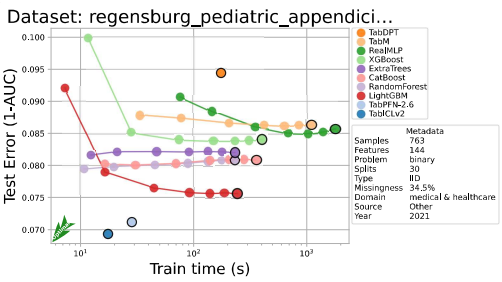

Figure G.116: **regensburg_pediatric_appendicitis**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**1450.3 ± 120.2**	**1450.4 ± 120.5**	**1450.4 ± 120**
RandomForest	**1129.3 ± 130.2**	**1128.8 ± 135.7**	**1127.1 ± 135**
ExtraTrees	**1197.7 ± 190.7**	**1183.6 ± 193.1**	**1182.2 ± 191**
CatBoost	**1059.2 ± 129.4**	**1021.1 ± 143.5**	**1018.7 ± 131**
LightGBM	**1027.3 ± 136.4**	**1007.3 ± 136.9**	**999.0 ± 141.**
XGBoost	**1048.2 ± 138.4**	**1011.4 ± 107.5**	**1009.2 ± 112**
RealMLP	**1029.4 ± 242.0**	**1006.1 ± 216.9**	**993.7 ± 217.**
TabM	**1110.7 ± 103.4**	**1090.7 ± 86.2**	**1021.3 ± 104**
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	**2151.7 ± 169.8**	-	-

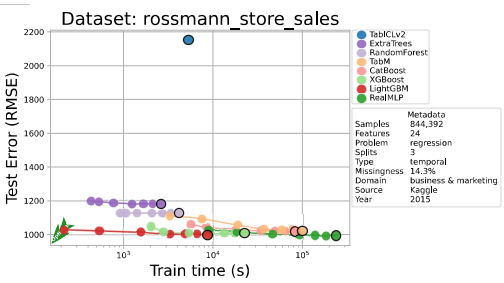

Figure G.117: **rossmann_store_sales**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.184 ± 0.003	0.183 ± 0.004	0.183 ± 0.004
RandomForest	0.225 ± 0.008	0.169 ± 0.007	0.167 ± 0.006
ExtraTrees	0.236 ± 0.008	0.178 ± 0.005	0.176 ± 0.005
CatBoost	0.162 ± 0.005	0.161 ± 0.005	0.161 ± 0.005
LightGBM	0.163 ± 0.005	0.161 ± 0.005	0.160 ± 0.005
XGBoost	0.163 ± 0.005	0.160 ± 0.005	0.160 ± 0.005
RealMLP	0.160 ± 0.005	0.159 ± 0.005	0.159 ± 0.005
TabM	0.161 ± 0.005	0.159 ± 0.005	0.159 ± 0.005
TabDPT	0.185 ± 0.005	-	-
TabPFN-2.6	**0.158 ± 0.005**	-	-
TabICLv2	0.183 ± 0.011	-	-

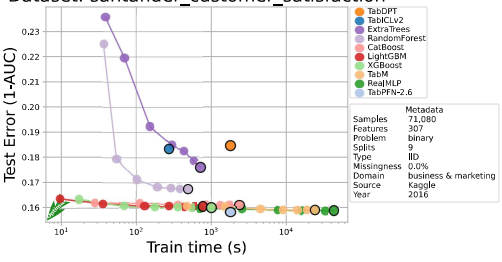

Figure G.118: **santander_customer_satisfaction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.140 ± 0.002	0.140 ± 0.002	0.140 ± 0.002
RandomForest	0.141 ± 0.002	0.141 ± 0.002	0.143 ± 0.002
ExtraTrees	0.166 ± 0.001	0.175 ± 0.002	0.138 ± 0.002
CatBoost	0.093 ± 0.002	0.092 ± 0.002	0.092 ± 0.002
LightGBM	0.097 ± 0.002	**0.091 ± 0.002**	**0.091 ± 0.002**
XGBoost	0.099 ± 0.002	0.093 ± 0.002	0.093 ± 0.002
RealMLP	0.113 ± 0.002	0.103 ± 0.001	0.102 ± 0.002
TabM	0.103 ± 0.002	0.103 ± 0.002	0.102 ± 0.002
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.116 ± 0.002	-	-

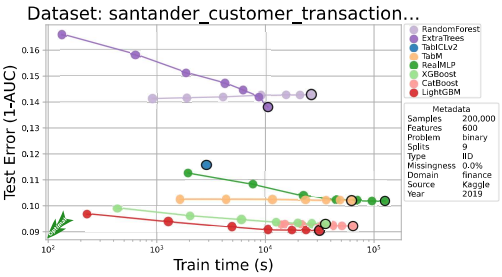

Figure G.119: **santander_customer_transaction_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.486 ± 0.019	1.413 ± 0.017	1.404 ± 0.015
RandomForest	1.341 ± 0.023	1.338 ± 0.019	1.337 ± 0.020
ExtraTrees	1.344 ± 0.017	1.345 ± 0.016	1.344 ± 0.016
CatBoost	1.350 ± 0.014	1.350 ± 0.015	1.348 ± 0.016
LightGBM	1.349 ± 0.020	1.348 ± 0.019	1.344 ± 0.019
XGBoost	1.346 ± 0.020	1.343 ± 0.023	1.342 ± 0.021
RealMLP	1.379 ± 0.017	1.362 ± 0.021	1.347 ± 0.019
TabM	1.363 ± 0.018	1.363 ± 0.018	1.360 ± 0.018
TabDPT	1.445 ± 0.022	-	-
TabPFN-2.6	1.329 ± 0.019	-	-
TabICLv2	**1.323 ± 0.020**	-	-

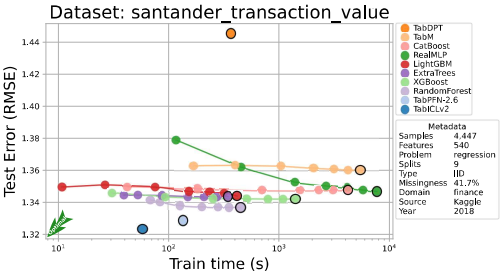

Figure G.120: **santander_transaction_value**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	3.757 ± 0.485	2.666 ± 0.232	2.722 ± 0.259
RandomForest	2.189 ± 0.247	2.141 ± 0.217	2.138 ± 0.216
ExtraTrees	2.064 ± 0.183	2.065 ± 0.173	2.054 ± 0.177
CatBoost	2.111 ± 0.204	2.098 ± 0.217	2.083 ± 0.205
LightGBM	2.236 ± 0.201	2.046 ± 0.181	2.058 ± 0.180
XGBoost	2.308 ± 0.228	2.115 ± 0.208	2.115 ± 0.202
RealMLP	2.110 ± 0.221	2.105 ± 0.210	2.086 ± 0.200
TabM	2.101 ± 0.189	2.111 ± 0.195	2.083 ± 0.194
TabDPT	2.413 ± 0.224	-	-
TabPFN-2.6	**1.976 ± 0.173**	-	-
TabICLv2	2.136 ± 0.171	-	-

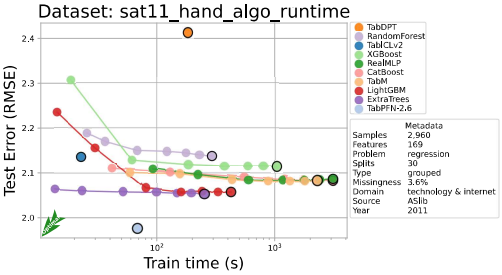

Figure G.121: **sat11_hand_algo_runtime**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.579 ± 0.360**	**0.276 ± 0.021**	**0.277 ± 0.031**
RandomForest	**0.273 ± 0.021**	**0.273 ± 0.020**	**0.267 ± 0.019**
ExtraTrees	**0.262 ± 0.018**	**0.262 ± 0.019**	**0.260 ± 0.018**
CatBoost	**0.252 ± 0.018**	**0.255 ± 0.018**	**0.254 ± 0.017**
LightGBM	**0.254 ± 0.016**	**0.253 ± 0.016**	**0.251 ± 0.018**
XGBoost	**0.254 ± 0.017**	**0.250 ± 0.018**	**0.250 ± 0.017**
RealMLP	**0.252 ± 0.017**	**0.248 ± 0.013**	**0.247 ± 0.015**
TabM	**0.258 ± 0.015**	**0.255 ± 0.013**	**0.254 ± 0.013**
TabDPT	**0.261 ± 0.022**	-	-
TabPFN-2.6	**0.243 ± 0.016**	-	-
TabICLv2	**0.285 ± 0.037**	-	-

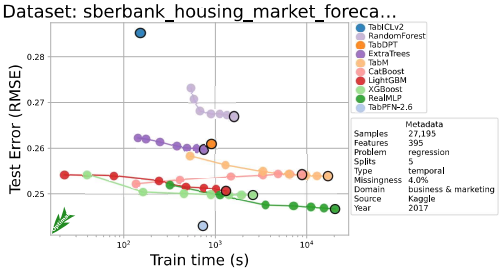

Figure G.122: **sberbank_housing_market_forecasting**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.147 ± 0.003	0.147 ± 0.003	0.136 ± 0.003
RandomForest	0.085 ± 0.002	0.077 ± 0.003	0.077 ± 0.003
ExtraTrees	0.131 ± 0.002	0.085 ± 0.002	0.081 ± 0.002
CatBoost	0.075 ± 0.002	0.074 ± 0.002	0.074 ± 0.002
LightGBM	0.088 ± 0.002	0.088 ± 0.002	0.084 ± 0.002
XGBoost	0.074 ± 0.002	0.074 ± 0.002	0.073 ± 0.002
RealMLP	0.104 ± 0.004	0.087 ± 0.002	0.088 ± 0.002
TabM	0.097 ± 0.003	0.085 ± 0.002	0.085 ± 0.002
TabDPT	0.085 ± 0.002	-	-
TabPFN-2.6	**0.069 ± 0.002**	-	-
TabICLv2	0.071 ± 0.003	-	-

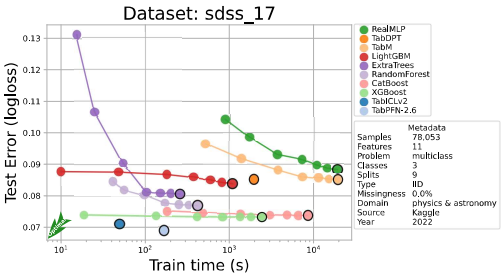

Figure G.123: **sdss_17**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.242 ± 0.024	0.241 ± 0.023	0.238 ± 0.023
RandomForest	0.249 ± 0.022	0.236 ± 0.027	0.235 ± 0.025
ExtraTrees	0.251 ± 0.024	0.235 ± 0.026	0.226 ± 0.024
CatBoost	0.225 ± 0.022	0.228 ± 0.022	0.228 ± 0.022
LightGBM	0.247 ± 0.025	0.231 ± 0.021	0.230 ± 0.021
XGBoost	0.240 ± 0.025	0.231 ± 0.020	0.230 ± 0.020
RealMLP	0.239 ± 0.026	0.238 ± 0.024	0.232 ± 0.023
TabM	0.234 ± 0.024	0.234 ± 0.026	0.233 ± 0.024
TabDPT	**0.222 ± 0.023**	-	-
TabPFN-2.6	**0.220 ± 0.024**	-	-
TabICLv2	**0.227 ± 0.035**	-	-

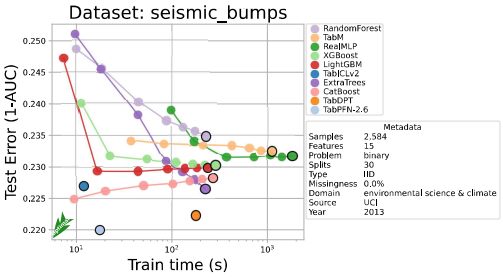

Figure G.124: **seismic_bumps**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.304	0.280	0.280
RandomForest	0.196	0.196	0.195
ExtraTrees	0.205	0.205	0.202
CatBoost	0.190	0.188	0.188
LightGBM	0.187	0.179	0.178
XGBoost	0.185	0.181	0.181
RealMLP	0.186	0.183	0.182
TabM	0.181	0.181	0.178
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	0.201	-	-

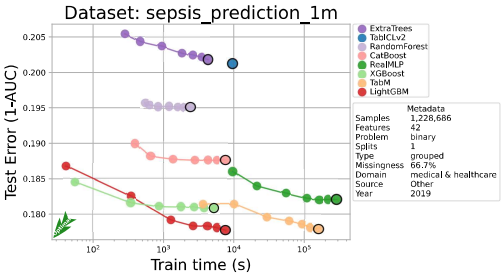

Figure G.125: **sepsis_prediction_1m**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.294 ± 0.001	0.294 ± 0.002	0.294 ± 0.002
RandomForest	0.311 ± 0.002	0.294 ± 0.002	0.294 ± 0.002
ExtraTrees	0.311 ± 0.002	0.294 ± 0.001	0.294 ± 0.002
CatBoost	0.294 ± 0.002	0.293 ± 0.002	0.293 ± 0.002
LightGBM	0.297 ± 0.002	**0.293 ± 0.002**	**0.293 ± 0.002**
XGBoost	0.297 ± 0.002	0.294 ± 0.001	0.294 ± 0.002
RealMLP	**0.293 ± 0.002**	**0.293 ± 0.002**	**0.293 ± 0.002**
TabM	0.294 ± 0.002	0.294 ± 0.002	0.293 ± 0.002
TabDPT	0.296 ± 0.002	-	-
TabPFN-2.6	**0.293 ± 0.002**	-	-
TabICLv2	0.301 ± 0.007	-	-

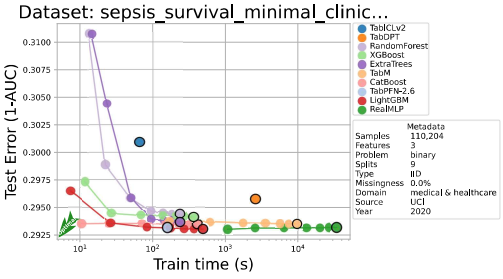

Figure G.126: **sepsis_survival_minimal_clinical_records**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	1.416 ± 0.062	1.420 ± 0.067	1.409 ± 0.069
RandomForest	**1.344 ± 0.073**	**1.342 ± 0.075**	**1.341 ± 0.074**
ExtraTrees	1.360 ± 0.091	1.360 ± 0.091	1.355 ± 0.091
CatBoost	**1.315 ± 0.068**	**1.315 ± 0.068**	**1.315 ± 0.068**
LightGBM	**1.321 ± 0.065**	**1.312 ± 0.069**	**1.315 ± 0.071**
XGBoost	**1.323 ± 0.067**	**1.317 ± 0.070**	**1.316 ± 0.068**
RealMLP	**1.349 ± 0.098**	**1.332 ± 0.089**	**1.334 ± 0.091**
TabM	**1.439 ± 0.199**	**1.447 ± 0.210**	**1.413 ± 0.148**
TabDPT	-	-	-
TabPFN-2.6	-	-	-
TabICLv2	**1.348 ± 0.081**	-	-

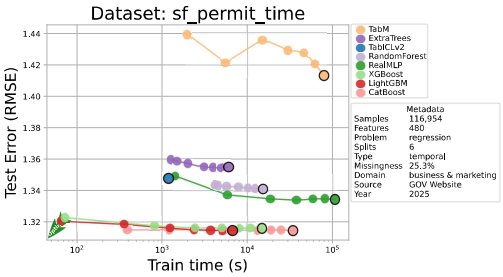

Figure G.127: **sf_permit_time**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	**0.228 ± 0.036**	0.231 ± 0.036	**0.227 ± 0.036**
RandomForest	0.266 ± 0.032	0.258 ± 0.035	0.259 ± 0.035
ExtraTrees	0.253 ± 0.034	0.240 ± 0.037	0.241 ± 0.037
CatBoost	0.246 ± 0.036	0.244 ± 0.036	0.244 ± 0.035
LightGBM	0.257 ± 0.033	0.239 ± 0.035	0.239 ± 0.035
XGBoost	0.278 ± 0.031	0.261 ± 0.034	0.260 ± 0.034
RealMLP	0.239 ± 0.036	0.241 ± 0.036	0.240 ± 0.036
TabM	0.245 ± 0.036	0.250 ± 0.034	0.247 ± 0.036
TabDPT	**0.229 ± 0.038**	-	-
TabPFN-2.6	0.240 ± 0.035	-	-
TabICLv2	0.230 ± 0.037	-	-

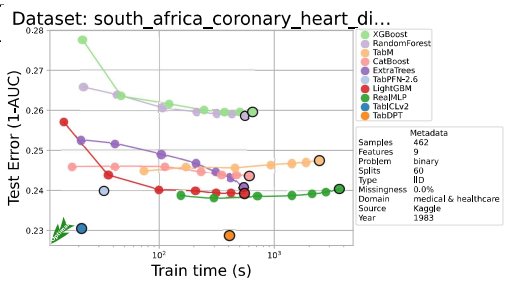

Figure G.128: **south_africa_coronary_heart_disease**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.170 ± 0.013	0.150 ± 0.010	0.149 ± 0.009
RandomForest	0.327 ± 0.008	0.180 ± 0.015	0.177 ± 0.014
ExtraTrees	0.403 ± 0.009	0.174 ± 0.016	0.172 ± 0.014
CatBoost	0.111 ± 0.014	0.114 ± 0.015	0.109 ± 0.015
LightGBM	0.110 ± 0.014	0.105 ± 0.014	0.105 ± 0.014
XGBoost	0.117 ± 0.017	0.107 ± 0.015	0.107 ± 0.015
RealMLP	0.128 ± 0.019	0.105 ± 0.014	0.104 ± 0.014
TabM	0.112 ± 0.015	0.112 ± 0.015	0.111 ± 0.015
TabDPT	0.200 ± 0.023	-	-
TabPFN-2.6	**0.098 ± 0.016**	-	-
TabICLv2	**0.096 ± 0.016**	-	-

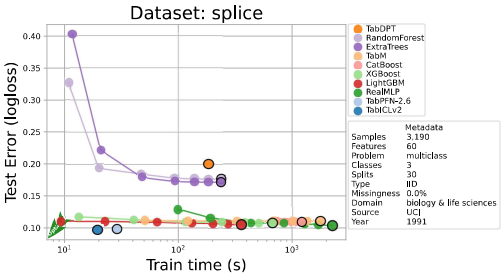

Figure G.129: **splice**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	2.768 ± 0.175	2.769 ± 0.175	2.729 ± 0.175
RandomForest	2.712 ± 0.174	2.676 ± 0.179	2.676 ± 0.177
ExtraTrees	2.681 ± 0.167	2.698 ± 0.171	2.683 ± 0.170
CatBoost	2.699 ± 0.183	2.696 ± 0.177	2.694 ± 0.176
LightGBM	2.713 ± 0.178	2.707 ± 0.173	2.694 ± 0.174
XGBoost	2.763 ± 0.175	2.705 ± 0.182	2.704 ± 0.181
RealMLP	2.687 ± 0.180	2.689 ± 0.183	2.680 ± 0.183
TabM	2.692 ± 0.175	2.683 ± 0.177	2.682 ± 0.176
TabDPT	**2.666 ± 0.178**	-	-
TabPFN-2.6	**2.661 ± 0.181**	-	-
TabICLv2	**2.657 ± 0.190**	-	-

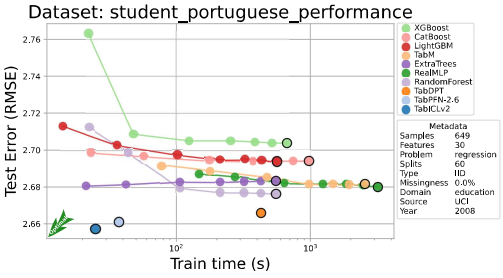

Figure G.130: **student_portuguese_performance**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	17.43 ± 0.09	17.42 ± 0.09	17.26 ± 0.08
RandomForest	9.65 ± 0.08	9.46 ± 0.11	9.48 ± 0.10
ExtraTrees	9.46 ± 0.13	9.34 ± 0.12	9.34 ± 0.12
CatBoost	9.33 ± 0.14	9.33 ± 0.13	9.31 ± 0.13
LightGBM	9.36 ± 0.11	9.34 ± 0.12	9.26 ± 0.12
XGBoost	9.41 ± 0.11	9.31 ± 0.12	9.31 ± 0.13
RealMLP	9.46 ± 0.17	9.38 ± 0.15	9.22 ± 0.16
TabM	9.44 ± 0.17	9.20 ± 0.18	9.21 ± 0.17
TabDPT	**8.93 ± 0.14**	-	-
TabPFN-2.6	9.11 ± 0.16	-	-
TabICLv2	**8.87 ± 0.17**	-	-

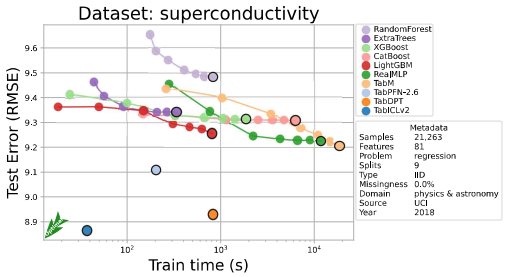

Figure G.131: **superconductivity**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.063 ± 0.015	0.060 ± 0.015	0.060 ± 0.014
RandomForest	0.062 ± 0.012	0.062 ± 0.012	0.063 ± 0.012
ExtraTrees	0.059 ± 0.010	0.059 ± 0.010	0.060 ± 0.010
CatBoost	0.054 ± 0.012	0.054 ± 0.012	0.054 ± 0.012
LightGBM	0.059 ± 0.015	0.052 ± 0.012	0.052 ± 0.013
XGBoost	0.058 ± 0.015	0.054 ± 0.013	0.054 ± 0.013
RealMLP	0.060 ± 0.012	0.057 ± 0.013	0.054 ± 0.013
TabM	0.056 ± 0.012	0.054 ± 0.012	0.054 ± 0.011
TabDPT	0.056 ± 0.010	-	-
TabPFN-2.6	**0.048 ± 0.009**	-	-
TabICLv2	0.051 ± 0.009	-	-

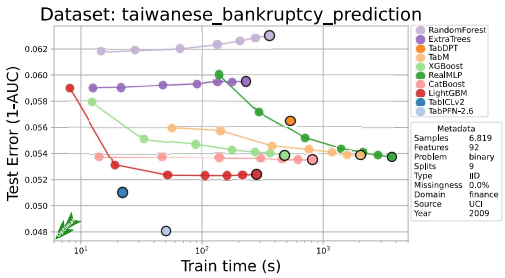

Figure G.132: **taiwanese_bankruptcy_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	11.94 ± 1.46	11.63 ± 1.52	**11.34 ± 1.54**
RandomForest	11.85 ± 1.53	11.46 ± 1.49	11.48 ± 1.48
ExtraTrees	11.39 ± 1.52	**11.24 ± 1.60**	**11.23 ± 1.60**
CatBoost	**11.08 ± 1.70**	**11.09 ± 1.68**	**11.08 ± 1.67**
LightGBM	11.15 ± 1.64	**11.10 ± 1.60**	**11.08 ± 1.60**
XGBoost	**11.13 ± 1.71**	11.16 ± 1.68	11.15 ± 1.68
RealMLP	**11.17 ± 1.47**	**11.23 ± 1.55**	**11.09 ± 1.57**
TabM	11.29 ± 1.42	**11.14 ± 1.52**	**11.15 ± 1.51**
TabDPT	12.01 ± 1.45	-	-
TabPFN-2.6	12.30 ± 1.66	-	-
TabICLv2	11.79 ± 1.52	-	-

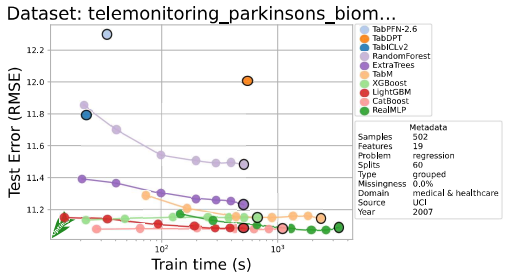

Figure G.133: **telemonitoring_parkinsons_biomedical_voice_measurements**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.164 ± 0.044	0.169 ± 0.043	0.170 ± 0.045
RandomForest	0.022 ± 0.022	0.024 ± 0.027	0.024 ± 0.025
ExtraTrees	0.021 ± 0.012	0.014 ± 0.007	0.014 ± 0.007
CatBoost	0.019 ± 0.012	**0.014 ± 0.010**	0.016 ± 0.015
LightGBM	0.028 ± 0.024	0.027 ± 0.021	0.021 ± 0.015
XGBoost	0.022 ± 0.010	0.024 ± 0.013	0.022 ± 0.011
RealMLP	0.027 ± 0.025	0.024 ± 0.019	0.016 ± 0.012
TabM	0.030 ± 0.026	0.032 ± 0.026	0.025 ± 0.017
TabDPT	0.018 ± 0.011	-	-
TabPFN-2.6	0.013 ± 0.007	-	-
TabICLv2	**0.011 ± 0.005**	-	-

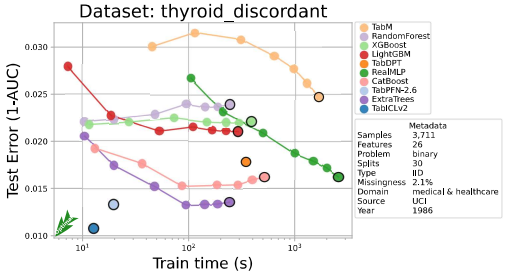

Figure G.134: **thyroid_discordant**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.166 ± 0.023	0.163 ± 0.024	0.163 ± 0.024
RandomForest	0.051 ± 0.010	**0.040 ± 0.011**	**0.040 ± 0.010**
ExtraTrees	0.058 ± 0.011	0.055 ± 0.011	0.051 ± 0.010
CatBoost	0.042 ± 0.008	0.042 ± 0.009	0.041 ± 0.009
LightGBM	0.041 ± 0.009	0.042 ± 0.010	0.042 ± 0.010
XGBoost	**0.040 ± 0.009**	**0.040 ± 0.009**	0.041 ± 0.009
RealMLP	0.054 ± 0.011	0.042 ± 0.009	0.042 ± 0.009
TabM	0.047 ± 0.011	0.044 ± 0.011	0.043 ± 0.010
TabDPT	**0.039 ± 0.010**	-	-
TabPFN-2.6	0.041 ± 0.009	-	-
TabICLv2	**0.040 ± 0.009**	-	-

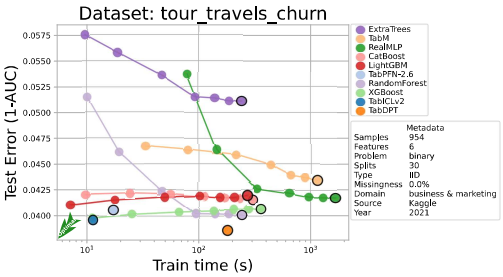

Figure G.135: **tour_travels_churn**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.031 ± 0.003	0.025 ± 0.003	0.025 ± 0.003
RandomForest	0.032 ± 0.005	0.034 ± 0.006	0.033 ± 0.005
ExtraTrees	0.033 ± 0.005	0.033 ± 0.005	0.034 ± 0.005
CatBoost	0.019 ± 0.008	0.008 ± 0.005	0.008 ± 0.005
LightGBM	0.023 ± 0.005	**0.002 ± 0.000**	**0.002 ± 0.000**
XGBoost	0.025 ± 0.004	0.019 ± 0.004	0.019 ± 0.004
RealMLP	0.010 ± 0.004	**0.006 ± 0.005**	**0.006 ± 0.005**
TabM	0.004 ± 0.001	0.003 ± 0.001	0.003 ± 0.001
TabDPT	0.015 ± 0.003	-	-
TabPFN-2.6	0.014 ± 0.002	-	-
TabICLv2	0.009 ± 0.002	-	-

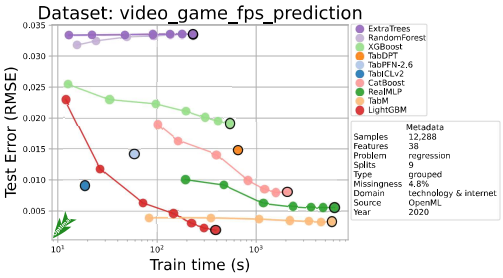

Figure G.136: **video_game_fps_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.432 ± 0.019	0.434 ± 0.017	0.423 ± 0.015
RandomForest	0.248 ± 0.029	0.236 ± 0.033	0.236 ± 0.032
ExtraTrees	0.227 ± 0.036	0.227 ± 0.036	0.228 ± 0.036
CatBoost	0.220 ± 0.028	0.221 ± 0.029	0.218 ± 0.027
LightGBM	0.224 ± 0.026	**0.226 ± 0.030**	0.222 ± 0.028
XGBoost	0.230 ± 0.028	0.224 ± 0.031	0.223 ± 0.029
RealMLP	**0.214 ± 0.030**	0.214 ± 0.034	**0.208 ± 0.032**
TabM	0.222 ± 0.027	**0.221 ± 0.033**	**0.217 ± 0.034**
TabDPT	0.309 ± 0.023	-	-
TabPFN-2.6	0.263 ± 0.018	-	-
TabICLv2	0.246 ± 0.019	-	-

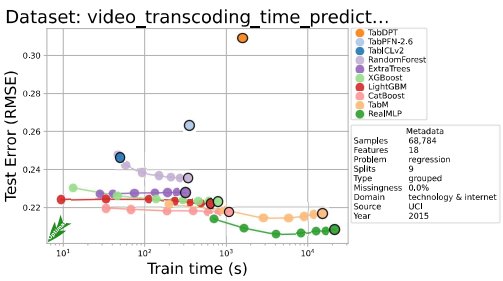

Figure G.137: **video_transcoding_time_prediction**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.356 ± 0.022	0.356 ± 0.025	0.354 ± 0.024
RandomForest	0.294 ± 0.025	0.261 ± 0.020	0.259 ± 0.019
ExtraTrees	0.295 ± 0.023	0.268 ± 0.022	0.258 ± 0.019
CatBoost	0.255 ± 0.019	0.239 ± 0.022	0.239 ± 0.021
LightGBM	0.258 ± 0.020	0.257 ± 0.021	0.257 ± 0.020
XGBoost	0.260 ± 0.022	0.256 ± 0.020	0.256 ± 0.020
RealMLP	0.259 ± 0.032	0.239 ± 0.021	0.241 ± 0.022
TabM	0.247 ± 0.026	0.238 ± 0.026	0.238 ± 0.025
TabDPT	**0.223 ± 0.021**	-	-
TabPFN-2.6	**0.221 ± 0.022**	-	-
TabICLv2	**0.221 ± 0.023**	-	-

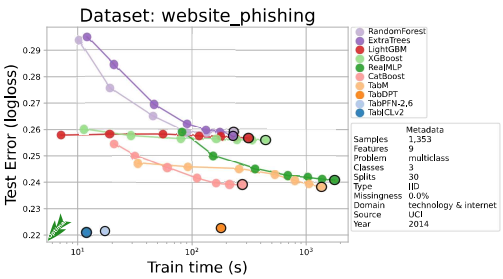

Figure G.138: **website_phishing**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.160 ± 0.005	0.160 ± 0.005	0.159 ± 0.005
RandomForest	0.165 ± 0.001	0.157 ± 0.002	0.157 ± 0.001
ExtraTrees	0.172 ± 0.001	0.157 ± 0.001	0.155 ± 0.001
CatBoost	0.128 ± 0.001	0.127 ± 0.001	0.127 ± 0.001
LightGBM	0.133 ± 0.001	0.125 ± 0.001	0.125 ± 0.001
XGBoost	0.130 ± 0.001	0.125 ± 0.001	0.125 ± 0.001
RealMLP	0.126 ± 0.001	0.123 ± 0.001	0.122 ± 0.001
TabM	0.124 ± 0.001	0.123 ± 0.001	**0.122 ± 0.001**
TabDPT	0.190 ± 0.002	-	-
TabPFN-2.6	0.126 ± 0.001	-	-
TabICLv2	0.143 ± 0.002	-	-

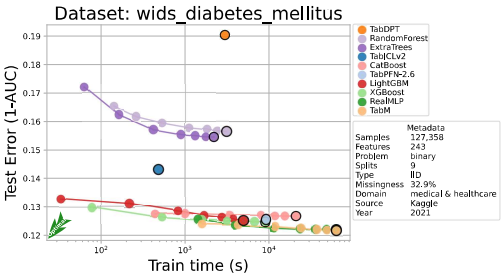

Figure G.139: **wids_diabetes_mellitus**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.730 ± 0.006	0.730 ± 0.006	0.729 ± 0.006
RandomForest	0.619 ± 0.009	0.601 ± 0.008	0.605 ± 0.008
ExtraTrees	0.614 ± 0.009	0.597 ± 0.010	0.598 ± 0.009
CatBoost	0.615 ± 0.009	0.599 ± 0.009	0.599 ± 0.009
LightGBM	0.623 ± 0.010	0.610 ± 0.010	0.610 ± 0.009
XGBoost	0.617 ± 0.008	0.605 ± 0.010	0.606 ± 0.010
RealMLP	0.617 ± 0.009	0.604 ± 0.007	0.594 ± 0.008
TabM	0.629 ± 0.009	0.615 ± 0.009	0.609 ± 0.008
TabDPT	**0.584 ± 0.008**	-	-
TabPFN-2.6	0.588 ± 0.007	-	-
TabICLv2	0.587 ± 0.008	-	-

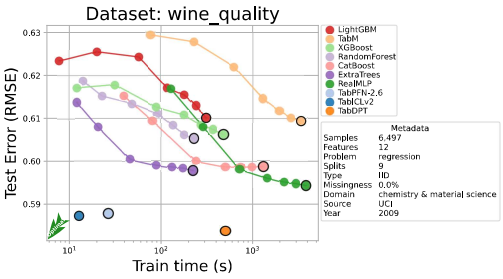

Figure G.140: **wine_quality**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	0.471 ± 0.018	0.424 ± 0.018	0.417 ± 0.019
RandomForest	0.448 ± 0.024	0.427 ± 0.023	0.429 ± 0.023
ExtraTrees	0.435 ± 0.023	0.419 ± 0.023	0.419 ± 0.023
CatBoost	0.395 ± 0.024	0.393 ± 0.023	0.393 ± 0.024
LightGBM	0.409 ± 0.021	0.379 ± 0.023	0.379 ± 0.023
XGBoost	0.405 ± 0.022	0.398 ± 0.022	0.395 ± 0.022
RealMLP	0.371 ± 0.023	0.365 ± 0.023	0.363 ± 0.022
TabM	0.389 ± 0.023	0.375 ± 0.023	0.374 ± 0.023
TabDPT	0.386 ± 0.027	-	-
TabPFN-2.6	**0.359 ± 0.019**	-	-
TabICLv2	0.385 ± 0.025	-	-

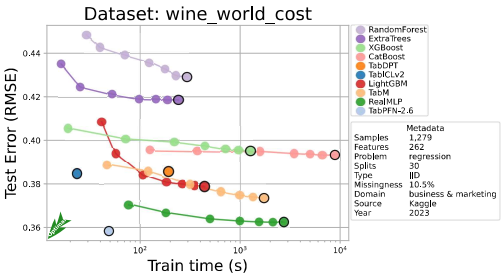

Figure G.141: **wine_world_cost**: per-method test error (left) and HPO Pareto trajectory (right).

	Default	Tuned	Tuned + Ens.
Linear	5.5 ± 0.2	5.0 ± 0.2	4.9 ± 0.2
RandomForest	4.3 ± 0.1	4.0 ± 0.1	4.0 ± 0.1
ExtraTrees	4.2 ± 0.1	4.0 ± 0.1	4.0 ± 0.1
CatBoost	**3.9 ± 0.1**	**3.9 ± 0.1**	**3.9 ± 0.1**
LightGBM	3.9 ± 0.1	**3.9 ± 0.1**	**3.9 ± 0.1**
XGBoost	4.0 ± 0.1	**3.9 ± 0.1**	**3.9 ± 0.1**
RealMLP	3.9 ± 0.1	**3.9 ± 0.1**	**3.9 ± 0.1**
TabM	**369.8 ± 1086.7**	**11.1 ± 14.8**	**3.9 ± 0.1**
TabDPT	4.3 ± 0.1	-	-
TabPFN-2.6	4.0 ± 0.2	-	-
TabICLv2	4.0 ± 0.1	-	-

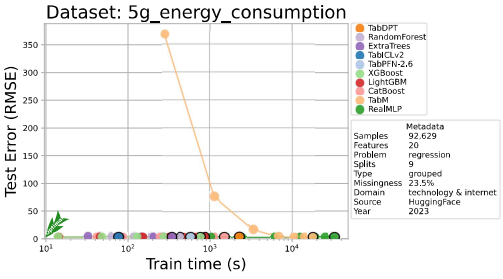

Figure G.142: **5g_energy_consumption**: per-method test error (left) and HPO Pareto trajectory (right).

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    Answer: [N/A]

    Justification: Our contributions do not involve LLMs in any components.

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